STRUCTURAL ALLOYS HANDBOOK

2-1/4 Cr - 1 Mo STEELS

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GENERAL

2-1/4Cr-1Mo steel is a low-carbon, low-alloy ferritic steel that has excellent creep-resistance properties at temperatures up to 1100 F. The alloy is used extensively in the utility industry for the fabrication of parts for steam-generating boilers and pressure vessels where temperatures typically range between 900 and 1100 F. The material also is used in the petrochemical industry, where thermal reforming and hydrocracking processes are carried out at high pressures and temperatures.

2-1/4Cr-1Mo steel is used primarily in the annealed and normalized-and-tempered heat-treated conditions. In these conditions the material has very good long-term creep-rupture properties between 1000 and 1100 F. For service conditions requiring lower temperatures and higher short-term tensile strengths, the alloy is given a quench-and-tempered heat treatment.

2-1/4Cr-1Mo steel has a number of attractive properties:

- It has been called the strongest low-alloy steel for high-temperature service, and is second only to austenitic stainless steels in terms of allowable stresses at temperatures exceeding 1050 F.
- The alloy is significantly less expensive than stainless steels.
- It has good to moderate corrosion, hydrogen attack, and oxidation resistance.
- It has high microstructural stability.
- The alloy has good formability and weldability.
- It has high bainitic hardenability, resulting in fully bainitic microstructure over a broad range of cooling rates.
- It can be heat treated to provide a range of tensile strengths.

A disadvantage of 2-1/4Cr-1Mo steel is that it is prone to temper embrittlement when exposed in the temperature range 600 to 900 F, or when subjected to a slow cool from 1100 to 600 F. A traditional guarantee of notch toughness was quarterline Charpy V-notch toughness of 40 ft-lb at 50 F, but this is being replaced by 40 ft-lb at 20 F (1). An associated effect of temper embrittlement is that the resistance to hydrogen embrittlement is reduced if the material is temper embrittled (2). Because of these implications, attempts are made to control the J*-factor [equal to (%Mn + %Si) x (%P + %Sn) x 10^4] to less than 200.

CORROSION

2-1/4Cr-1Mo steel has good resistance to water corrosion and low stress corrosion susceptibility to wetting steam conditions, provided that water chemistry is properly controlled (3). The material tends to form a fairly adherent oxide scale under high-temperature steam or water exposure, with an oxidation rate that is parabolic with time. 2-1/4Cr-1Mo steel is highly resistant to stress corrosion cracking (SCC) in a chloride environment (4), and is almost immune to SCC in a water environment contaminated with 5 percent NaOH (5). Liquid sodium tends to decarburize the surface of 2-1/4Cr-1Mo steels, leading to the possibility of strain ratcheting of the soft surface layer under cyclic thermal strains (5). At ultimate strength levels less than 100 ksi, 2-1/4Cr-1Mo steel is highly resistant to hydrogen-induced loss of fracture toughness (6). At elevated temperatures the material has good resistance to hydrogen attack, and has therefore been routinely used in hydroprocessing units for a number of years. Recently there have been concerns in the petrochemical industry regarding the rise in 50 percent shear-fracture transition temperature (FATT) arising from absorption of hydrogen after extended use in hydroprocessing environments (2). More attention is now being focused on the synergistic effects of temper embrittlement and hydrogen embrittlement, since the threshold stress intensity for crack growth in a hydrogen environment is found to decrease with increase in FATT (2).

HEAT TREATMENT

The transformation temperatures for 2-1/4Cr-1Mo steel are:

Ac_1	1480 F
Ac_3	1600 F
Ar_1	1330 F
Ar_3	1510 F

From the isothermal transformation diagram and the continuous-cooling transformation diagrams, it can be seen that there is a wide range of cooling rates over which the resulting microstructure will be bainitic.

The material is used in the annealed, normalized-and-tempered, and quenched-and-tempered conditions. Typical heat treatments are:

Annealed: Austenitize in the range 1600 to 1700 F (typically 1700 F) followed by furnace cooling. A 2-hour hold at 1300 F is sometimes used (isothermal annealing).

Normalized-and-tempered: Austenitize at 1675 to 1725 F, cool in air, temper at 1075 to 1420 F. Instead of air cooling, accelerated liquid spray cooling is sometimes used, particularly for sections in excess of 4-inch thickness.

Quenched-and-tempered: Austenitize at 1725 to 1800 F, quench in oil, temper at 1050 to 1250 F.

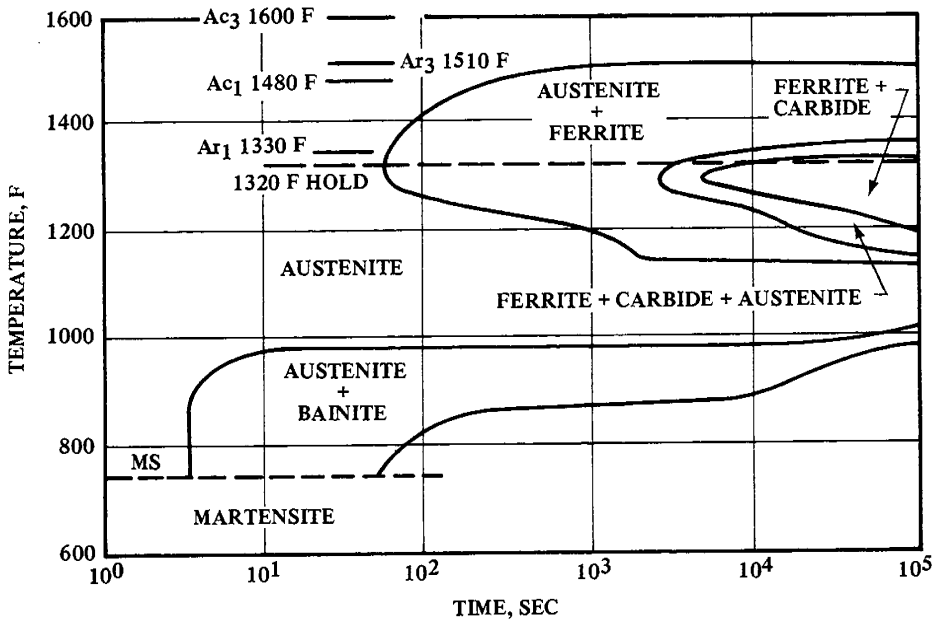
Typical holding periods at the desired temperatures are 1 hour per inch of section thickness.

The microstructure of the annealed steel is predominantly ferrite with dispersed carbides and pearlite and possibly bainite. Molybdenum carbides, mainly of the type M_2C, are responsible for imparting good creep-rupture properties to all of the heat-treated conditions. Normalized-and-tempered 2-1/4Cr-1Mo steel contains a mixture of tempered bainitic and proeutectoid ferrite. Quenching has the potential to produce martensite, although in thick sections slower cooling rates lead to bainitic microstructures. With regard to high-temperature properties, tempered bainite results in higher creep strength than tempered martensite or ferrite-pearlite aggregates for temperatures up to 1050 F and rupture times to 100,000 hours. For higher temperatures and times the ferrite-pearlite structure is the strongest (7).

HEAT TREATMENT

AUSTENITIZED AT 1860 F FOR 45 MIN, ASTM GRAIN SIZE 5-6

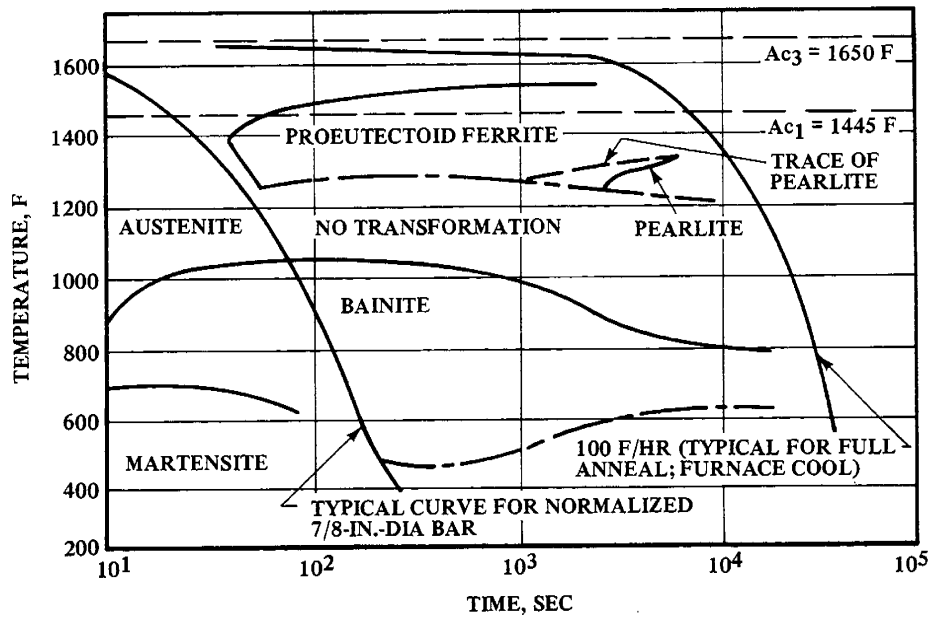
C	Mn	S	P	Si	Cr	Ni	Mo
0.10	0.42	0.013	0.018	0.25	2.16	0.27	0.96



ISOTHERMAL TRANSFORMATION DIAGRAM FOR 2-1/4-Cr-1Mo STEEL (8)

AUSTENITIZED AT 1690 F FOR 10 MIN, ASTM GRAIN SIZE 5.5
(HEATING RATE 18 F/MIN)

C	Si	Mn	P	S	Cr	Mo	Fe
0.09	0.26	0.44	0.008	0.010	2.25	0.99	BAL



CONTINUOUS-COOLING TRANSFORMATION DIAGRAM FOR 2-1/4Cr-1Mo STEEL (9)

2¼ Cr-1 Mo STEELS

SPECIFICATIONS

COMPOSITION, HEAT TREATMENTS, AND USES (10–20)

Specifications/Grades/Classes	Heat Treatments[a]	Minimum Tempering Temperature, F	Composition, weight percent (Maximum Unless a Range is Given)							Uses
			C	Mn	P	S	Si	Cr	Mo	
ASTM										
A217, Grade WC-9	N+T	1250	0.18	0.40–0.70	0.04	0.045	0.60	2.00–2.75	0.90–1.20[b]	Castings for valves, flanges, and fittings for elevated temperature service
A335, Grade P22	A, IA+T, N+T	1250	0.15	0.30–0.60	0.025	0.025	0.50	1.90–2.60	0.87–1.13	Seamless pipes for high temperature service
A356, Grade 9	N+T	1100	0.20	0.50–0.90	0.035	0.030	(d)	1.00–1.50	0.90–1.20	Heavy-walled castings
A387, Grade 22, Classes 1 & 2	A, N+T	1250	0.15	0.30–0.60	0.035	0.035	0.50	2.00–2.50	0.90–1.10	Plates for boilers and pressure vessels for elevated temperature service
A542, Classes 1, 2, 3, & 4	Q+T	1050	0.15	0.30–0.60	0.025	0.025	0.50	2.00–2.50	0.90–1.10	Fabrication of welded pressure vessels and other pressure equipment
ASME										
SA182, Grades F22 & F22a	N+T	1250	0.15	0.30–0.60	0.040	0.040	0.50	2.00–2.50	0.87–1.13	Pipe fittings, flanges, and valves
SA199, Grade T22	–	1250	0.15	0.30–0.60	0.030	0.030	0.50	1.90–2.60	0.87–1.13	Cold drawn seamless tube for heat exchangers and condensers
SA213, Grade T22	A, IA, N+T	1250	0.15	0.30–0.60	0.030	0.030	0.50	1.90–2.60	0.87–1.13	Seamless tubes for boilers and superheaters
SA217, Grade WC9	N+T	1250	0.18	0.40–0.70	0.040	0.045	0.60	2.00–2.75	0.90–1.20[c]	
SA336, Grade 22, Class 1	N+T, Q+T	1100	0.15	0.30–0.60	0.030	0.030	0.50	2.00–2.50	0.90–1.10	Forgings for boilers and pressure vessels
SA336, Grade 22, Class 2	N+T, Q+T	1250	0.15	0.30–0.60	0.030	0.030	0.50	2.00–2.50	0.90–1.10	Forgings for boilers and pressure vessels
SA542, Classes 1, 2, 3, & 4	Q+T	1050	0.15	0.30–0.60	0.035	0.035	0.50	2.00–2.50	0.90–1.10	Fabrication of welded pressure vessels and other pressure equipment

(a) A = anneal, IA = isothermal anneal, N+T = normalize and temper, Q+T = quench and temper.
(b) Cu 0.50 max, Ni 0.50 max, V 0.10 max, total (Cu + Ni + V) 1.00 max.
(c) Cu 0.50 max, Ni 0.50 max, W 0.10 max, total (Cu + Ni + W) 1.00 max.
(d) 0.20–0.60.

SPECIFICATION TENSILE PROPERTIES FOR 2-1/4Cr-1Mo STEELS (10–13,16,18,21)

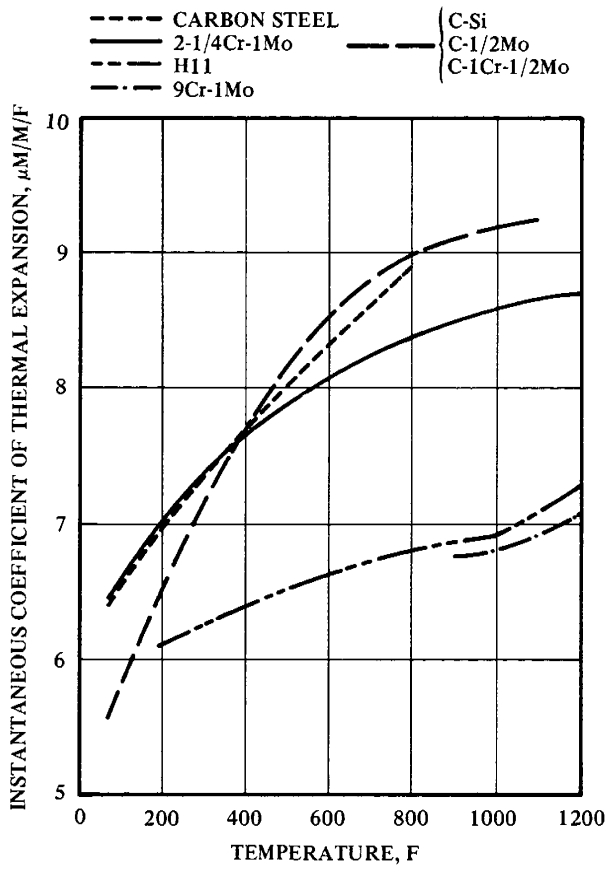
Specifications/ Grades/Classes	Ultimate Tensile Strength, ksi	Tensile Yield Strength (Min), ksi	Elongation (Min), in 2 In., percent	Reduction in Area (Min), percent
ASTM				
A217, Grade WC-9	70–95	40	20	35
A335, Grade P22	60[a]	30	30	–
A356, Grade 9	85[a]	55	20	35
A387, Grade 22, Class 1	60–85	30	18	40
A387, Grade 22, Class 2	75–100	45	18	40
A542, Class 1	105–125	85	14	–
A542, Class 2	115–135	100	13	–
A542, Class 3	95–115	75	20	–
A542, Class 4	85–105	60	20	–
ASME				
SA182, Grade F22	75[a]	45	20	30
SA182, Grade F22a	60	30	20	35
SA199, Grade T22	60	25	30	–
SA213, Grade T22	60	30	30	–
SA217, Grade WC9	70–95	40	20	35
SA336, Class F22	75–100	45	19	40
SA336, Class F22a	60–85	30	20	45
SA387, Grade 22, Class 1	60–85	30	18	40
SA387, Grade 22, Class 2	75–100	45	18	40
SA542, Class 1	105–125	85	14	–
SA542, Class 2	115–135	100	13	–
SA542, Class 3	95–115	75	20	–
SA542, Class 4	85–105	60	20	–

(a) Minimum.

PHYSICAL PROPERTIES

COEFFICIENT OF LINEAR THERMAL EXPANSION OF 2-1/4Cr-1Mo STEELS (22,23)

Temperature, F	Instantaneous Expansion, 10^{-6} in./in./F	Mean Expansion From 70 F, 10^{-6} in./in./F
70	6.45	–
100	6.60	6.50
200	6.90	6.70
300	7.35	6.90
400	7.65	7.07
500	7.90	7.23
600	8.10	7.38
700	8.25	7.50
800	8.40	7.62
900	8.50	7.72
1000	8.61	7.83
1100	8.67	7.89
1200	8.72	7.94



COEFFICIENTS OF LINEAR THERMAL EXPANSION FOR CARBON AND LOW-ALLOY STEELS AT THE INDICATED TEMPERATURES (23)

These are not mean values of the coefficient over a range of temperatures.

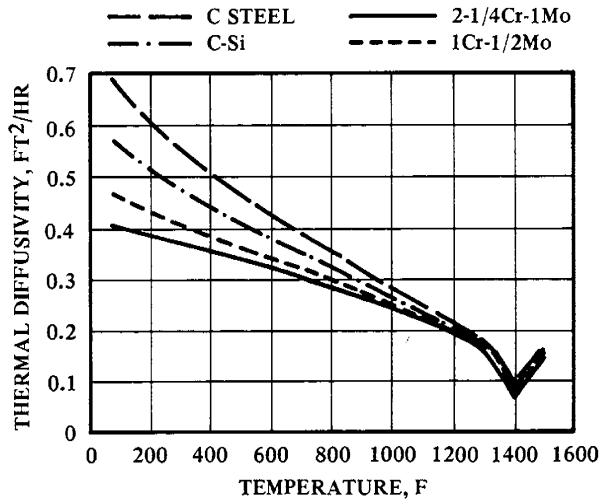
THERMAL CONDUCTIVITY OF 2-1/4Cr-1Mo STEEL (22)

Temperature, F	Thermal Conductivity, Btu ft/hr-ft^2 F
70	20.9
100	21.0
200	21.3
300	21.5
400	21.5
500	21.4
600	21.1
700	20.7
800	20.2
900	19.7
1000	19.1
1100	18.5
1200	18.0

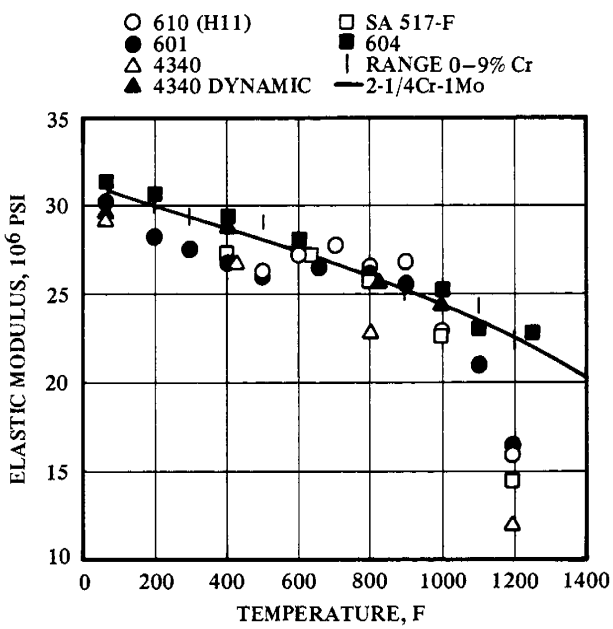
PHYSICAL PROPERTIES

THERMAL DIFFUSIVITY OF 2-1/4Cr-1Mo STEEL (23)

Temperature, F	Thermal Diffusivity, ft^2/hr
70	0.408
100	0.397
200	0.385
300	0.371
400	0.357
500	0.341
600	0.323
700	0.305
800	0.285
900	0.264
1000	0.241
1100	0.217
1200	0.192



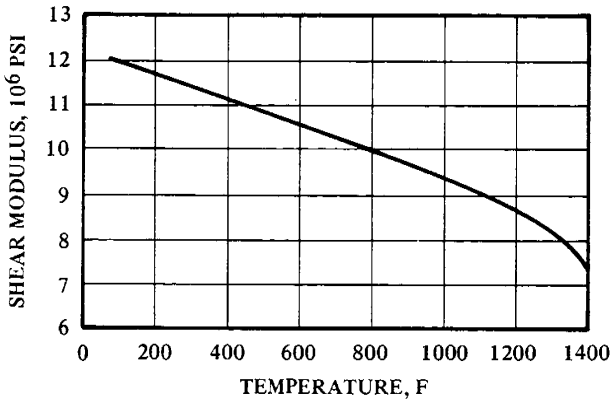
THERMAL DIFFUSIVITY (THERMAL CONDUCTIVITY DIVIDED BY PRODUCT OF DENSITY AND SPECIFIC HEAT) OF CARBON AND LOW-ALLOY STEELS AT INDICATED TEMPERATURES (23)



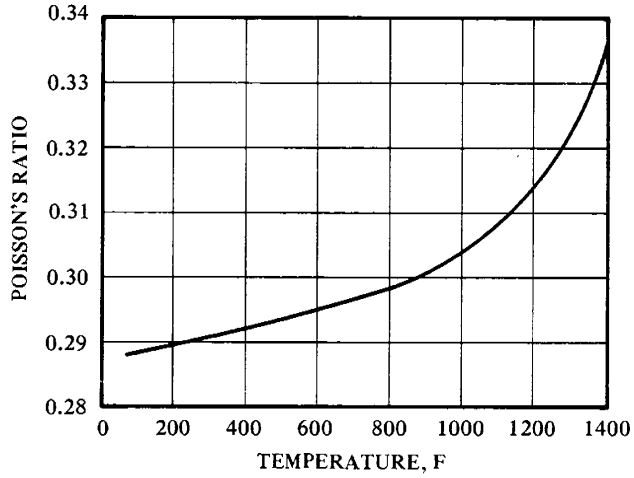
EFFECT OF TEMPERATURE ON MODULUS OF ELASTICITY FOR SEVERAL STEELS COMMONLY USED AT ELEVATED TEMPERATURES (23)

Dynamic measurements were made by determining the material frequencies of test specimens; static measurements were made during tensile testing.

PHYSICAL PROPERTIES



EFFECT OF TEMPERATURE ON SHEAR MODULUS OF 2-1/4Cr-1Mo STEEL (23)



EFFECT OF TEMPERATURE ON POISSON'S RATIO OF 2-1/4Cr-1Mo STEEL (23)

MODULUS OF ELASTICITY OF 2-1/4Cr-1Mo STEEL (23)

Temperature, F	Modulus of Elasticity (E), 10^6 psi
70	30.6
200	29.8
300	29.4
400	28.8
500	28.3
600	27.7
700	27.1
800	26.3
900	25.6
1000	24.6
1100	23.7
1200	22.5

TYPICAL BRINELL HARDNESS OF NORMALIZED 2-1/4Cr-1Mo STEEL[a] (24)

Austenitizing Temperature, F	Hardness, BHN
Annealed	160
1200/1450	155
1500	202
1550	241
1600	285
1650	309
1700	309
1750	311
1800	302
1850	297
1900	285
1950	285
2000	285

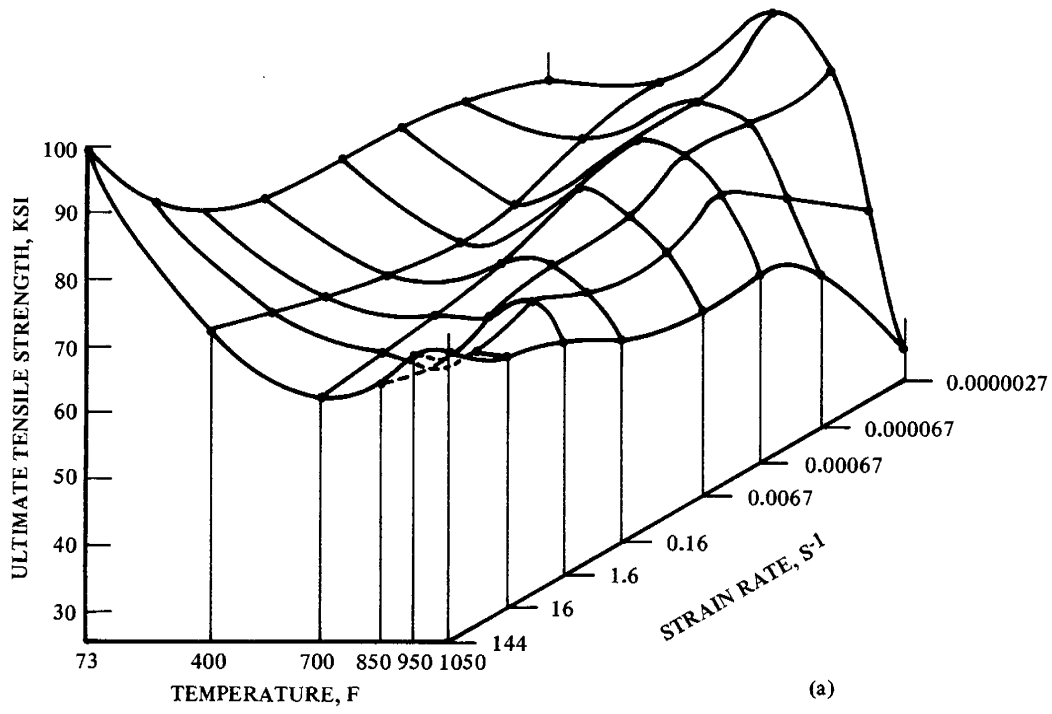
(a) One-inch diameter specimens air cooled from the indicated austenitizing temperatures.

SPECIFICATION TENSILE PROPERTIES FOR 2-1/4Cr-1Mo STEELS (10–13,16,18,21)

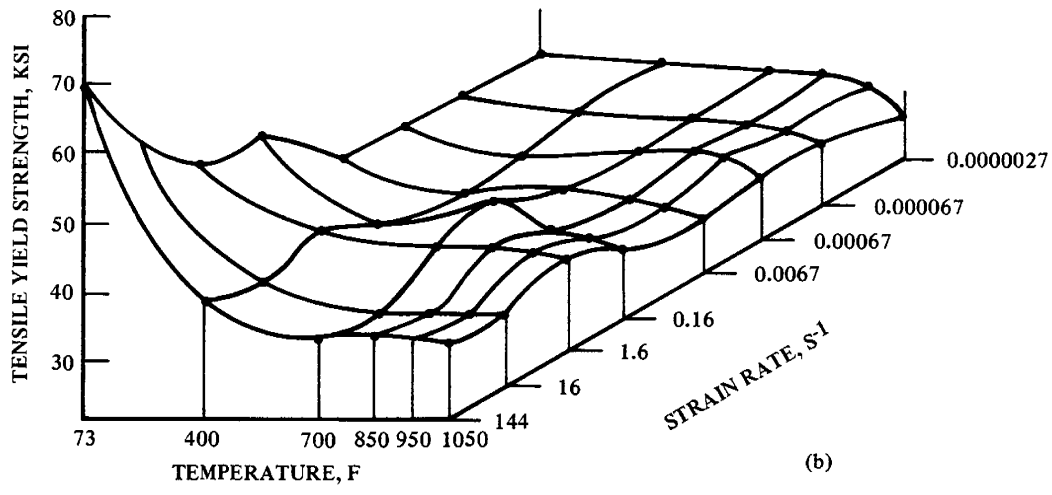
Specifications/ Grades/Classes	Ultimate Tensile Strength, ksi	Tensile Yield Strength (Min), ksi	Elongation (Min), in 2 In., percent	Reduction in Area (Min), percent
ASTM				
A217, Grade WC-9	70–95	40	20	35
A335, Grade P22	60[a]	30	30	–
A356, Grade 9	85[a]	55	20	35
A387, Grade 22, Class 1	60–85	30	18	40
A387, Grade 22, Class 2	75–100	45	18	40
A542, Class 1	105–125	85	14	–
A542, Class 2	115–135	100	13	–
A542, Class 3	95–115	75	20	–
A542, Class 4	85–105	60	20	–
ASME				
SA182, Grade F22	75[a]	45	20	30
SA182, Grade F22a	60	30	20	35
SA199, Grade T22	60	25	30	–
SA213, Grade T22	60	30	30	–
SA217, Grade WC9	70–95	40	20	35
SA336, Class F22	75–100	45	19	40
SA336, Class F22a	60–85	30	20	45
SA387, Grade 22, Class 1	60–85	30	18	40
SA387, Grade 22, Class 2	75–100	45	18	40
SA542, Class 1	105–125	85	14	–
SA542, Class 2	115–135	100	13	–
SA542, Class 3	95–115	75	20	–
SA542, Class 4	85–105	60	20	–

(a) Minimum.

TENSILE



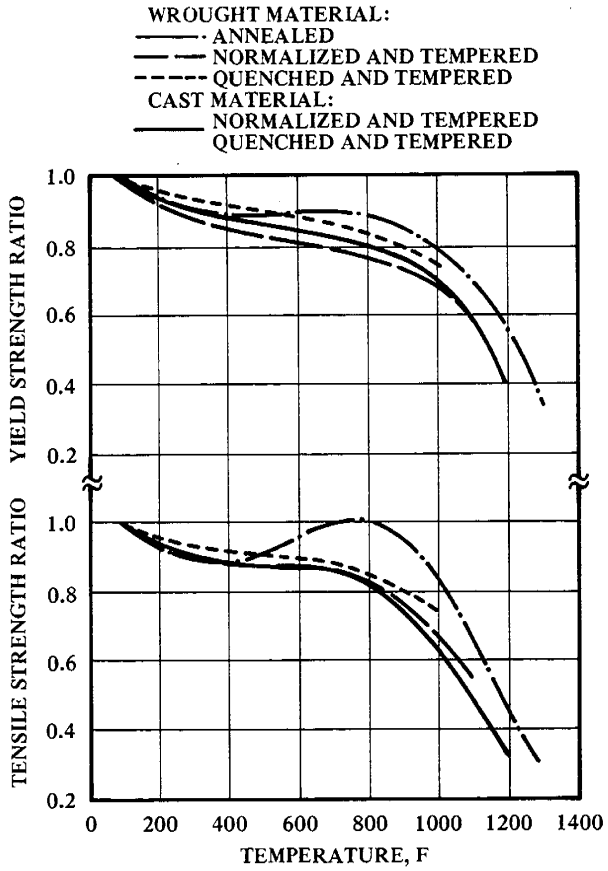
(a)



(b)

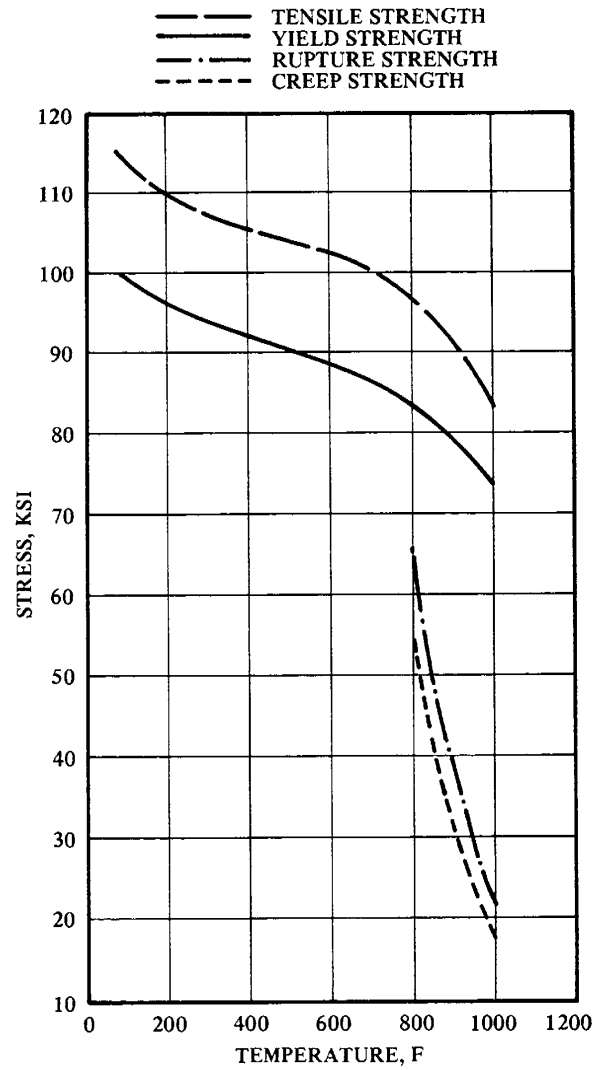
EFFECT OF TEMPERATURE AND STRAIN RATE ON TENSILE AND YIELD STRENGTH OF 2-1/4Cr-1Mo STEEL (23)

TENSILE



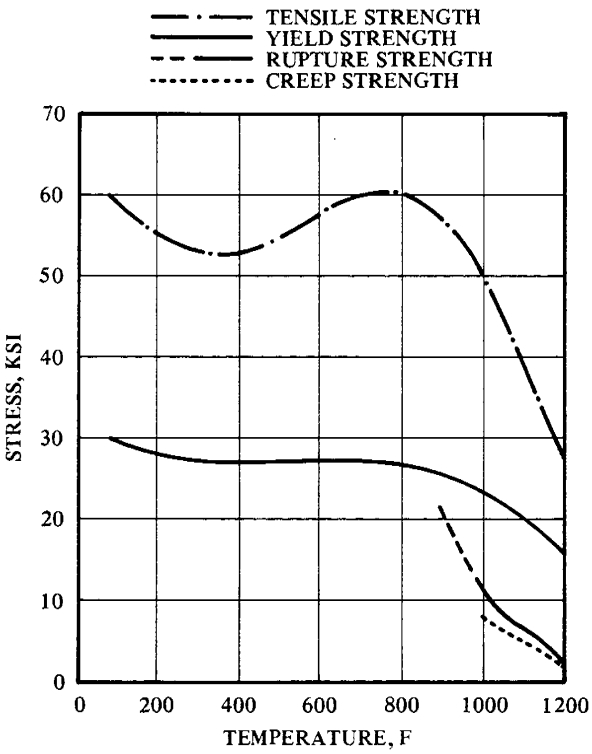
COMPARISON OF THE TEMPERATURE DEPENDENCIES OF YIELD AND TENSILE STRENGTHS, EXPRESSED AS STRENGTH RATIOS, FOR ANNEALED, NORMALIZED AND TEMPERED, AND QUENCHED AND TEMPERED WROUGHT MATERIAL AND NORMALIZED AND TEMPERED AND QUENCHED AND TEMPERED CAST MATERIAL (25)

Average data for yield and tensile strengths.



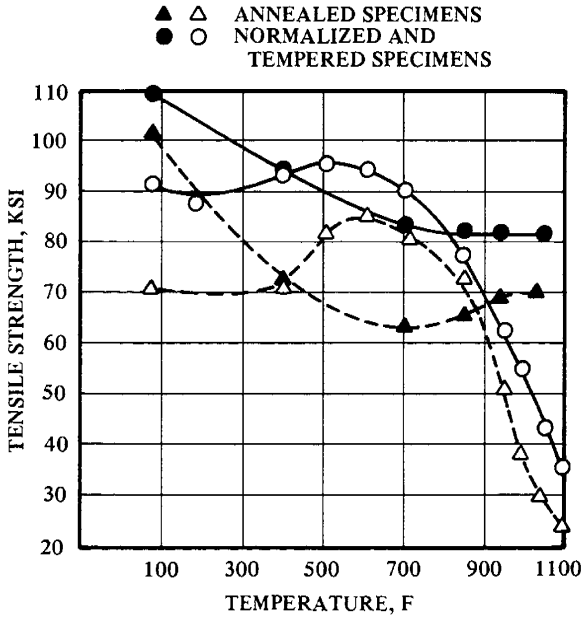
EFFECT OF TEMPERATURE ON YIELD STRENGTH, TENSILE STRENGTH, CREEP STRENGTH (0.01 PERCENT PER 1000 HOURS) AND RUPTURE STRENGTH (100,000 HOURS) OF QUENCHED AND TEMPERED MATERIAL (25)

Yield and tensile strengths have been adjusted to 100 and 115 ksi at 75 F; creep and rupture strengths represent averages of data adjusted to a specified tensile strength of 115 ksi at 75 F.



EFFECT OF TEMPERATURE ON YIELD STRENGTH, TENSILE STRENGTH, CREEP STRENGTH (0.01 PERCENT PER 1000 HOURS) AND RUPTURE STRENGTH (100,000 HOURS) OF ANNEALED 2-1/4Cr-1Mo STEEL IN WROUGHT OR CAST FORM (25)

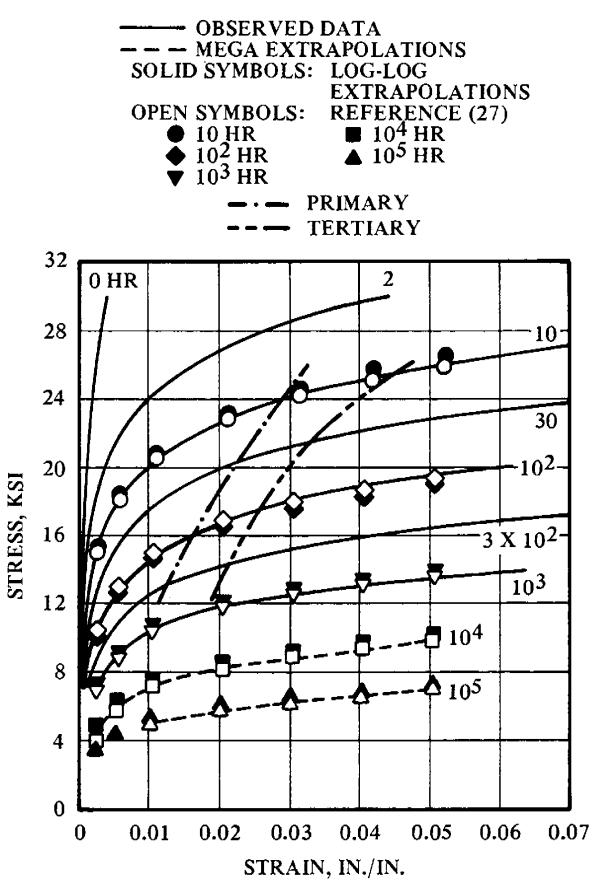
Yield and tensile strengths have been adjusted to 30 and 60 ksi at 75 F. Creep and rupture strengths are averages of available data.



EFFECT OF STRAIN RATE ON ELEVATED-TEMPERATURE TENSILE STRENGTH OF 2-1/4Cr-1Mo STEEL (30)

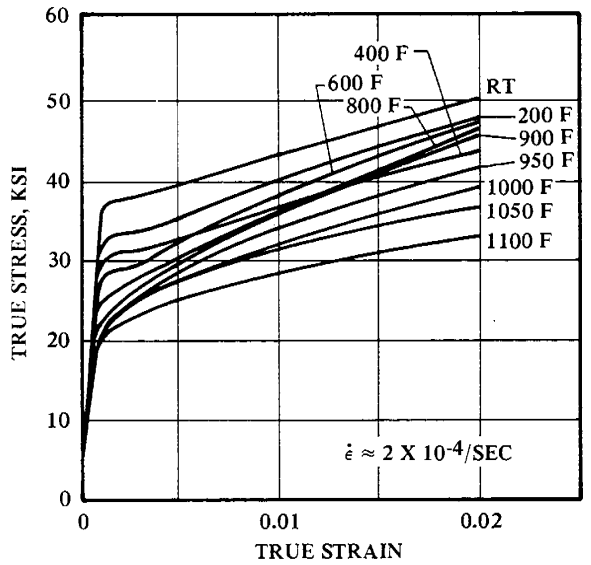
Annealed specimens (triangle symbols) and normalized and tempered (circles) specimens of 2-1/4Cr-1Mo steel were tested at strain rates of 2.67 μin./in./sec (open symbols) and 144 in./in./sec (filled symbols) at the indicated temperatures.

SA336, GRADE F22 FORGED BAR; NORMALIZED AT 1690 F FOR 2 HR; TEMPERED AT 1275 F FOR 2 HR; FURNACE COOLED

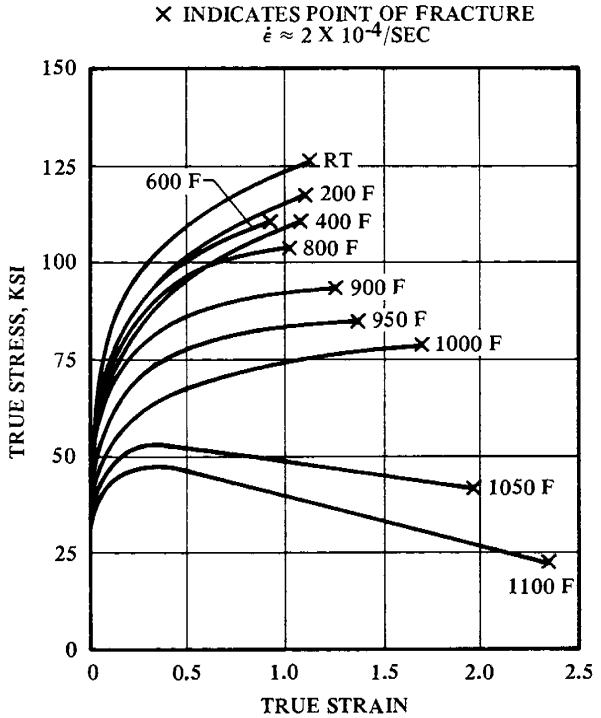


ISOCHRONOUS STRESS-STRAIN CURVES FOR 2-1/4Cr-1Mo STEEL AT 1050 F (26)

MATERIAL: 2-1/4Cr-1Mo STEEL (ASTM A335, GRADE P22) PIPE, ANNEALED + 1700 F for 2 HR, FURNACE COOL AT 50 F/HR TO 1100 F, AIR COOL TO RT
AVERAGE HARDNESS: 143 BHN
MICROSTRUCTURE: PEARLITE ISLANDS IN FERRITE MATRIX



TRUE-STRESS/TRUE-STRAIN CURVES TO 2 PERCENT TRUE STRAIN AS A FUNCTION OF TEMPERATURE FOR 2-1/4Cr-1Mo PIPE (28)



TRUE-STRESS/TRUE-STRAIN CURVES TO FRACTURE AS A FUNCTION OF TEMPERATURE FOR 2-1/4Cr-1Mo PIPE (28)

CONSTITUTIVE EQUATIONS

Tensile Properties (29)

Expected values are shown

$$PL = 151.26 - 0.1015\ T$$
$$YS = 263 - 0.315\ T + (9.39 \times 10^{-4})\ T^2 - (1.08 \times 10^{-6})\ T^3$$
$$UTS = 527 - 1.03\ T + (4.54 \times 10^{-3})\ T^2 - (5.83 \times 10^{-6})\ T^3$$
$$TFS = 1270 - 6.53\ T + 0.0266\ T^2 - (3.09 \times 10^{-5})\ T^3$$

where

PL = proportional limit
T = temperature, C
YS = yield strength, MPa
UTS = ultimate tensile strength, MPa
TFS = true fracture stress, MPa.

The strength values above correspond to the expected values. Assuming a normal distribution of the dependent variables, lower and upper bound values can be obtained, depending upon the level of confidence that is desired.

Data were collected from the literature for 2-1/4Cr-1Mo with the following typical annealing treatments:

(a) Austenitize at 927 C for 1 hr; furnace cool to 450 C; air cool to RT
(b) Isothermal anneal: austenitize at 927 C for 1 hr; furnace cool to 704 C; hold 2 hr; furnace cool to RT.

RT UTS ≥ 414 MPa (60 ksi)
RT YS ≥ 207 MPa (30 ksi).

Strain rates were between 1.7×10^{-6}/sec and 1.7×10^{-3}/sec.

Creep Properties (29)

Only materials with ultimate tensile strength in the range of 490 to 520 MPa were used. Temperature range was 454 to 593 C. The equation can be used to determine isochronous stress-strain curves

$$\epsilon_c = \frac{t}{A + Bt} + \dot{\epsilon}_m t \quad ,$$

where

ϵ_c = engineering creep strain, percent
t = creep time, hr
$\dot{\epsilon}_m$ = minimum creep rate, percent/hr
A, B = functions of stress and temperature.

The advantage of the equation is that it allows determination of primary creep rates from the generally known secondary creep rates.

$$\log_{10} A = 12.26 - 3.348 \times 10^{-6}\ T^2 + 9.353 \times 10^{-4}\ \sigma^2 - 1.167 \times 10^{-4}\ T\sigma$$
$$\log_{10} B = -52.19 + 8.682 \times 10^{-2}\ T - 3.368 \times 10^{-5}\ T^2 - 1.152\ (\log_{10}\sigma)^2$$
$$\log_{10} \dot{\epsilon}_m = -30.04 + 1.513 \times 10^{-2}\ T + 2.001 \times 10^{-3}\ T\ (\log_{10}\sigma)^2$$

where

Temperature (T) is in degrees Rankine (R)
Stress (σ) is in ksi.

The equation is not valid for σ greater than 65 ksi and temperatures greater than 650 C.

An alternate equation for $\dot{\epsilon}_m$ is:

$$\log \dot{\epsilon}_m = 1.979 + 6.961 \times 10^{-5}\ \tau\ T - 0.097\ \tau + 0.01459\ \tau \log \sigma,$$

where

Temperature (T) is in degrees Kelvin (K)
Stress (σ) is in MPa
RT UTS (τ) is in MPa.

The advantage of the equation is that it can be used to determine both the average and the minimum secondary creep rate, based on average and minimum ultimate tensile strength data.

Time to Tertiary Creep (t_3) (33)

$$t_3 = 0.28\ t_r^{1.07}$$

where

Tertiary creep (t_3) is in hours
Time to rupture (t_r) is in hours.

Rupture Life

Note: This equation is for annealed material.

$$\log t_r = -31.45 + \frac{3.932 \times 10^4}{T} - 19.75 \log \sigma + 0.01642\ T \log \sigma$$

where

Rupture life (t_r) is in hours
Temperature (T) is in degrees Kelvin (K)
Stress (σ) is in MPa.

Fatigue (29)

$$\log N_f = A + B \log \Delta\epsilon_t + C\ (\log \Delta\epsilon_t)^2 + D\ (\log \Delta\epsilon_t)^3$$

where

N_f = number of cycles to failure
$\Delta\epsilon_t$ = total strain range, percent.

Valid for strain rates greater than 4×10^{-3}/sec.

Temperature, F	A	B	C	D
800	3.578	-2.358	3.506	-4.197
1000	3.302	-2.388	3.521	-2.577
1100	3.153	-1.803	2.613	-3.738

Impact (Charpy V Notch) (29)

$$E = A\left\{1 - \exp^{-B(T-C)^2}\right\}$$

where

E = energy absorbed at fracture, joules
T = temperature, C
A,B,C = constants, dependent on the particular lot of material.

A ranged from 82 to 227
B ranged from 2.2×10^{-5} to 6.29×10^{-4}
C ranged from -90 to 4.

TRUE STRESS-STRAIN CURVES[a] (26)

$$\sigma = \sigma_{pl}\left(\frac{\epsilon_p}{\epsilon_o} + 1\right)^{\lambda} + \sigma_{\infty}\left[1 - \exp\left(\epsilon_p/\epsilon_c\right)\right]$$

where σ = true stress and ϵ_p = true plastic strain (see table for other values).

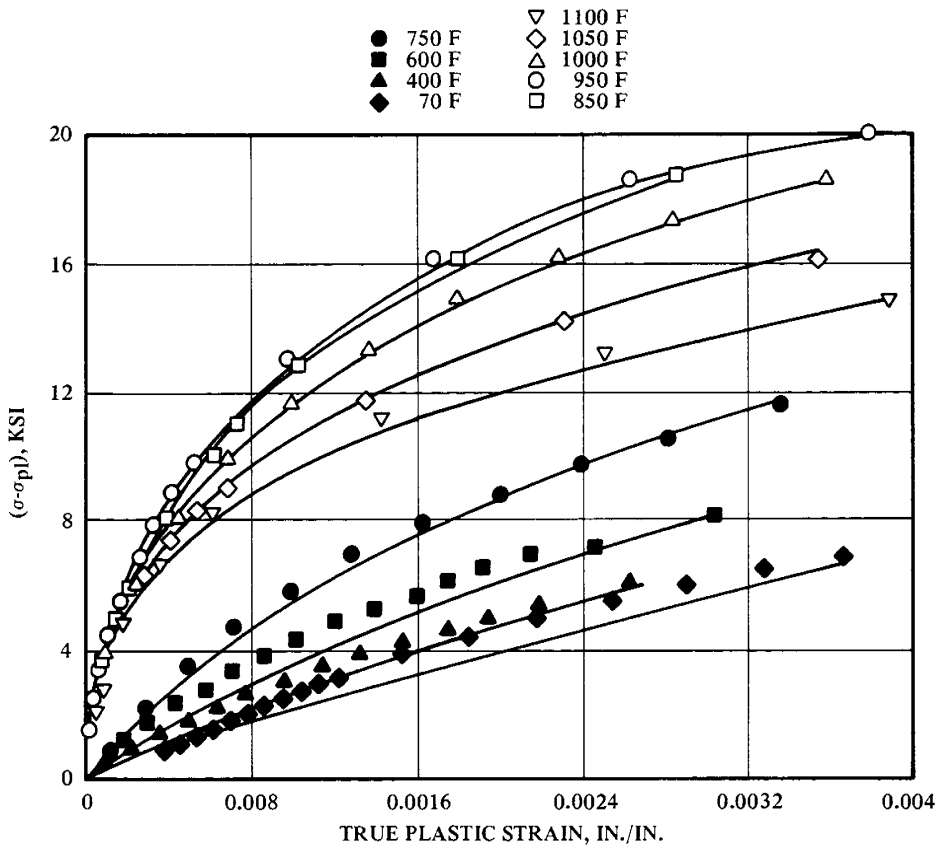
(a) For 2-1/4Cr-1Mo steel normalized 2 hr at 1690 F, AC, tempered 2 hr at 1275 F, FC; specimens were tensile tested at a plastic strain rate of 0.01/minute.

OPTIMUM CONSTANTS FOR STRESS-STRAIN EQUATIONS (26)

Temperature		σ_{pl}		$\epsilon_p^T \times 10^{-6}$	$\epsilon_o \times 10^{-6}$	λ	σ_∞		$\epsilon_c \times 10^{-6}$
F	C	ksi	MPa				ksi	MPa	
70	21	39.6	273.0	1030	4155	0.240	0	0	0
400	204	30.9	213.0	1254	2060	0.211	0	0	0
600	316	28.4	195.8	875	1345	0.214	0	0	0
750	399	26.3	181.3	777	645	0.202	0	0	0
850	454	16.0	110.3	0	45	0.186	0	0	0
950	510	15.0	103.4	0	23	0.152	4.8	33.1	4844
1000	538	14.5	100.0	0	24	0.138	5.8	40.0	2733
1050	566	14.0	96.5	0	15	0.120	4.8	33.1	3514
1100	593	13.0	89.6	0	10	0.104	4.4	30.3	1838

ϵ_p^T is yield point transition strain.
Total strain is given by $\epsilon = (\sigma/E) + \epsilon_p^T + \epsilon_p$.

CONSTITUTIVE EQUATIONS



LOW-STRAIN COMPARISON OF CALCULATED AND OBSERVED TRUE STRESS VERSUS TRUE PLASTIC STRAIN IN 2-1/4Cr-1Mo STEEL AS A FUNCTION OF TEMPERATURE (26)

CONSTITUTIVE EQUATIONS

CREEP (Alternative Version) (26)

$$\epsilon_c = \frac{b}{\log t_o - \log A} - \frac{b}{\log t + t_o - \log A}$$

where

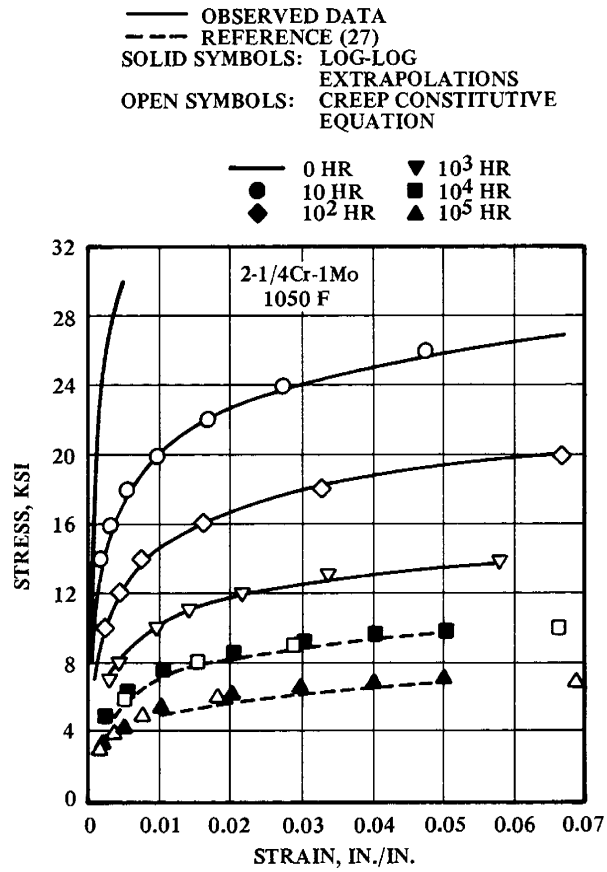
ϵ_c = creep strain, and t = creep time, hr
b, t, and A are stress- and temperature-dependent parameters
$\log A$ = AI + AS log S
$\log t_o$ = TI + TS log A when $S > S_{pl}$
$\log t_o$ = TI + TS · AI + TS · AS log S_{pl} when $S < S_{pl}$
$\log b$ = BI + BS log A

where

S_{pl} = the proportional limit
t_r = time to rupture, hr ≃ A/1.5
t_t = time to tertiary creep ≃ 0.25 t_r
S = applied stress, psi
$\dot{\epsilon}_m t_r$ ≅ 0.15 in./in.
$\dot{\epsilon}_m$ = secondary creep rate, in./in./hr
t_r = time to rupture, hr

CREEP CONSTITUTIVE EQUATION CONSTANTS (26)

Temperature, F	AI	AS	TI	TS	BI	BS
850	40.609	-7.936	-5.0	1.0	0.1493	-0.3880
950	38.241	-7.936	-2.8	1.0	-0.5678	-0.2870
1000	29.582	-6.114	-2.0	1.0	-0.5599	-0.2477
1050	27.450	-5.784	-1.7	1.0	-0.7614	-0.2017
1100	26.721	-5.784	-1.7	1.0	-1.0988	-0.1331



ISOCHRONOUS STRESS-STRAIN CURVES FOR 2-1/4Cr-1Mo STEEL AT 1050 F SHOWING CREEP CONSTITUTIVE EQUATION CALCULATIONS (26)

CONSTITUTIVE EQUATIONS

Constitutive Equations for Normalized-and-Tempered 2-1/4Cr-1Mo Steel (38)

$$\log_{10}\sigma y = 2.621 - 6.435 \times 10^{-4}\,T + 1.948 \times 10^{-6}\,T^2 - 2.382 \times 10^{-9}\,T^3$$

$$\log_{10}\sigma u = 2.780 - 1.102 \times 10^{-3}\,T + 4.704 \times 10^{-6}\,T^2 - 6.084 \times 10^{-9}\,T^3$$

where

σy = yield strength, MPa
σu = ultimate tensile strength, MPa
T = degree Kelvin.

Rupture Equation:

$$\log_{10}t_r = C_h - 3.656 \log_{10}\sigma - 0.02564\,T - 1.052 \times 10^{-5}\,T\,\sigma$$

where

t_r = rupture life, hr
σ = applied stress, MPa
T = degree Kelvin
C_h = 33.41; based on average data for steels from USA, GB, West Germany, Japan, France
= 32.83; minimum value based on 95-percent confidence limit (average – 1.65 times standard deviation).

EQUATION FOR STRESS-STRAIN RESPONSE (41)

Material: 2-1/4Cr-1Mo Steel, 1-in. thick plate
Heat Treatment: 1/2 hr at 1700 F, FC to 1330 F, hold 2 hr AC to RT
For fully reversed loading

$$\Delta\sigma = K\left(\frac{\Delta\epsilon_p}{2}\right)^n$$

where $\Delta\epsilon_p$ = Plastic strain range, percent
$\Delta\sigma$ = Stress range, ksi

Temperature, F	K, ksi	n
70	180	0.11
200	210	0.18
400	241	0.26
600	180	0.17
800	207	0.20
900	269	0.22
950	209	0.21
1000	168	0.17
1100	84	0.08

FATIGUE

Effect of Elevated Temperature on Strain Controlled Fatigue Behavior of Annealed 2-1/4Cr-1Mo Steel (29)

Strain rate greater than 4×10^{-2}/sec
Completely reversed fatigue

Best Fit Equations

800 F:

$$\log_{10}N_f = 3.578 - 2.358 \log \Delta\epsilon_t + 3.506 (\log \Delta\epsilon_t)^2 - 4.197 (\log \Delta\epsilon_t)^3$$

1000 F:

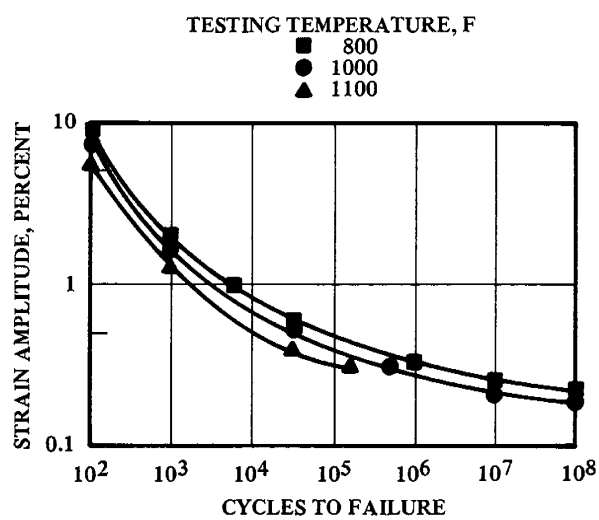
$$\log_{10}N_f = 3.302 - 2.388 \log \Delta\epsilon_t + 3.521 (\log \Delta\epsilon_t)^2 - 2.577 (\log \Delta\epsilon_t)^3$$

1100 F:

$$\log_{10}N_f = 3.153 - 1.803 \log \Delta\epsilon_t + 2.613 (\log \Delta\epsilon_t)^2 - 3.738 (\log \Delta\epsilon_t)^3$$

where

N_f = number of cycles to failure
$\Delta\epsilon_t$ = total strain range, percent.



EFFECT OF ELEVATED TEMPERATURE ON STRAIN CONTROLLED FATIGUE BEHAVIOR OF ANNEALED 2-1/4Cr-1Mo STEEL (29)

Strain rate greater than 4×10^{-3}/sec. Completely reversed fatigue.

STRESS—RELAXATION

STRESS-RELAXATION EQUATION (30)

$$\log_e\left(\frac{\sigma_A}{\sigma}\right) = \frac{A}{(1+m)}\, t^{(1+m)}$$

where

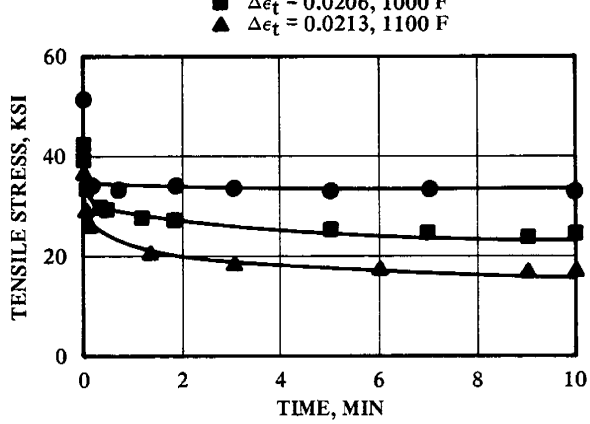
σ_A = initial stress, ksi
σ = instantaneous stress, ksi
t = time, min
A, m = constants.

CONSTANTS FOR STRESS-RELAXATION EQUATION[a]

Temperature, F	A	m
1100	0.1085	-0.7679
	0.1446	-0.7380
	0.1099	-0.8007
1000	0.0677	-0.8148
	0.0595	-0.8285
	0.0643	-0.7646
900	0.0093	-0.9782
	0.0188	-0.9583

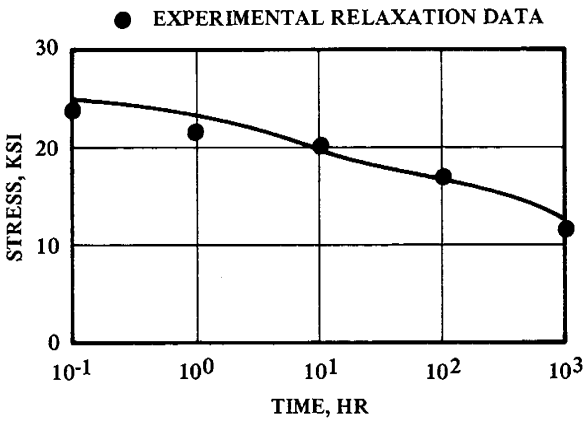
(a) Isothermally annealed material.

SA335 GRADE P22; ANNEALED OR ISOTHERMALLY ANNEALED
$\Delta\epsilon_t$ = TOTAL STRAIN



STRESS-RELAXATION DATA FOR SPECIMENS HELD AT PEAK TENSILE STRAIN DURING FATIGUE TESTING WITH HOLD PERIODS (30)

ANNEALED 2-1/4Cr-1Mo STEEL
950 F (510 C)



COMPARISON OF CALCULATED RELAXATION CURVE WITH ACTUAL EXPERIMENTAL RELAXATION DATA AT 510 C (29)

Curve calculated using

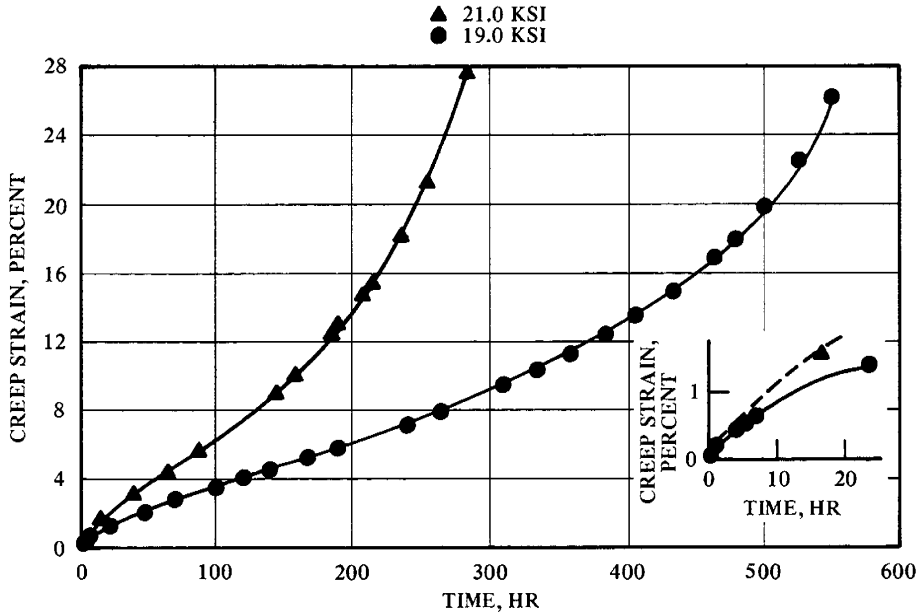
$$t = \frac{(B\epsilon_c - A\dot{\epsilon}_m - 1) \pm [(A\dot{\epsilon}_m + 1 - B\epsilon_c)^2 + 4A\epsilon_c B\dot{\epsilon}_m]^{1/2}}{2B\dot{\epsilon}_m}$$

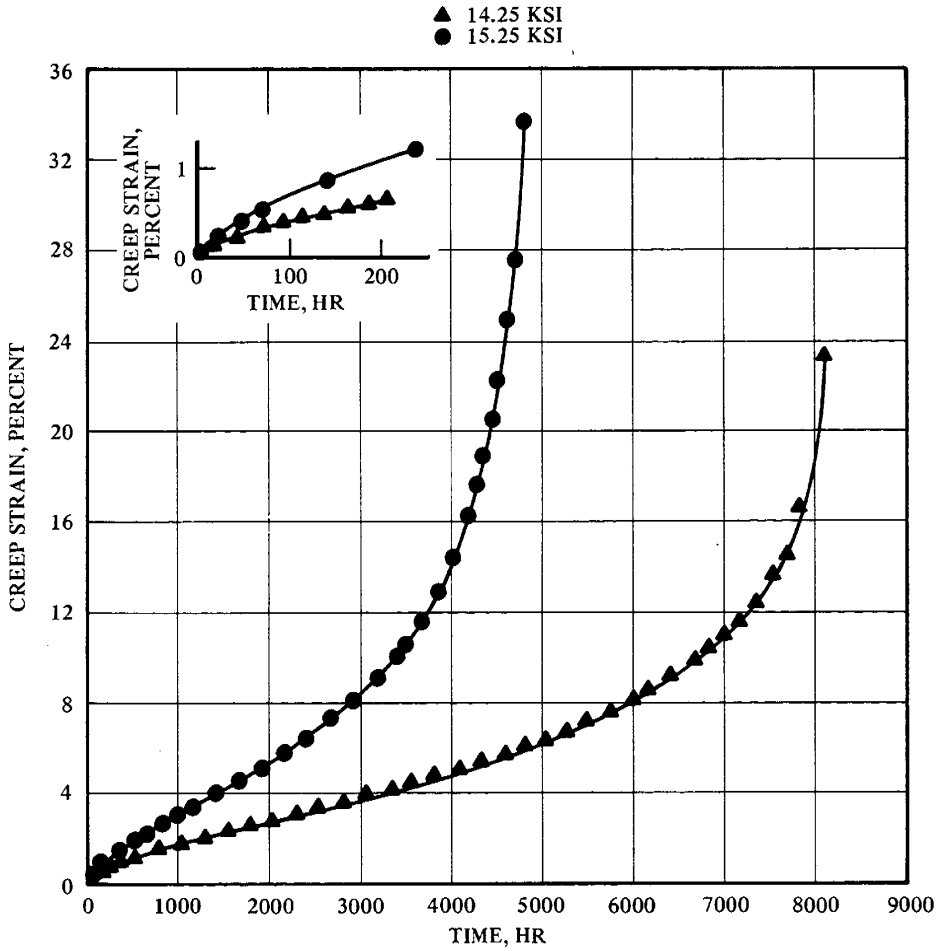
as given in an earlier section from (29).

Refer to creep properties on page 14 of this chapter.

CREEP

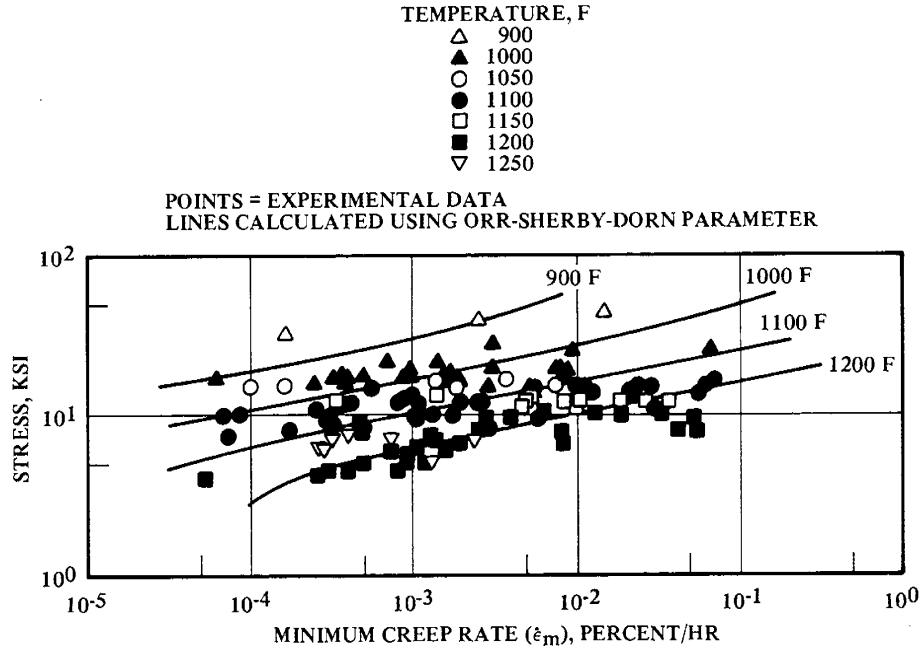
ASTM A335, GRADE P22; FULLY ANNEALED



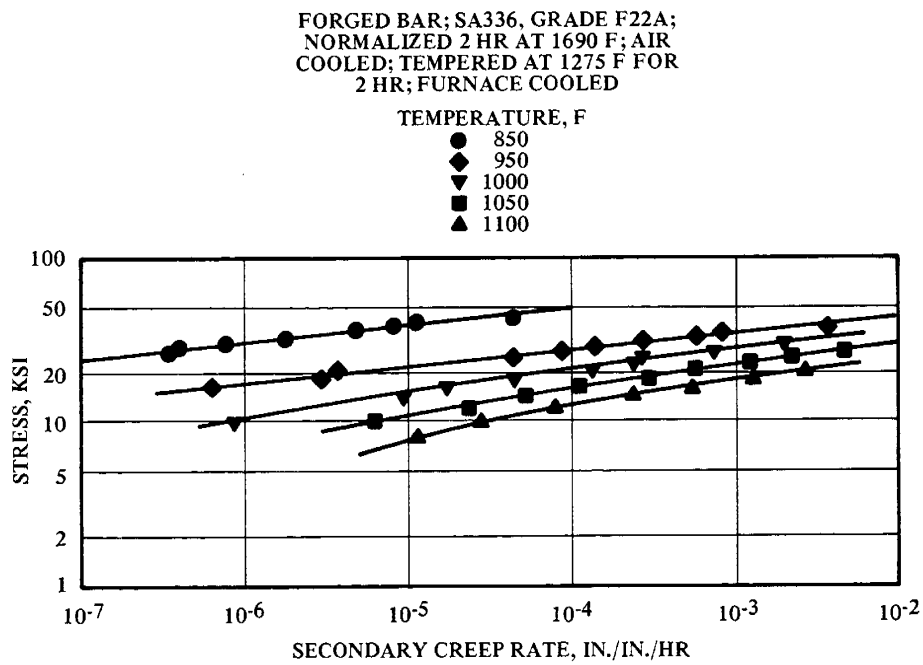


CREEP-STRAIN CURVES FOR 2-1/4Cr-1Mo PIPE AT 1050 F (28)

CREEP

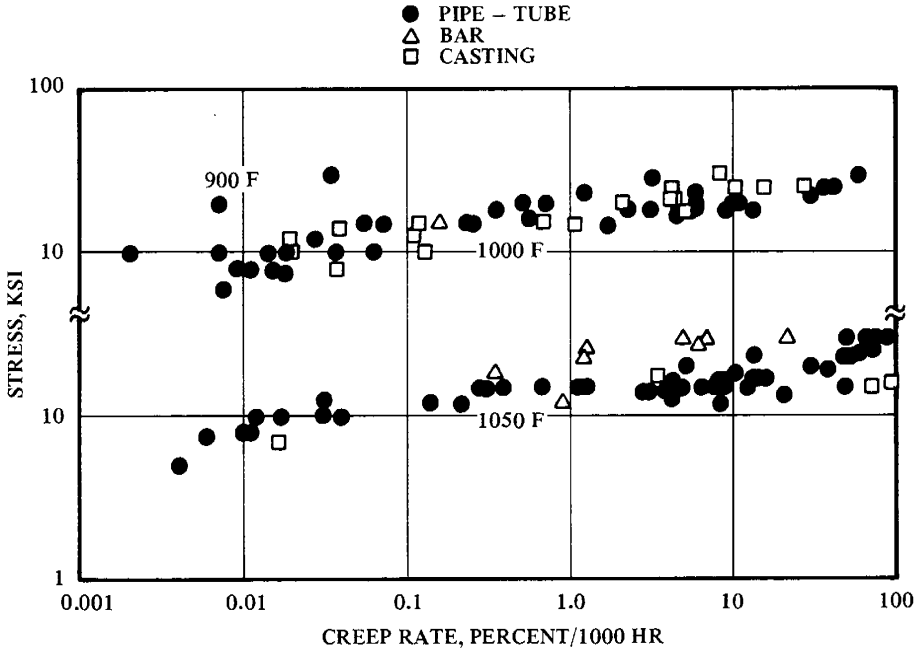


STRESS VERSUS MINIMUM CREEP RATE ($\dot{\varepsilon}_m$) ISOTHERMALS FOR SEVERAL HEATS OF ANNEALED AND ISOTHERMALLY ANNEALED 2-1/4Cr-1Mo STEEL (31)

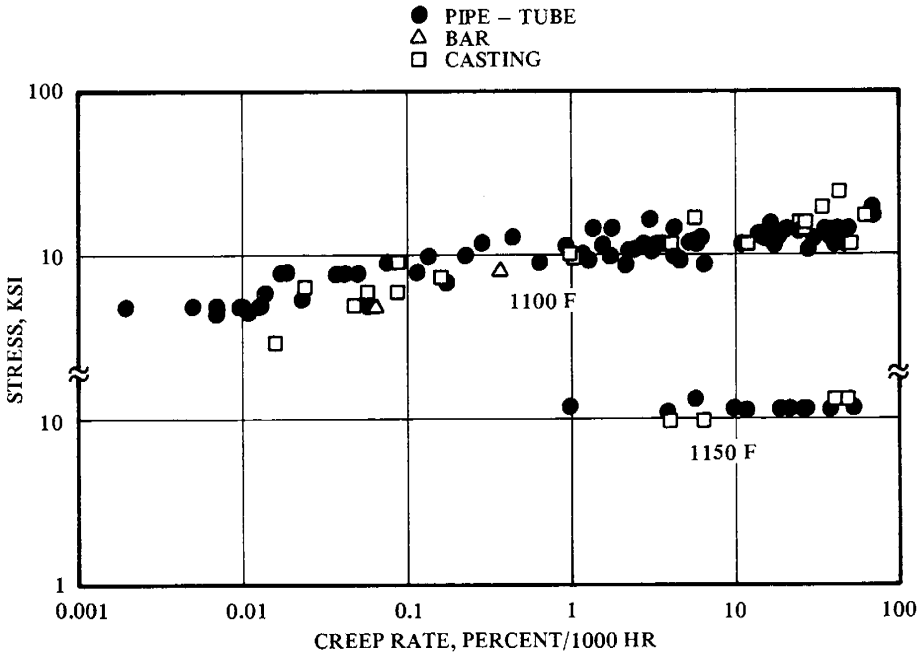


SECONDARY CREEP RATE OF 2-1/4Cr-1Mo STEEL AS A FUNCTION OF STRESS AND TEMPERATURE (26)

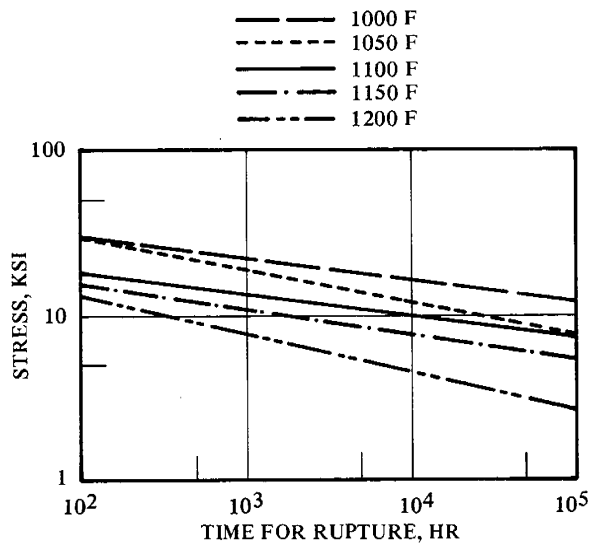
CREEP



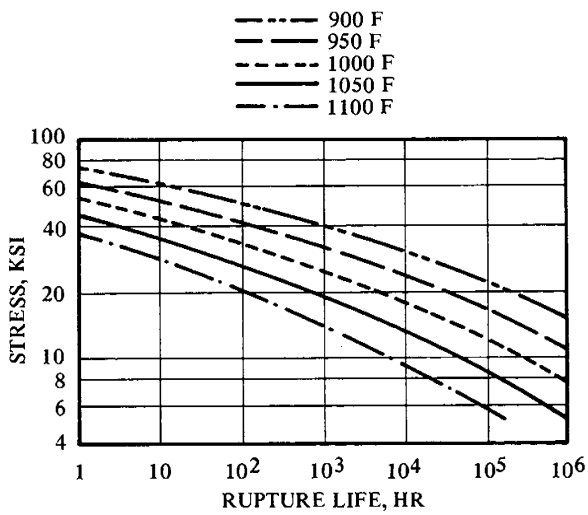
STRESS VERSUS SECONDARY CREEP RATE FOR ANNEALED 2-1/4Cr-1Mo STEEL (32)



STRESS VERSUS SECONDARY CREEP RATE FOR ANNEALED 2-1/4Cr-1Mo STEEL (32)

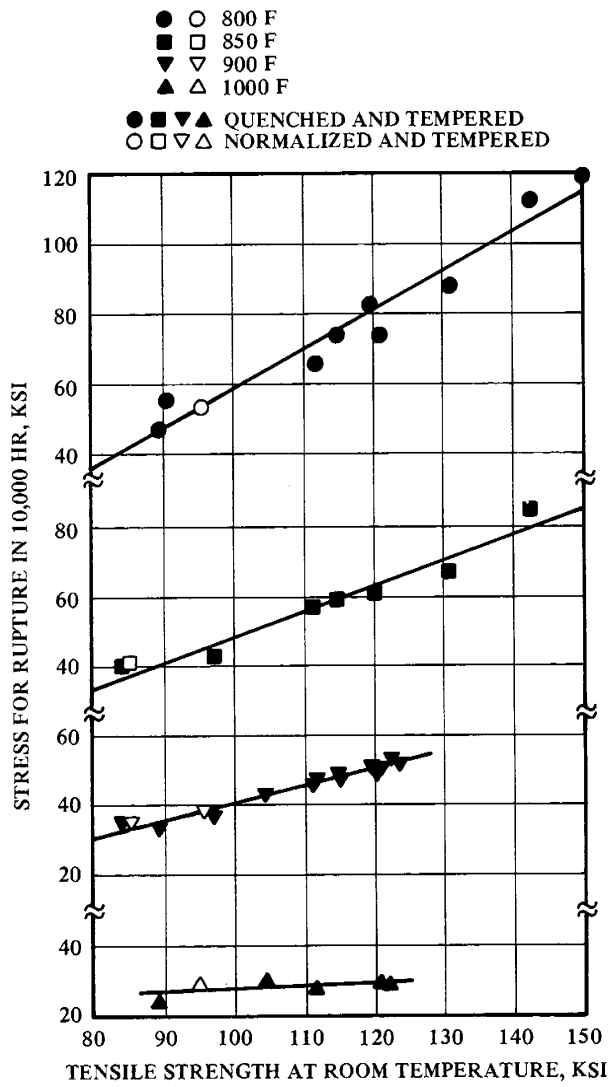


FAMILY OF REGRESSION LINES FOR ISOTHERMAL LOG STRESS VERSUS LOG TIME-FOR-RUPTURE SCATTER BANDS, EXTENDED TO 100,000 HOURS FOR ANNEALED 2-1/4Cr-1Mo STEEL (32)



AVERAGE STRESS-RUPTURE VALUES FOR NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL BASED ON THE RUPTURE EQUATION BELOW (33):

$$\log_{10} t_r = 33.41 - 3.65 \log_{10}\sigma - 0.02564T - 1.052 \times 10^{-5}\, T\sigma$$



VARIATION OF RUPTURE STRENGTH (10,000 HOURS) OF NORMALIZED-AND-TEMPERED AND QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo STEEL (32)

where

t_r = rupture time, hr
σ = applied stress, MPa
T = degrees Kelvin

and, the value 33.41 is based on average data for steels from USA, Great Britain, West Germany, Japan, and France.

ESTIMATED CREEP AND CREEP-RUPTURE STRENGTHS OF NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL[a] (33)

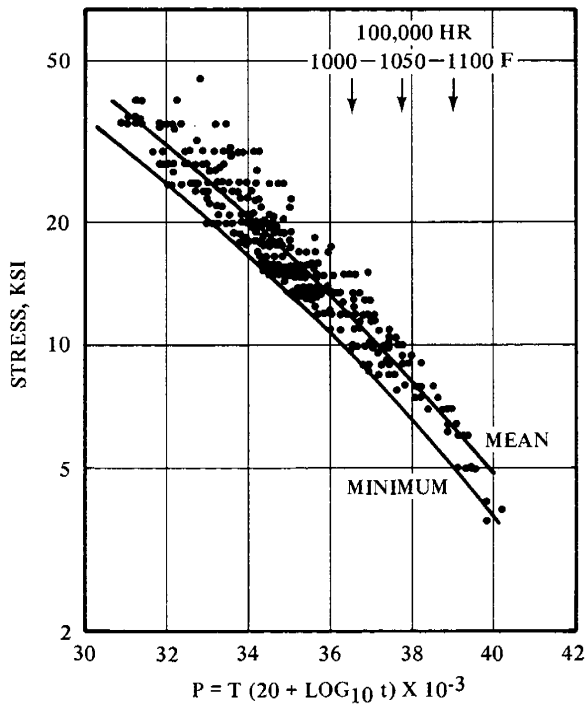
Temperature, F	10^5 Hr Creep-Rupture Strength, MPa (ksi)				10^{-5} Percent/Hr Average Creep Strength, MPa (ksi)
	Average[b]	Minimum[c]	0.67 x Average	0.8 x Minimum	
800	240 (34.7)	200 (29.1)	160 (23.2)	160 (23.3)	201 (29.2)
850	189 (27.4)	155 (22.5)	127 (18.4)	124 (18.0)	156 (22.6)
900	145 (21.0)	117 (16.9)	97 (14.0)	94 (13.5)	117 (17.0)
950	108 (15.7)	85 (12.4)	72 (10.5)	68 (9.9)	86 (12.5)
1000	79 (11.5)	61 (8.8)	53 (7.7)	49 (7.0)	61 (8.9)
1050	56 (8.2)	42 (6.2)	38 (5.5)	34 (5.0)	43 (6.2)

(a) Material conforming to ASTM A387, Grade 22, Class 2.
(b) Estimated from average lot constant.
(c) Estimated from a "minimum" lot constant given by the average minus 1.65 $\sqrt{V_B + V_W}$.

ESTIMATED ALLOWABLE DESIGN STRESS VALUES FOR NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL[a] (33)

Temperature, F	Allowable Stress Value, ksi			
	Current ASME Code Section VIII, Division I for SA-387, Grade 22		Analysis[b]	Controlling Criterion[c]
	Class 1[d]	Class 2		
75	15.0	–	18.7	1
100	15.0	–	18.7	1
200	15.0	–	18.7	1
300	15.0	–	18.1	2
400	15.0	–	18.1	2
500	15.0	–	18.1	2
600	15.0	–	18.1	2
700	15.0	17.2	18.1	2
800	15.0	16.9	17.8	2
850	14.4	16.4	17.2	2
900	13.1	15.8	13.6	3
950	11.0	11.0	9.9	3
1000	7.8	7.6	7.1	3
1050	5.8	5.8	4.9	3

(a) Material conforming to ASTM A387, Grade 22, Class 2.
(b) Estimated from the analysis of (33). Note, however, that allowable stresses can only be established by the appropriate ASME Code bodies and that factors other than data (such as service experience) can influence those values.
(c) 1 = one-fourth of the room-temperature ultimate tensile strength.
2 = 1.1 x one-fourth of the ultimate tensile strength at temperature.
3 = 80 percent of the minimum stress for rupture in 100,000 hr.
(d) Also Code Case N-47 for maximum allowable stress.

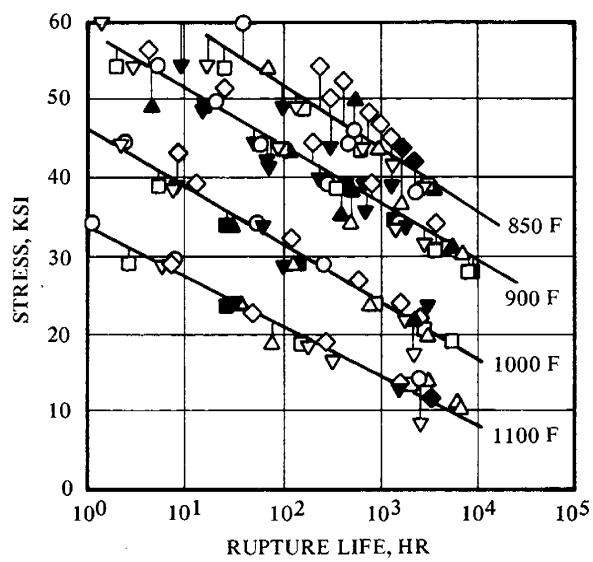


VARIATION OF LARSON-MILLER PARAMETER WITH STRESS FOR RUPTURE OF ANNEALED MATERIAL (33)

T = temperature, R; t = time, hr

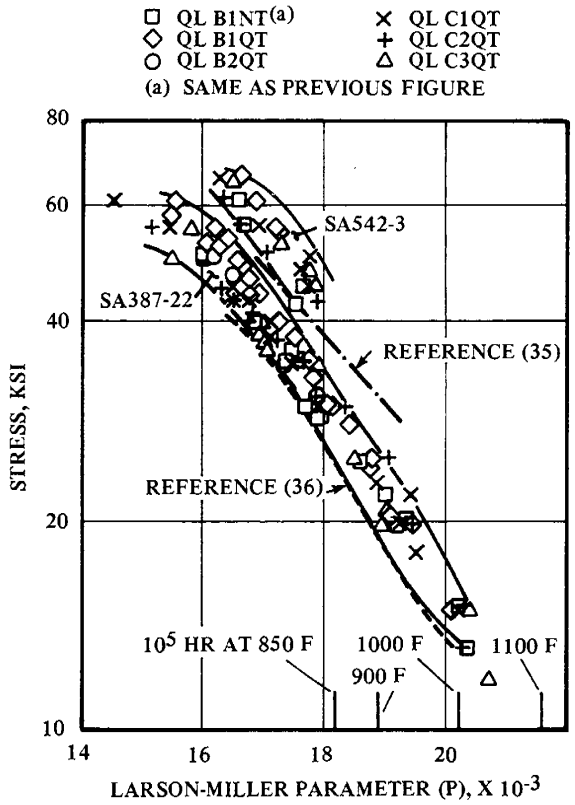
□ QL B1NT[a]
■ CL B1NT
◇ QL B1QT
◆ CL B1QT
○ QL B2QT
▼ QL C1QT
▽ QL C2QT
▲ QL C3QT

(a) QL INDICATES QUARTER LINE OF PLATE
CL INDICATES CENTER LINE OF PLATE
NT INDICATES NORMALIZED AND TEMPERED
QT INDICATES QUENCHED AND TEMPERED



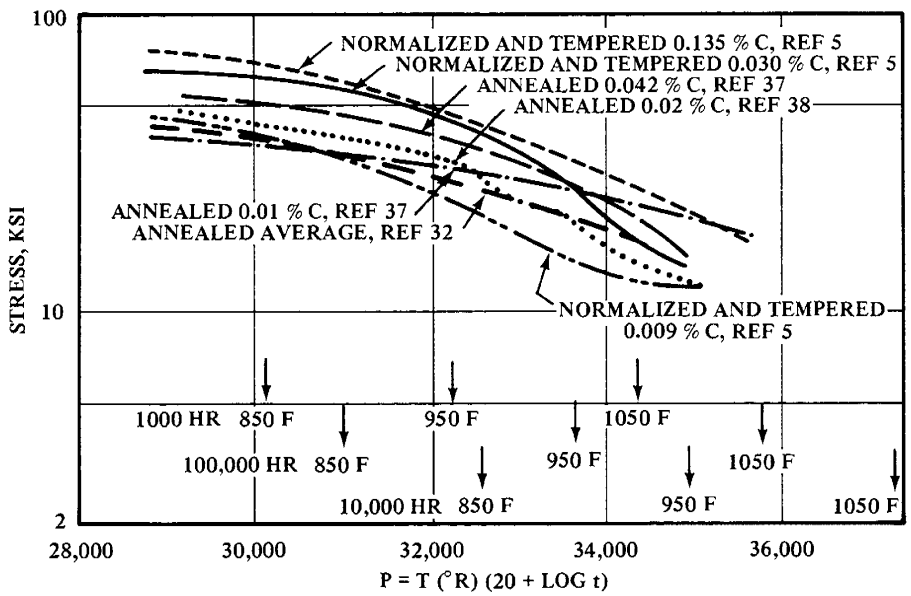
STRESS-RUPTURE DATA FOR NORMALIZED-AND-TEMPERED AND QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo PLATES HEAT TREATED TO SA387-22, CLASS 2 PROPERTIES (34)

Thick plates (260 to 300 mm).



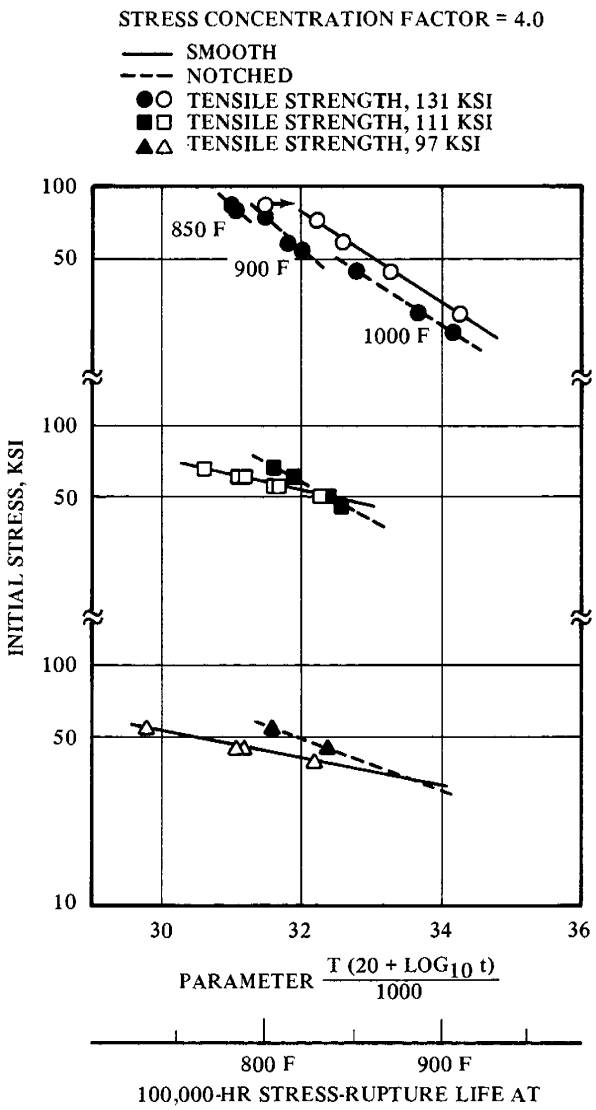
STRESS-RUPTURE DATA PLOTTED AS A FUNCTION OF THE LARSON-MILLER PARAMETER FOR QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo STEEL HEAT TREATED TO TWO DIFFERENT STRENGTH LEVELS (34)

$P = T (20 + LOG_{10} t_r)$; T = temperature, R;
t_r = rupture time, hr



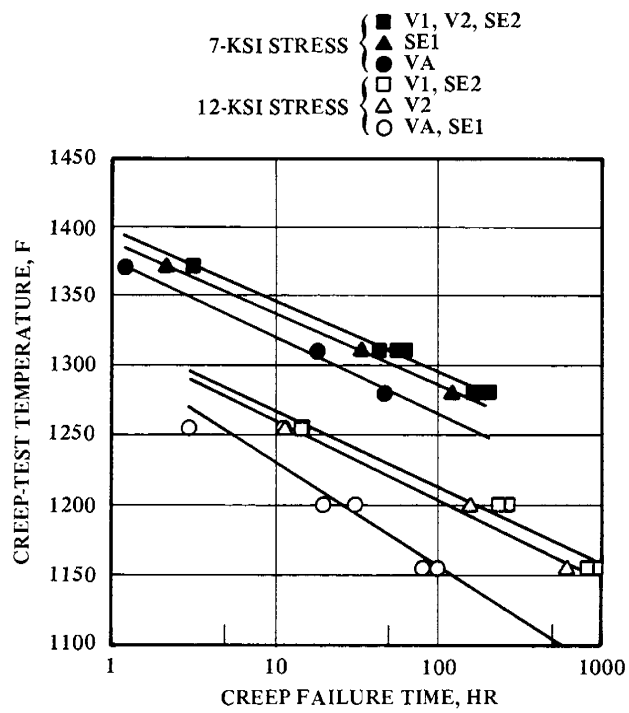
COMPARISON OF LARSON-MILLER PARAMETERS FOR 2-1/4Cr-1Mo STEEL (5)

(T = temperature, R; t = time, hr)



EFFECT OF NOTCHES ON CREEP-RUPTURE PROPERTIES OF 2-1/4Cr-1Mo STEEL QUENCHED AND TEMPERED TO VARIOUS STRENGTH LEVELS (39)

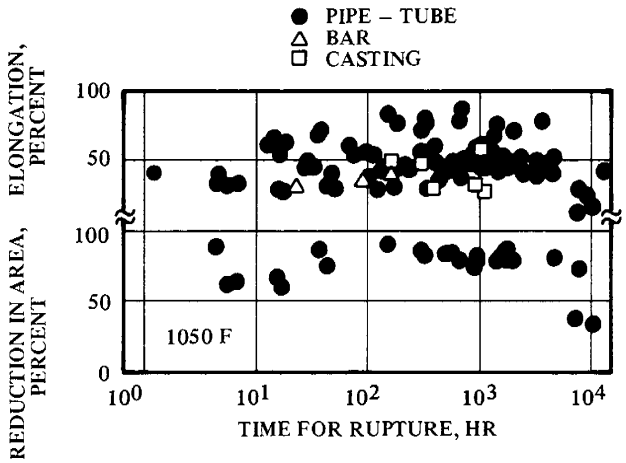
Note: (T = temperature, R; t = time, hr).



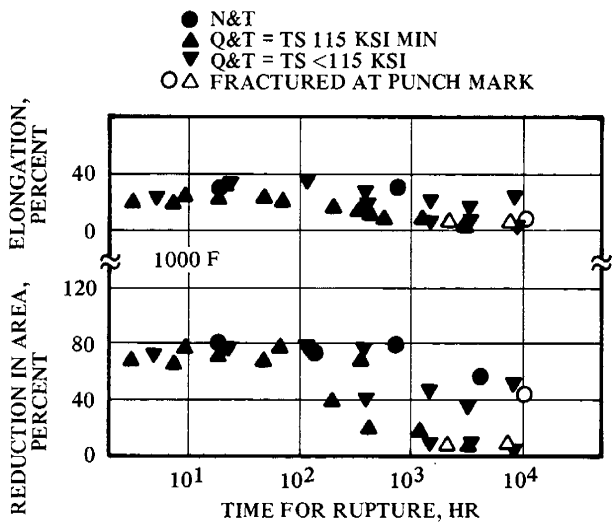
RESULTS OF ISOSTRESS CREEP TESTS ON SERVICE-EXPOSED 2-1/4Cr-1Mo STEEL (40)

Material Conditions:

- V1 = Virgin material
- V2 = Virgin + aged at 1375 F for 200 hr (to simulate 30 years at 1015 F), reheat treated at 1700 F for 2 hr (to restore virgin microstructure)
- VA = Virgin + aged at 1375 F for 200 hr
- SE1 = Service exposed at 1015 F for 30 years
- SE2 = Service exposed + reheat treated at 1700 F for 2 hr.



VARIATION OF RUPTURE DUCTILITY WITH TIME FOR RUPTURE FOR ANNEALED 2-1/4Cr-1Mo STEEL (32)

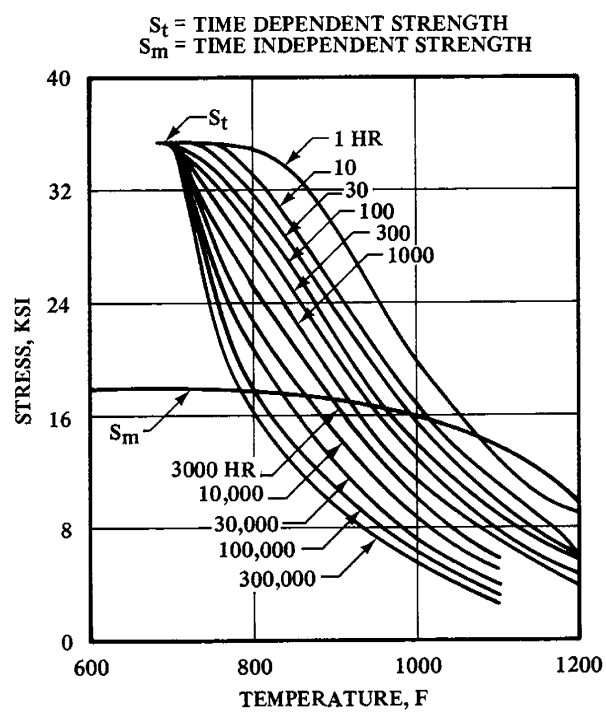


VARIATION OF RUPTURE DUCTILITY WITH TIME FOR RUPTURE FOR NORMALIZED-AND-TEMPERED AND QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo STEEL (32)

ALLOWABLE STRESS-INTENSITY VALUES FOR 2-1/4Cr-1Mo STEEL[a] (22)

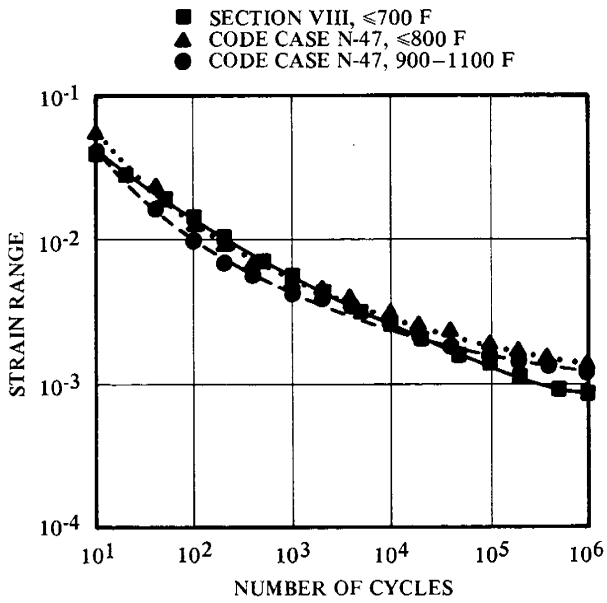
Temperature, F	1 Hr	10 Hr	30 Hr	10^2 Hr	3 x 10^2 Hr	10^3 Hr	3 x 10^3 Hr	10^4 Hr	3 x 10^4 Hr	10^5 Hr	3 x 10^5 Hr
700	–	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9
750	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9
800	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	17.9	16.1
850	17.6	17.6	17.6	17.6	17.6	17.6	17.6	17.6	16.3	14.0	12.3
900	17.2	17.2	17.2	17.2	17.2	17.2	16.5	14.4	12.5	10.9	9.6
950	16.7	16.7	16.7	16.7	16.3	14.8	13.2	11.3	9.7	8.4	7.3
1000	15.9	15.9	15.5	14.2	13.1	11.9	10.4	8.7	7.5	6.3	5.2
1050	14.9	13.8	12.5	11.2	10.2	9.3	7.9	6.7	5.7	4.7	4.0
1100	13.6	11.0	10.0	9.0	8.2	7.2	6.2	5.0	4.1	3.3	2.7
1150	10.8	8.8	8.0	7.2	6.3	5.4	–	–	–	–	–
1200	9.0	6.2	6.1	5.9	5.1	4.1	–	–	–	–	–

(a) SA387, Grade 22, Class 1.



ALLOWABLE STRESSES AS A FUNCTION OF TEMPERATURE FOR 2-1/4Cr-1Mo STEEL (ASME SA387, GRADE 22, CLASS 1 MATERIAL) (22)

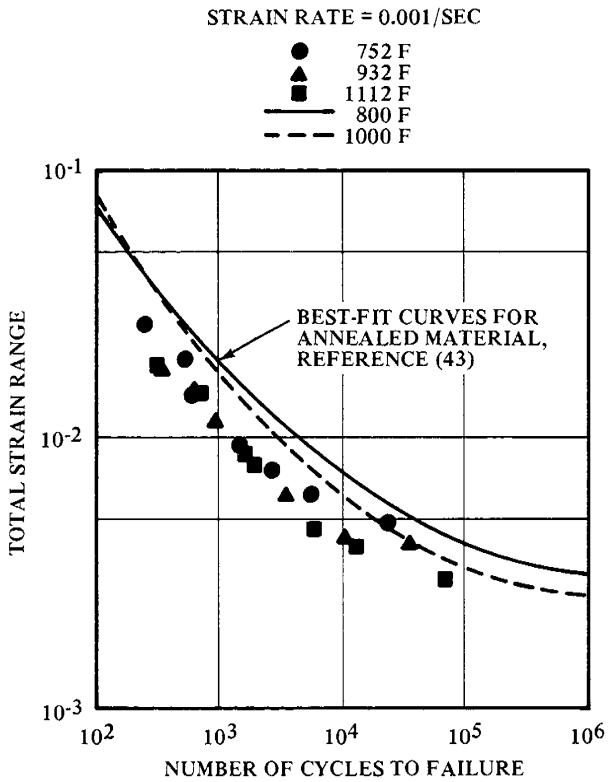
FATIGUE



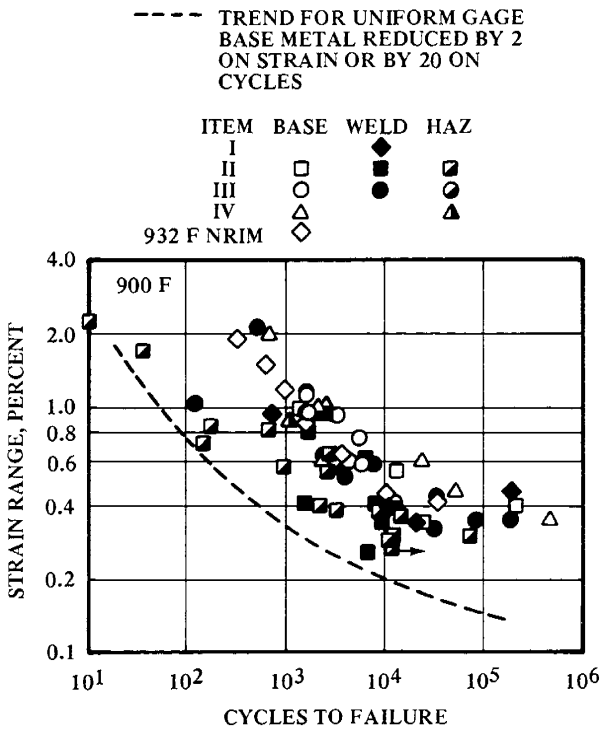
COMPARISON OF DESIGN FATIGUE CURVES FOR 2-1/4Cr-1Mo STEEL (42)

Stress ranges may be obtained by multiplying the strain range by the elastic modulus at the appropriate temperature.

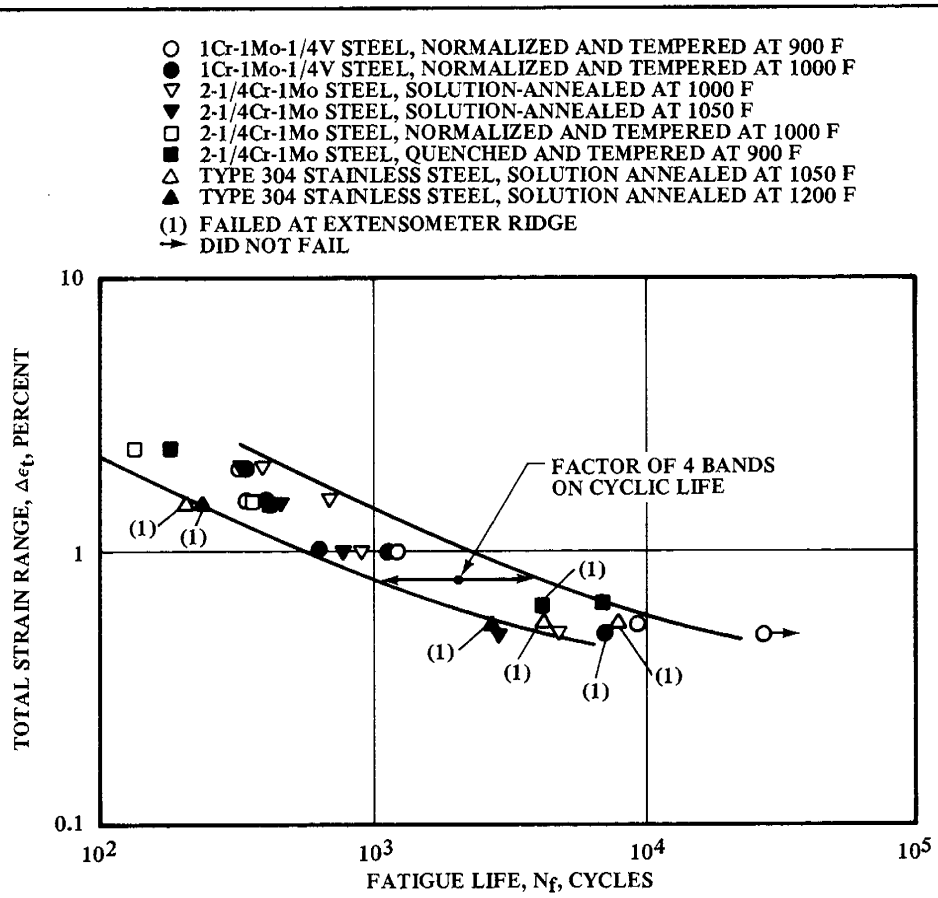
The design curves were obtained by applying a factor of 2 on stress amplitude or a factor of 20 on cycles to failure on actual minimum data.



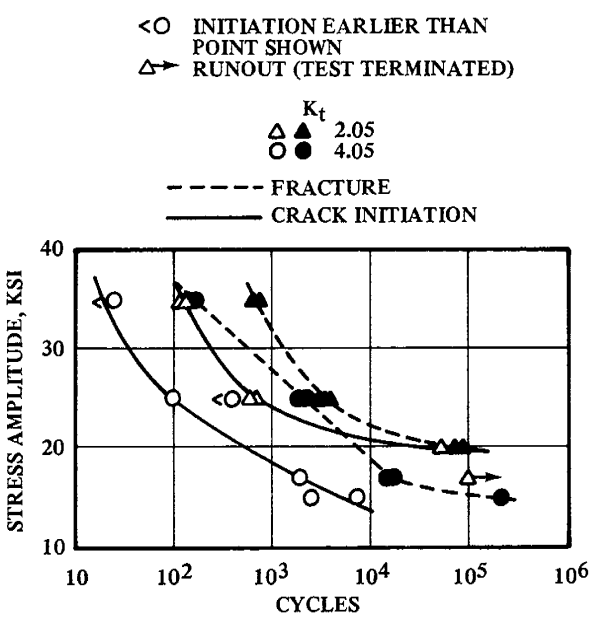
FATIGUE DATA FOR NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL (43)



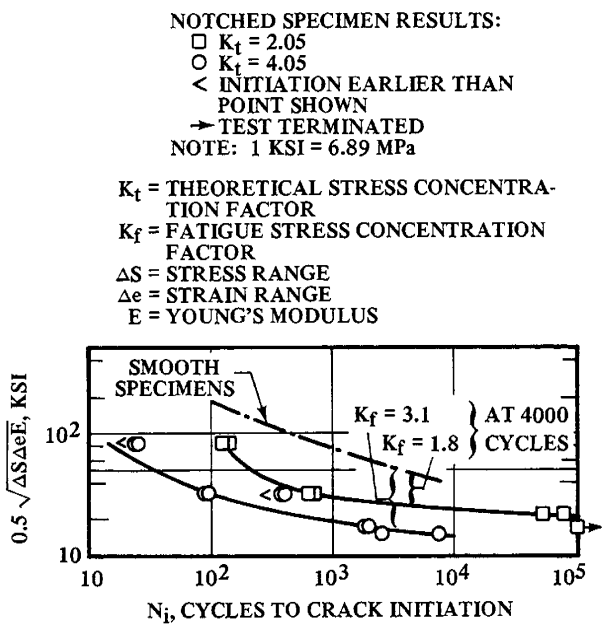
FATIGUE DATA FOR NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL BASE METAL AND WELDMENTS (44)



LOW-CYCLE FATIGUE OF QUENCHED AND TEMPERED, AND ANNEALED 2-1/4Cr-1Mo STEEL AND SELECTED OTHER MATERIALS (45)

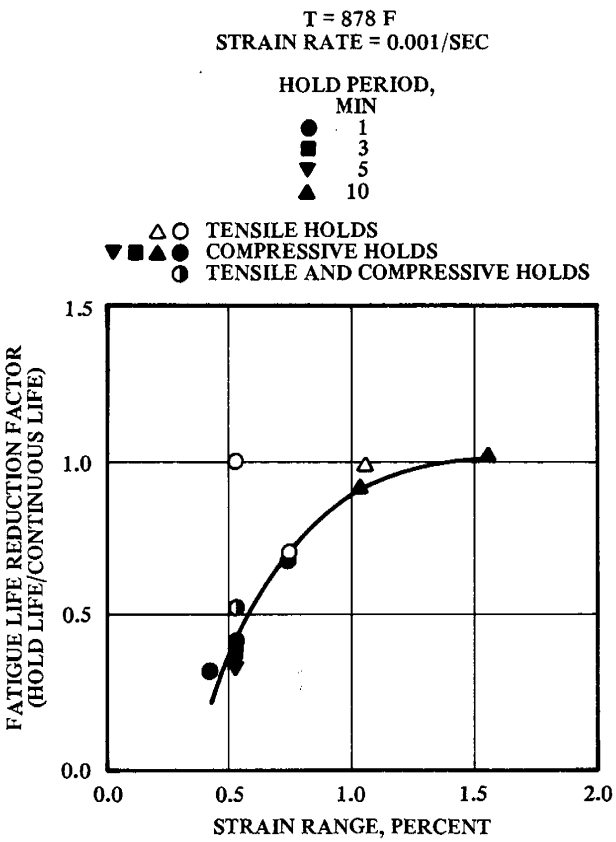


RESULTS OF NOTCHED-SPECIMEN LOW-CYCLE FATIGUE TESTS ON ANNEALED 2-1/4Cr-1Mo STEEL AT 1000 F (46)

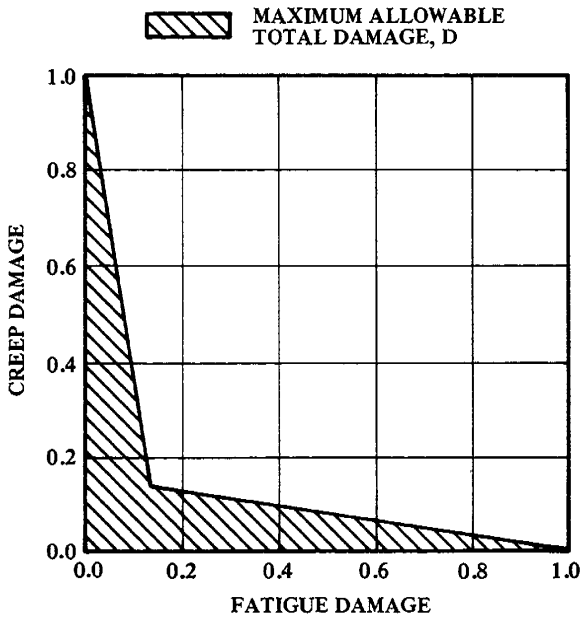


COMPARISON OF SMOOTH AND NOTCHED SPECIMEN LOW-CYCLE FATIGUE RESULTS FOR ANNEALED 2-1/4Cr-1Mo STEEL AT 1000 F (46)

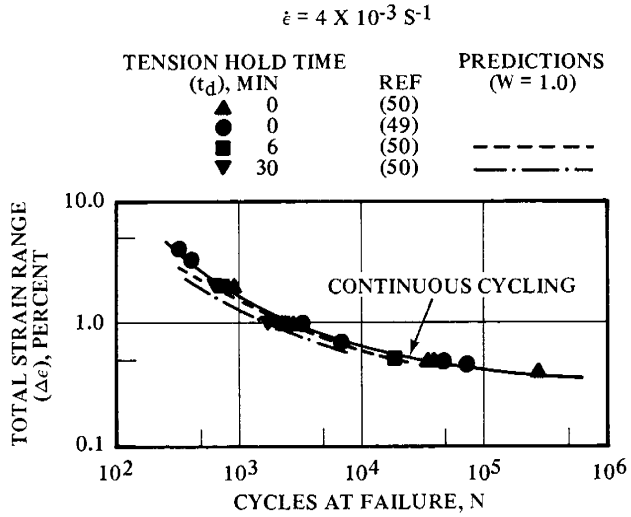
FATIGUE



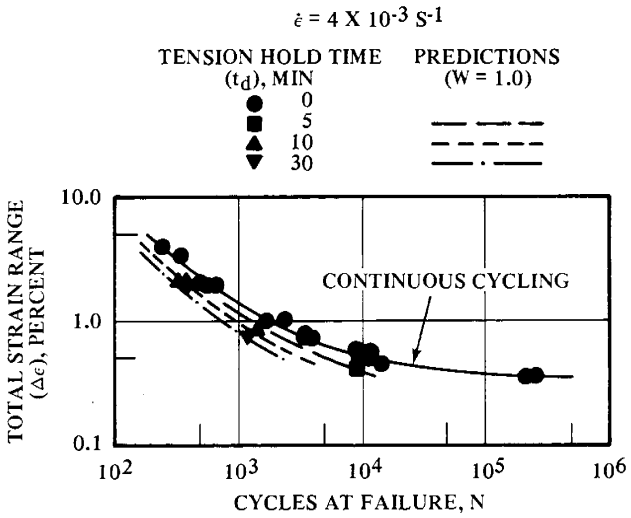
EFFECT OF TYPE AND LENGTH OF HOLD PERIOD AND LEVEL OF STRAIN RANGE ON THE CYCLIC LIFE OF NORMALIZED AND TEMPERED 2-1/4Cr-1Mo STEEL (47)



CREEP-FATIGUE DAMAGE ENVELOPE FOR 2-1/4Cr-1Mo Steel (22,42)



a. AT 900 F



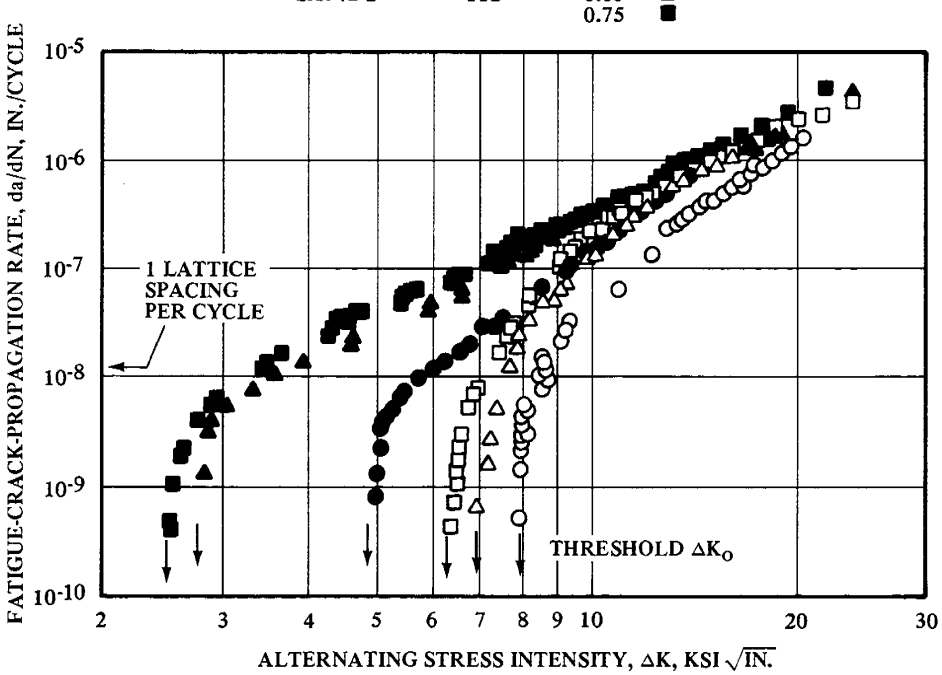
b. AT 1100 F

EFFECT OF HOLD TIME ON FATIGUE LIFE OF ANNEALED 2-1/4Cr-1Mo STEEL (48)

FATIGUE CRACK GROWTH

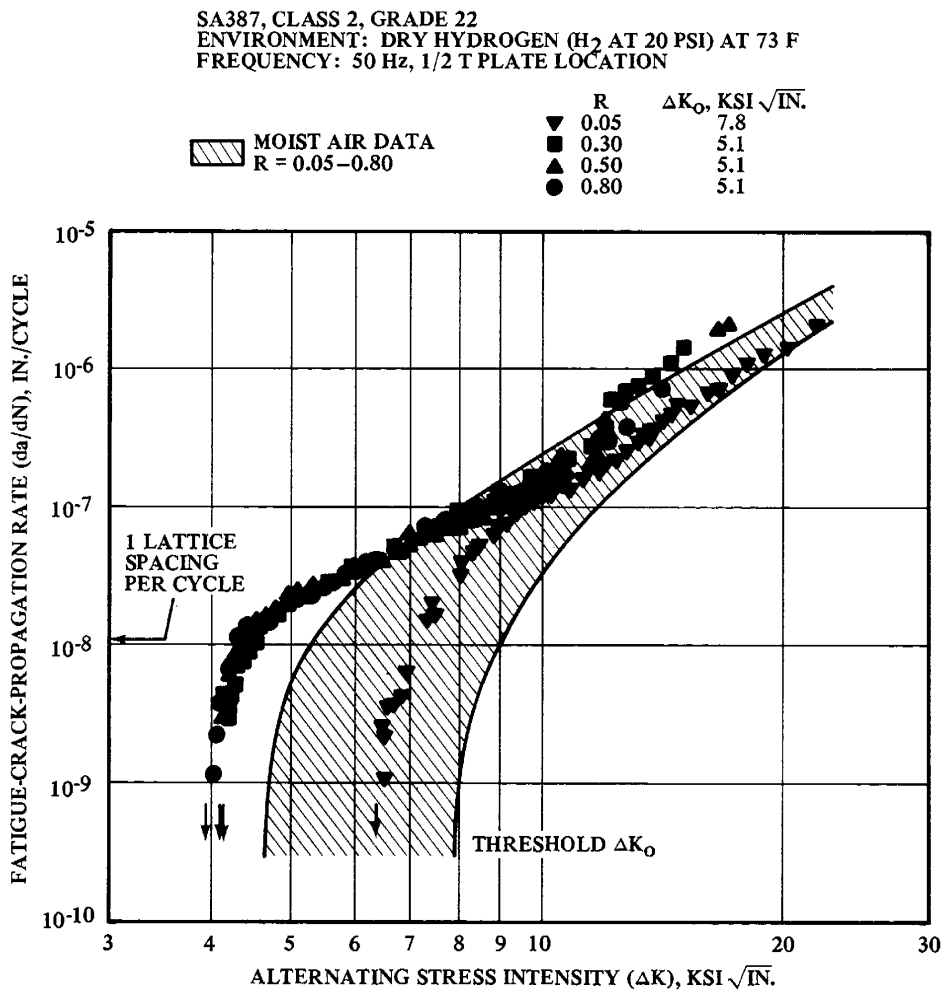
AS-RECEIVED MATERIAL, 1/2 T SAMPLES
AMBIENT TEMPERATURE MOIST AIR (30 PERCENT RH)
FREQUENCY = 50 Hz

STEEL	YIELD STRENGTH, KSI	R	
SA387	42	0.05	○
		0.80	●
SA542-3	70	0.05	△
		0.75	▲
SA542-2	112	0.05	□
		0.75	■



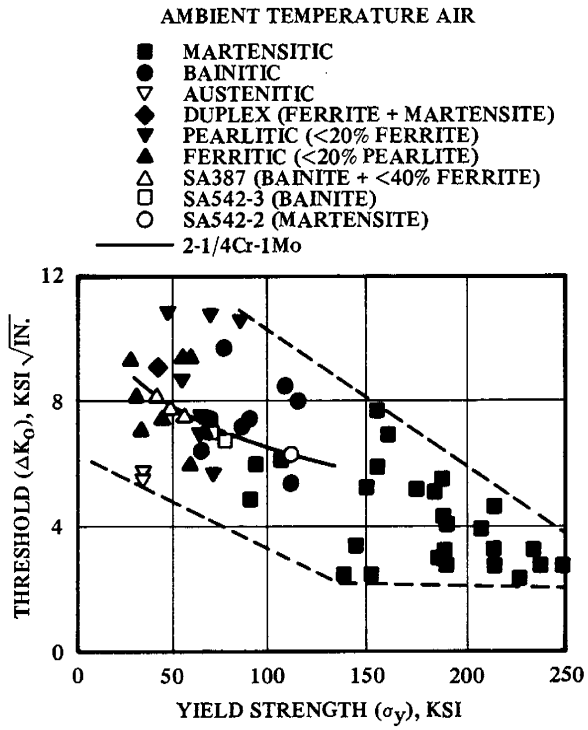
VARIATION OF FATIGUE-CRACK-PROPAGATION RATES WITH ALTERNATING STRESS INTENSITY FOR 2-1/4Cr-1Mo STEELS TESTED AT 50 Hz IN MOIST AIR AT LOW (R = 0.05) AND HIGH (R = 0.75 TO 0.80) LOAD RATIOS (51)

FATIGUE CRACK GROWTH

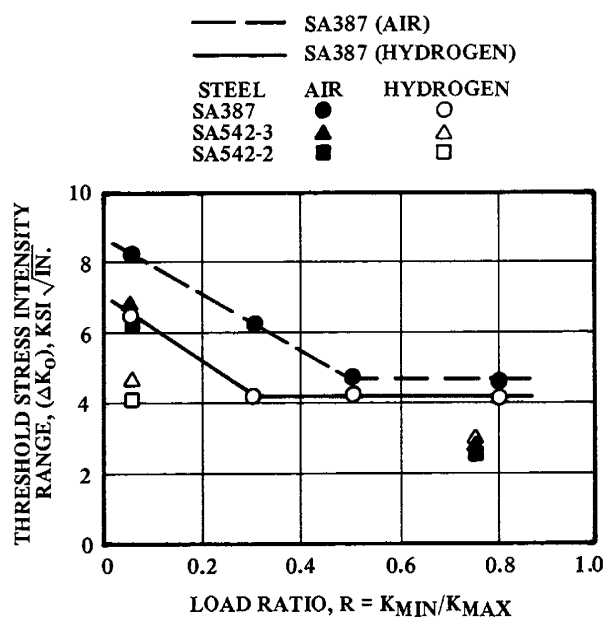


VARIATION OF FATIGUE CRACK PROPAGATION RATES, da/dN, WITH ALTERNATING STRESS INTENSITY, ΔK, IN 2-1/4Cr-1Mo STEEL TESTED AT 50 Hz IN DEHUMIDIFIED GASEOUS HYDROGEN AS A FUNCTION OF LOAD RATIO, R (51)

FATIGUE CRACK GROWTH



VARIATION OF THRESHOLD ΔK_O VALUES AT R = 0 TO 0.05 WITH YIELD STRENGTH FOR A WIDE RANGE OF STEELS [12–14] SHOWING PRESENT DATA FOR 2-1/4Cr-1Mo PRESSURE VESSEL STEELS (51)

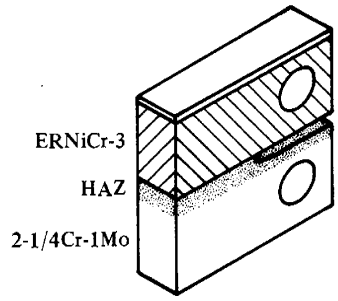


VARIATION OF THRESHOLD STRESS INTENSITIES, ΔK_O, WITH LOAD RATIO, R, FOR 2-1/4Cr-1Mo STEELS TESTED AT 50 Hz IN AIR AND HYDROGEN GAS (51)

Room-temperature data.

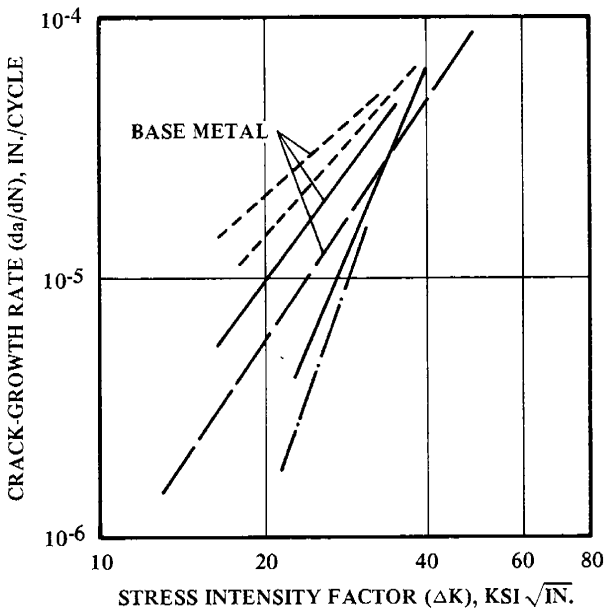
FATIGUE CRACK GROWTH

WELDED SPECIMEN ORIENTATION



TEMPERATURE = 950 F
R = 0.05

- - - - - - - 0.067 Hz
——— 0.67 Hz
— — 3.3–5.0 Hz
—·— 6.7 Hz



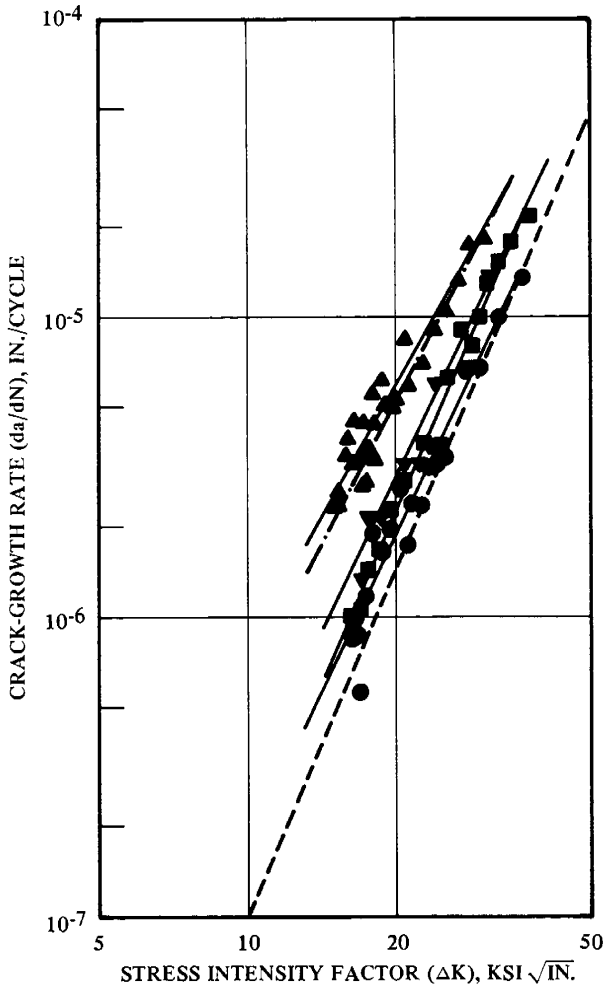
EFFECT OF FREQUENCY ON FATIGUE CRACK GROWTH RATE IN HEAT-AFFECTED ZONE IN AIR

Crack growth rate of base metal at the corresponding frequency is always higher.

TESTED IN AIR; R = 0.05

SPECIMEN	TEMP, F	FREQ., CPM
■ 10DTL	400	400
● 11DTL	700	400
▼ 12DTL	850	400
▲ 8DTL	1100	400

—·— 950 F
— — ROOM TEMPERATURE

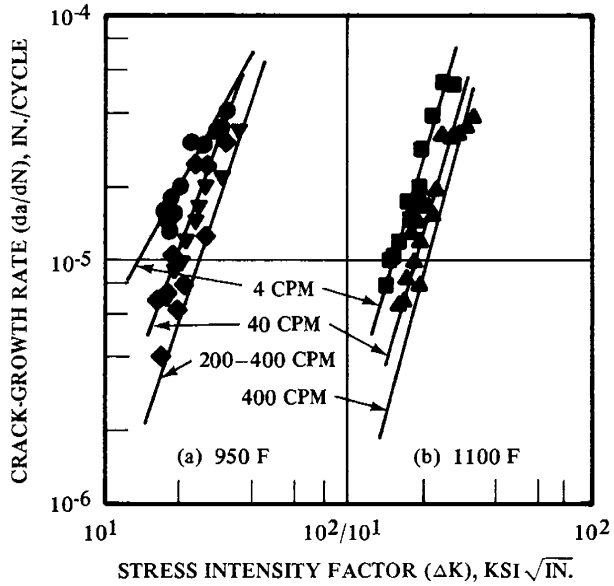


EFFECT OF TEMPERATURE ON THE FATIGUE-CRACK-GROWTH RATE OF ANNEALED 2-1/4Cr-1Mo STEEL (42,53)

FATIGUE CRACK GROWTH

TESTED IN AIR; R = 0.05

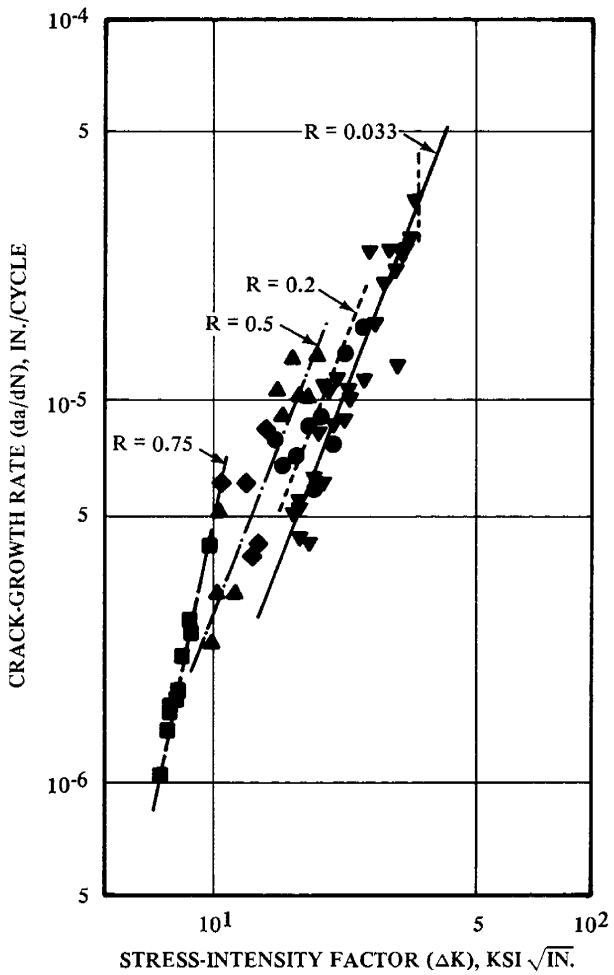
	FREQ., CPM	SPECIMEN	TEMP, F
●	4	9DTL	950
■	4	17ATL	1100
▲	40	8ATL	1100
▼	40	14DTL	950
◆	40	1PTL	950



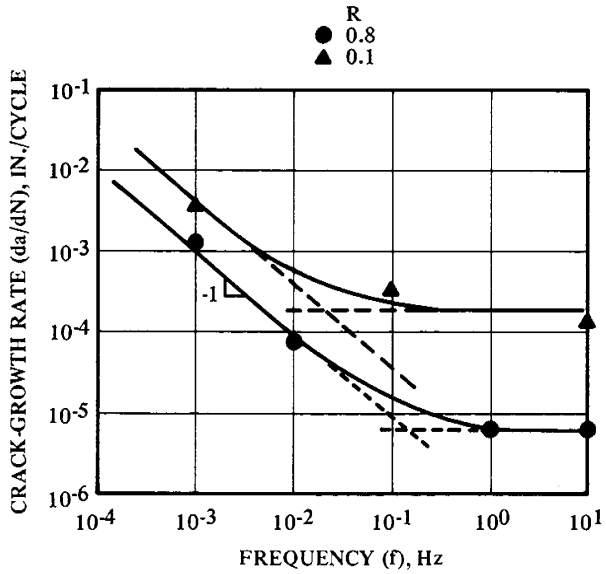
EFFECT OF CYCLIC FREQUENCY ON THE FATIGUE-CRACK-GROWTH RATE OF ANNEALED 2-1/4Cr-1Mo STEEL AT 950 AND 1100 F (42,53)

TESTED IN AIR AT 950 F; 40 TO 50 CPM

	SPECIMEN	RATIO, R
▼	4ALT	0.033
●	5ALT	0.2
◆	3CTL	0.5
▲	6CTL	0.5
■	9CTL	0.75

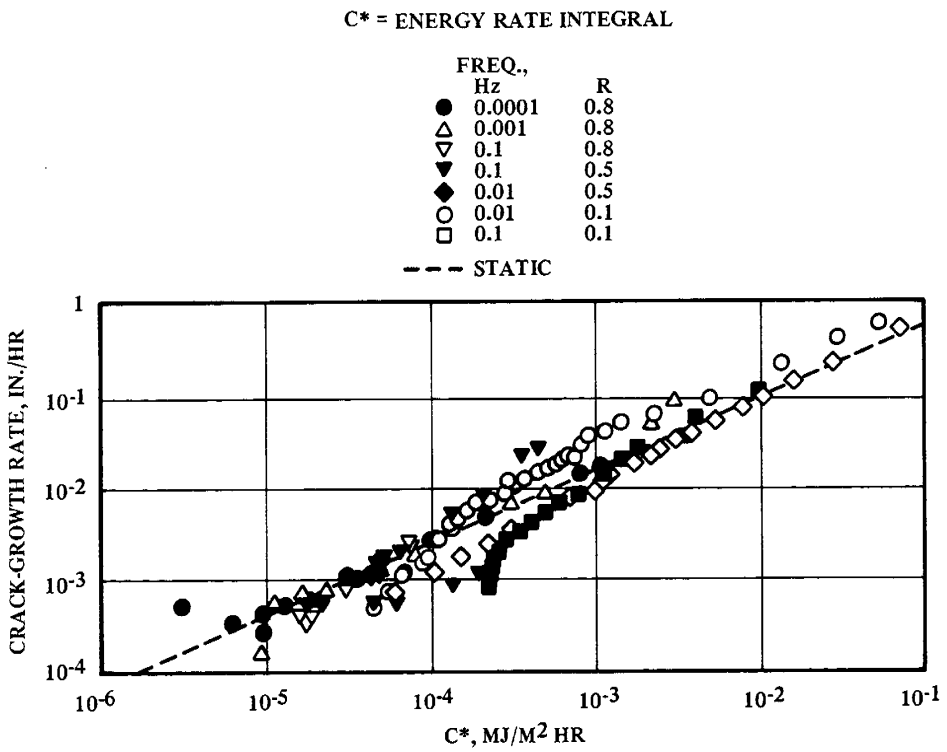


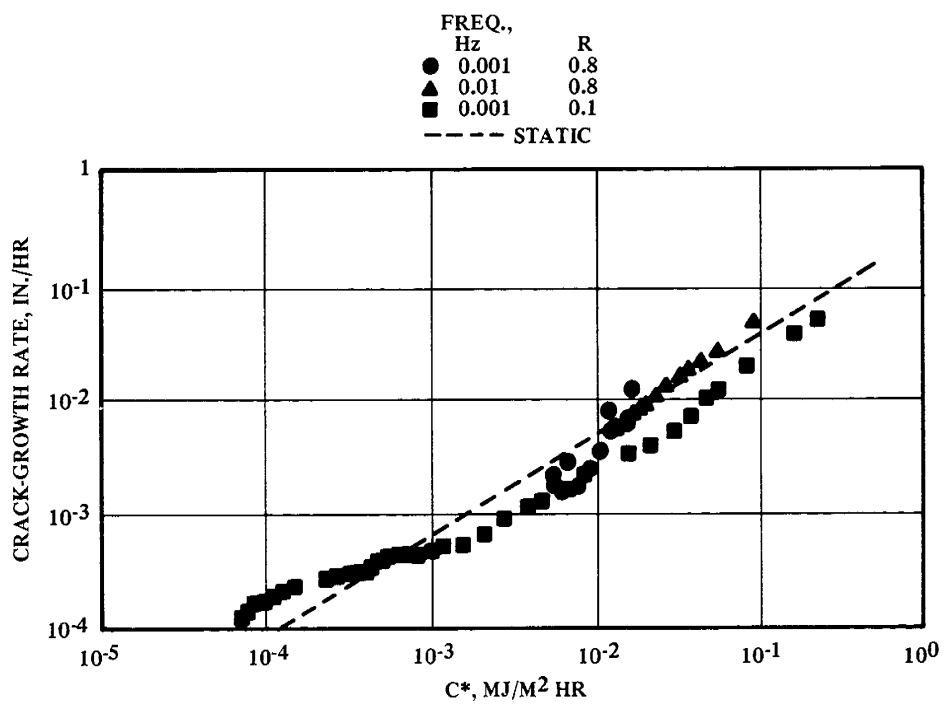
EFFECT OF STRESS RATIO ON THE FATIGUE-CRACK-GROWTH RATE OF ANNEALED 2-1/4Cr-1Mo STEEL AT 950 F (42,53)



DEPENDENCE OF CRACK-GROWTH/CYCLE FOR 2-1/4Cr-1Mo STEEL AT 1000 F AT A CONSTANT K_{MAX} of 45.5 KSI $\sqrt{IN.}$ (54)

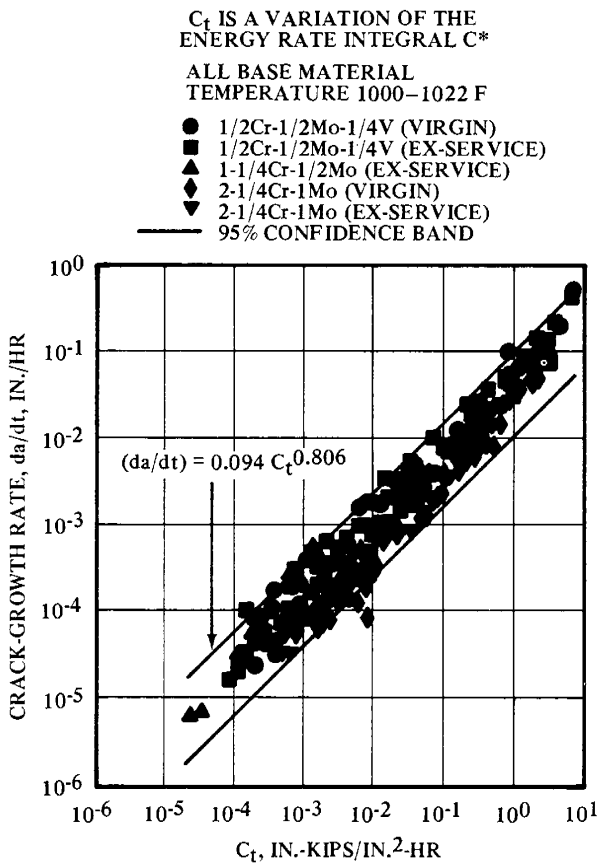
FATIGUE CRACK GROWTH



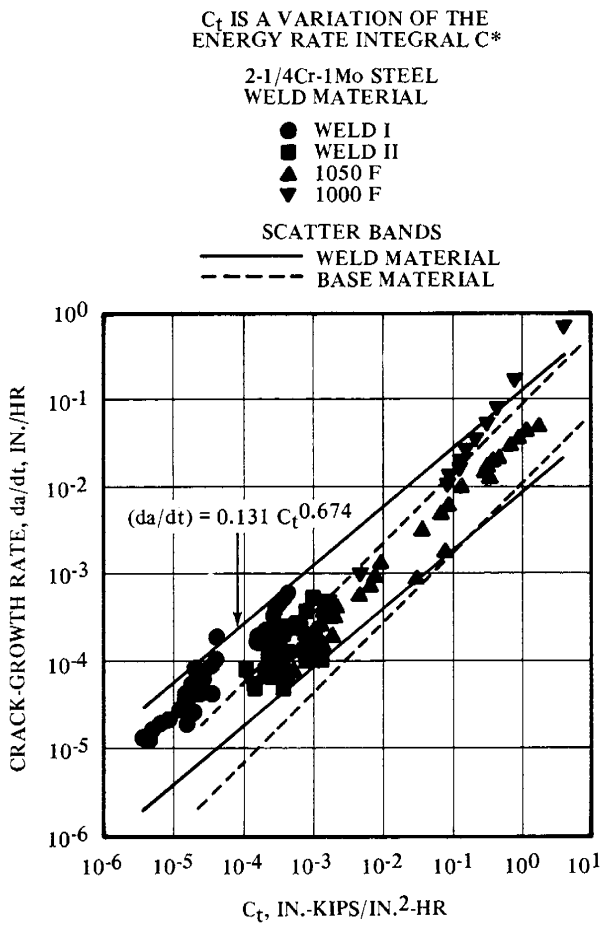


SENSITIVITY OF CRACK-GROWTH RATE AT LOW FREQUENCIES TO C* FOR 2-1/4Cr-1Mo STEEL AT 1000 F (54)

CREEP CRACK GROWTH



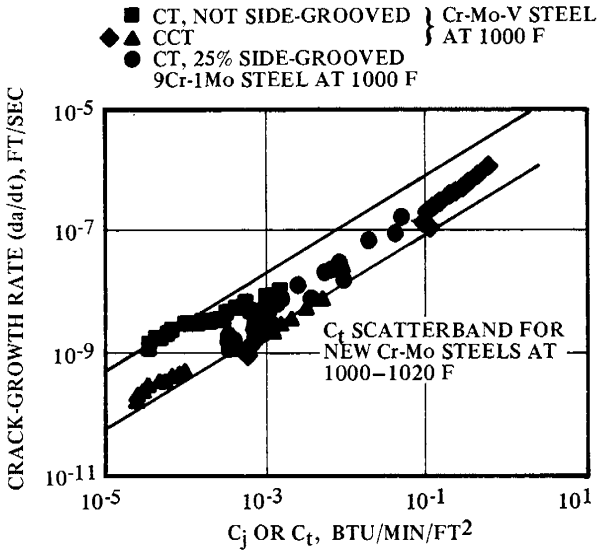
CREEP CRACK-GROWTH BEHAVIOR OF Cr-Mo AND Cr-Mo-V BASE MATERIAL. THE PLOT INCLUDES ALL EX-SERVICE AND NEW MATERIAL DATA (55)



CREEP CRACK-GROWTH BEHAVIOR OF 2-1/4Cr-1Mo WELD MATERIAL (55)

Welds I and II represent data from laboratory weldments, while the remainder are from weldments removed from ex-service components.

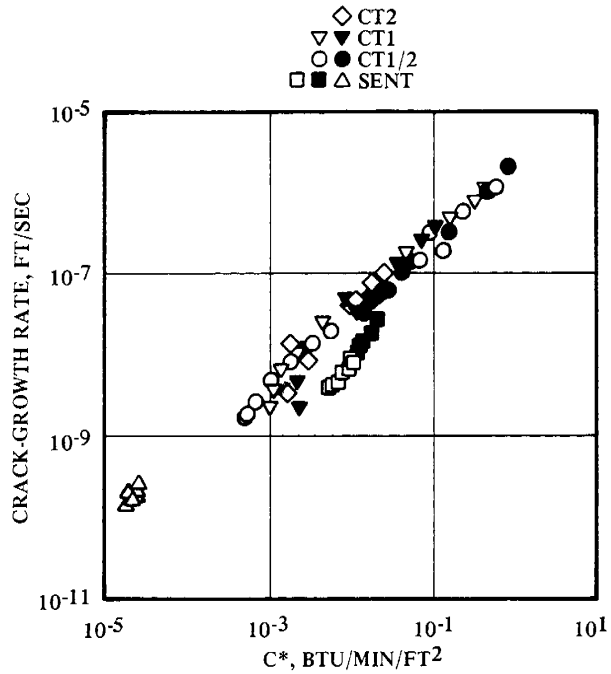
CREEP CRACK GROWTH



CREEP CRACK-GROWTH RATE FOR 2-1/4Cr-1Mo, 1-1/4Cr-1/2Mo, 1Cr-1Mo-1/4V and 9Cr-1Mo-V-Nb STEELS TESTED AT 1000 F (55,56)

The data points correspond to C_j and the solid lines correspond to the C_t parameter.

C_j and C_t are variations of the energy rate integral C^*.



MEASURED CRACK-GROWTH RATE IN 2-1/4Cr-1Mo STEEL (AS-PROCESSED MATERIAL) AT 1004 F DURING STEADY-STATE CREEP VERSUS C* (57)

WELDS—TENSILE

COMPARISON OF TENSILE PROPERTIES OF 2-1/4Cr-1Mo STEEL WELD METAL, TRANSVERSE WELD SPECIMENS, AND BASE METAL (58)

Carbon, weight percent	Temperature, F	Yield and Ultimate Tensile Strength, ksi					
		Weld Metal[a]		Transverse Weld[b]		Base Metal[c]	
		Yield	Ultimate	Yield	Ultimate	Yield	Ultimate
0.003	75	60.2	74.0	32.3	55.2	37.0	52.9
	1050	46.3	50.1	20.5	44.3	18.1	37.3
0.035	75	72.2	86.1	41.3	60.2	38.5	59.0
	1050	55.0	57.5	22.4	42.9	20.4	36.8
0.11	75	90.7	101.6	58.1	82.6	53.7	78.7
	1050	66.0	68.6	41.8	53.2	44.3	51.3

(a) Longitudinal all-weld specimens; multipass welds made by gas tungsten-arc process.
(b) Transverse weld specimens, containing weld metal, heat-affected zone, and base metal in the gage length.
(c) Normalized and tempered (austenitized 1 hr at 1700 F, air cooled; tempered 1 hr at 1300 F, air cooled).

TENSILE PROPERTIES OF 2-1/4Cr-1Mo WELD METAL PREPARED BY SHIELDED-METAL-ARC WELD PROCESS[a] (3)

Temperature, F	Weld Metal	Ultimate Tensile Strength, ksi	Yield Strength (0.2 Percent Offset), ksi	Elongation, percent	Reduction in Area, percent
80	0.02–0.03C	70.3	41.8	33.5	54.3
	0.04–0.05C	78.3	66.9	24.7	71.3
	0.08–0.09C	80.9	63.4	25.0	71.9
	0.12–0.13C	84.5	62.8	33.0	72.0
850	0.02–0.03C	53.1	35.8	28.5	75.0
	0.04–0.05C	60.1	52.1	23.9	71.7
	0.08–0.09C	61.4	51.8	26.0	75.3
	0.12–0.13C	63.0	52.9	27.0	62.5
950	0.02–0.03C	41.1	29.2	35.9	81.5
	0.04–0.05C	51.2	45.4	32.7	80.1
	0.08–0.09C	52.8	47.9	29.0	81.8
	0.12–0.13C	51.8	46.4	44.0	84.5
1050	0.02–0.03C	32.7	26.5	53.1	88.7
	0.04–0.05C	40.4	37.9	36.6	84.4
	0.08–0.09C	43.0	38.9	42.0	88.8
	0.12–0.13C	43.5	39.5	55.0	90.9

(a) Material postweld heat treated at 1300 F for 25 hours.

WELDS—TENSILE

COMPARISON OF TENSILE STRENGTHS OF SHIELDED METAL-ARC WELDS OF 2-1/4Cr-1Mo STEEL WITH ALLOWABLE STRESS VALUES SPECIFIED IN THE ASME CODE (3)

Code/Material	Allowable Stress for Base Metal, or 0.25 Times Ultimate Tensile Strength of Weld Metal, ksi, at Indicated Temperature			
	RT	850 F	950 F	1050 F
ASME Section VIII, Division 1, Class 1; also Code Case N-47 for 2-1/4Cr-1Mo steel	15.0	14.4	11.0	5.8
0.02–0.03C weld metal[a,b]	17.6	13.3	10.3	8.2
0.04–0.05C weld metal[a,b]	19.6	15.0	12.8	10.1
0.08–0.09C weld metal[a,b]	20.2	15.4	13.2	10.8
0.12–0.13C weld metal[a,b]	21.1	15.8	13.0	10.9

(a) The weld metals were postweld heat treated for 25 hours at 1300 F.

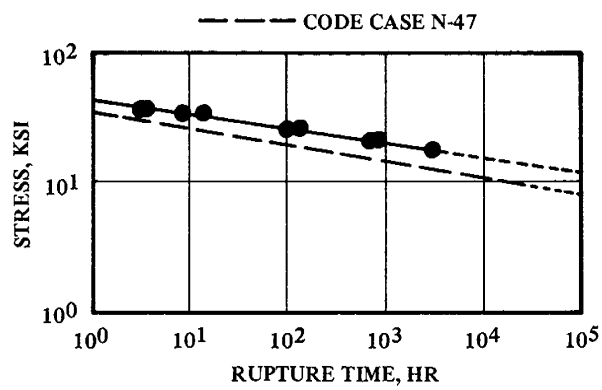
(b) The values quoted here are one-fourth of the measured tensile strengths. The reason is that the ASME Code applies a safety factor of 4 on the ultimate tensile strength to arrive at allowable stress values.

WELDS—CREEP

COMPARISON OF RUPTURE STRENGTHS FOR WELD METALS AND BASE METALS (59)

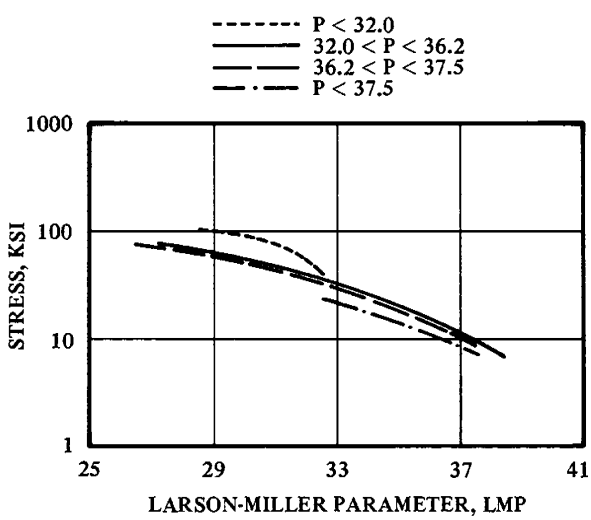
Material	Stress, ksi, at Temperature Indicated		
	850 F	950 F	1050 F
Weld Metals[a]			
Submerged arc weld	25	14.9	8.1
Electroslag weld (Q&T)[b]	24.3	15.7	8.7
Electroslag weld (N&T)[b]	24.1	12.9	6.8
Allowable Base-Metal Properties			
ASME Section VIII, Division 1, Class 2, Table UCS-23	16.4	11.0	5.8
Code Case N-47, maximum allowable for 100,000-hr use	14.0	8.4	4.7

(a) Data reported here are two-thirds of experimental values; the rationale is that the ASME Code allowable stresses are based partly on two-thirds of average stress for rupture in 100,000 hr.
(b) Q&T = quenched and tempered; N&T = normalized and tempered.



STRESS-RUPTURE DATA FOR 2-1/4Cr-1Mo WELD METAL AT 1050 F (59)

Submerged arc welds; postweld heat treated at 1300 F for 25 hr.



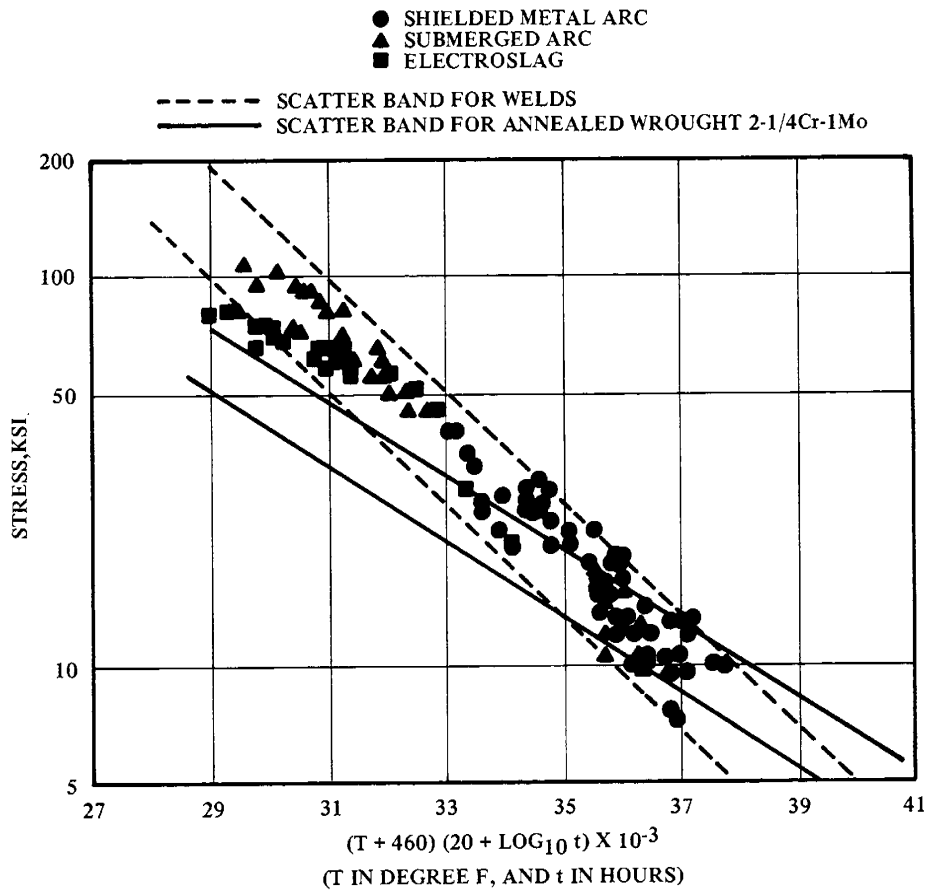
STRESS-RUPTURE BEHAVIOR OF 2-1/4Cr-1Mo SMA WELD METAL AS A FUNCTION OF TEMPER PARAMETER DETERMINED BY ANALYSIS OF LITERATURE DATA (3)

$P = (T + 460)(20 + \log_{10} t) \times 0.001$

where T is the tempering temperature (F), and t is the tempering time (hours).

$LMP = (T + 460)(20 + \log_{10} t_r) \times 0.001$
T = creep-rupture test temperature, F
t_r = time to rupture, hours

WELDS—CREEP

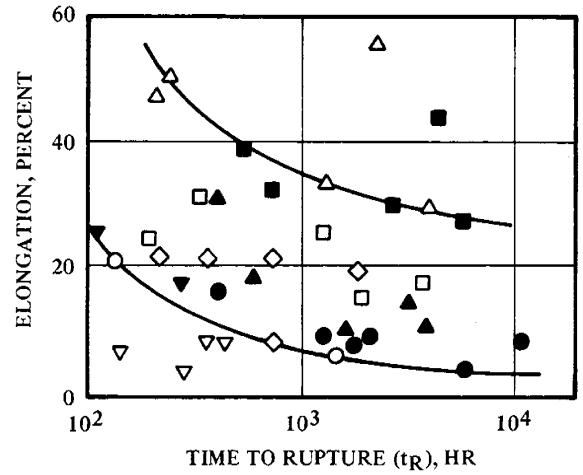


LARSON-MILLER PLOT FOR WELD METALS OF 2-1/4Cr-1Mo STEEL PREPARED BY VARIOUS WELDING TECHNIQUES (60)

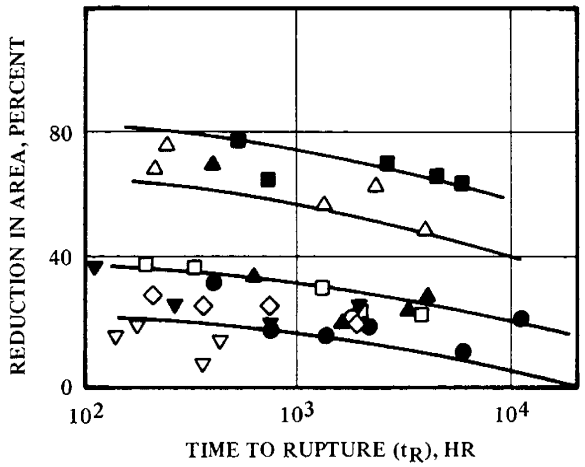
WELDS—CREEP

SMA = SHIELDED METAL ARC WELD
SA = SUBMERGED ARC WELD

TEST TEMP.		PROCESS	UTS, KSI
	○	SMA	83
	△	SMA	75
	▲	SMA	76
1100 F	□	SA	75
	■	SMA	80
	◇	SA	76
	●	SMA	95
	▼	SA	83
	▽	SA	131, 132



TEST TEMP.		PROCESS	UTS, KSI
	△	SMA	75
	▲	SMA	76
1100 F	□	SA	75
	■	SMA	80
	◇	SA	76
	●	SMA	95
	▼	SA	83
	▽	SA	131, 132

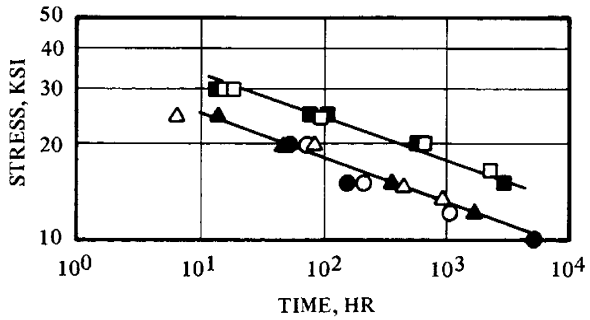


VARIATION OF RUPTURE DUCTILITY WITH TIME TO RUPTURE (60)

(a) NORMALIZED AND TEMPERED (1700 F FOR 1 HR, AIR COOL, 1300 F FOR 1 HR, AIR COOL)
(b) MULTIPASS WELD MADE BY GAS TUNGSTEN-ARC PROCESS; SPECIMENS CONTAINED WELD METAL, HAZ, AND BASE METAL IN THE GAGE LENGTH. CREEP CRACKS INITIATED AT PRIOR AUSTENITE GRAIN BOUNDARIES OF HAZ

○ ● 0.003 C
△ ▲ 0.035 C
□ ■ 0.110 C

□△○ BASE METAL[a]
■▲● TRANSVERSE WELDS[b]



A COMPARISON OF THE CREEP-RUPTURE STRENGTHS OF 2-1/4Cr-1Mo STEEL BASE METAL AND TRANSVERSE WELDS AT 1050 F (61)

CHARPY IMPACT DATA FOR QUENCHED AND TEMPERED 2-1/4Cr-1Mo PLATES AS A FUNCTION OF TEMPERING TIME AND TEMPERATURE

Source	(62)	(62)	(63)
Time-Temperature Parameter	33.4	36.4	36.55
Room Temperature Tensile Yield Strength, ksi	78.1	60.4	73.5
Upper Shelf Impact Energy, ft-lb	90	117	75
Impact Energy at -50 F, ft-lb	61	107	–
Impact Energy at -10 F, ft-lb	47	92	–
40 ft-lb Impact Transition Temperature, F	-8	-86	0
40-mil Lateral Expansion Transition Temperature, F	24	-56	–
50% Shear Fracture Appearance Transition Temperature, F	50	-12	–

Material: ASTM A542, Quenched and Tempered Plates, 3/4 inch thick.

Time-Temperature Parameter = T (20 + log t)
where
T = tempering temperature, R
t = tempering time, hr.

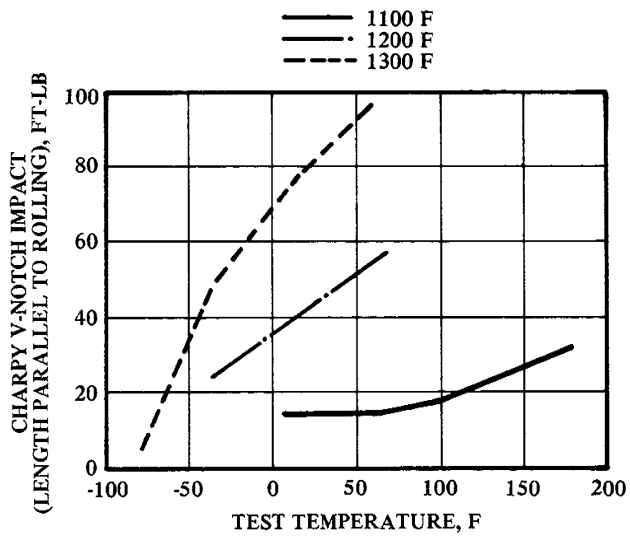
IMPACT PROPERTIES OF WELDMENTS IN 2-1/4Cr-1Mo STEEL (64)

Material	A	B	C	D
50 ft-lb Transition Temperature, F	84	14	96	56
35-mil Lateral Expansion Temperature, F	52	8	58	45
50% Shear Fracture Appearance Transition Temperature, F	90	66	86	66
Nil-Ductility Temperature, F	30	30	20	30
200 F Upper Shelf Energy, ft-lb	87	140	113	95

Composition: 0.11C, 0.41Mn, 0.014P, 0.026S, 0.19Si, 2.00Cr, 0.12Ni, 1Mo, <0.005Al, 0.013As, 0.007Sb, 0.012Sn, 0.33Cu.

A – Air melted; annealed 3 hr at 1700 F, furnace cooled at a maximum rate of 100 F/hr to less than 900 F
B – Same as A + postweld heat treatment for 4 hr at 1340 F, air cooled
C – Same as B + 1000 hr at 950 F, air cooled
D – Normalized 3 hr at 1700 F, air cooled + 4 hr at 1340 F, air cooled.

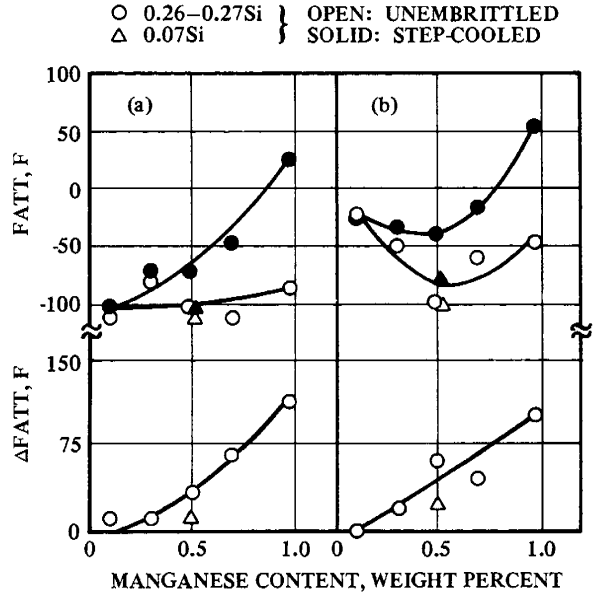
IMPACT



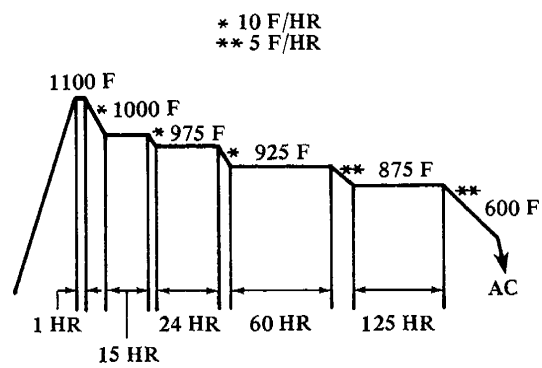
EFFECT OF TEMPERING TEMPERATURE ON IMPACT STRENGTH OF QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo STEEL (24)

Samples taken from center of plate, 7-3/16 inch thick, quenched and tempered 20 hours at indicated temperatures.

MICROSTRUCTURES:
(a) FULLY BAINITIC
(b) FERRITE PLUS PEARLITE AT 0.1% Mn, INCREASING TO FULLY BAINITIC FOR Mn LEVELS GREATER THAN OR EQUAL TO 0.5%. FATT IS 50% SHEAR-FRACTURE. APPEARANCE TRANSITION TEMPERATURE.



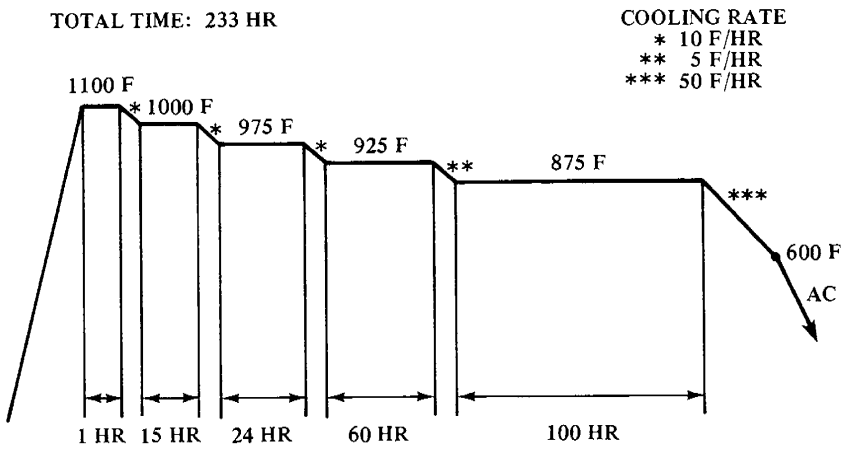
FATT IN UNEMBRITTLED AND STEP-COOLED CONDITIONS AND ΔFATT PLOTTED AGAINST MANGANESE CONTENT IN STEELS WITH TWO LEVELS OF SILICON CONTENT. AUSTENITIZING CONDITION IS 1742 F FOR 3 HR; COOLING RATES ARE (a) 117 F/MIN AND (b) 40 F/MIN. TEMPERING CONDITION IS 1275 F FOR 7-1/2 HR FOLLOWED BY AIR COOLING (2)

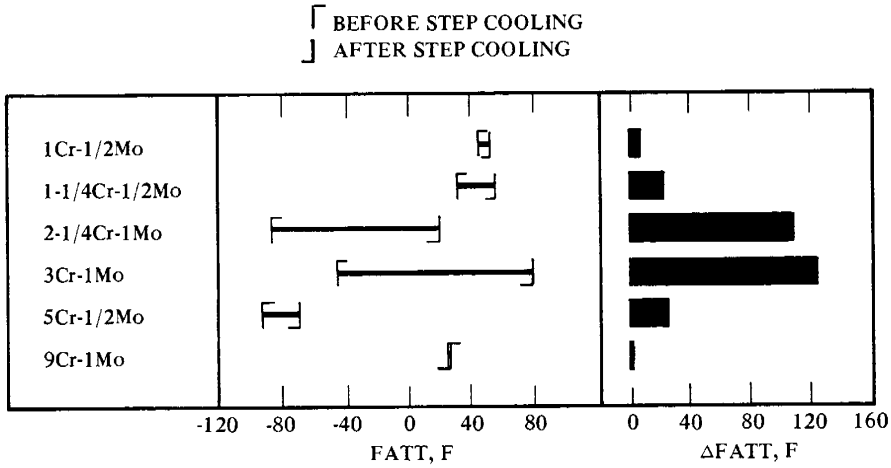


STEP-COOLING TREATMENT USED IN THE EXPERIMENT (2)

The step cooling promotes temper embrittlement; it is designed to simulate temper embrittlement corresponding to 1000 hours of exposure at 900 F.

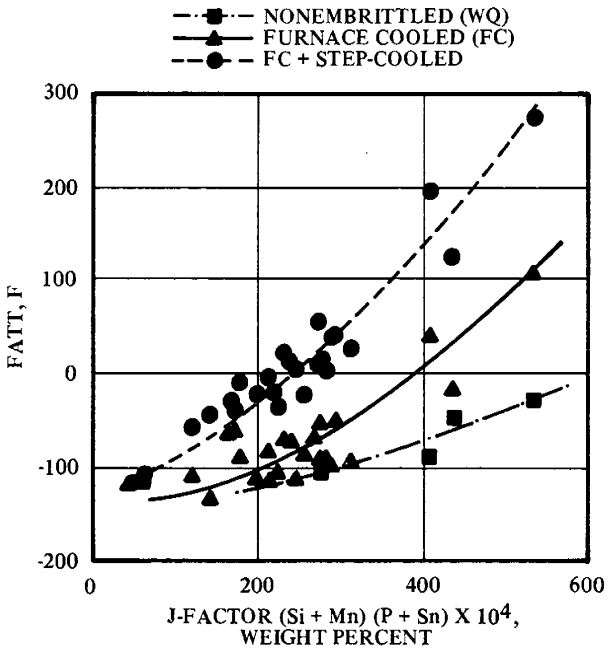
IMPACT





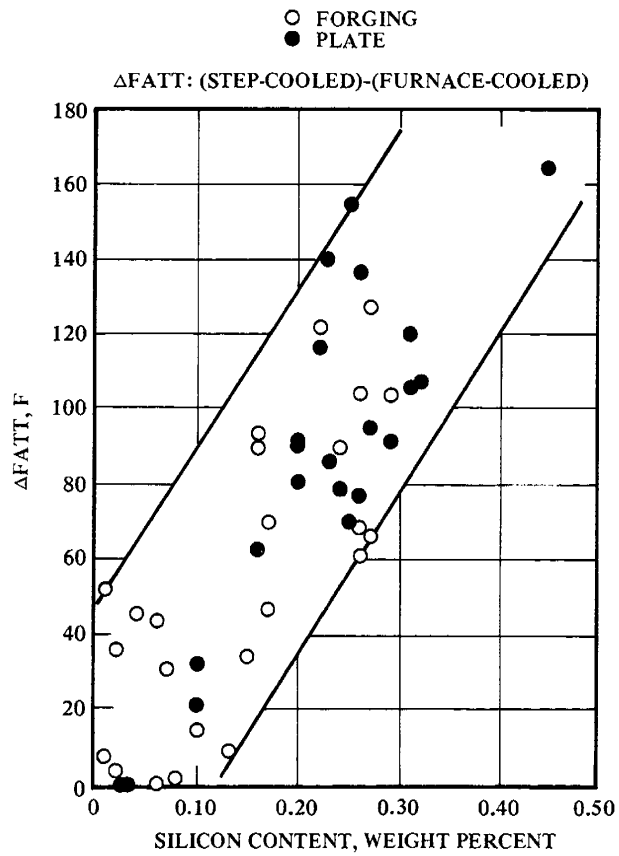
COMPARISON OF TEMPER EMBRITTLEMENT SUSCEPTIBILITY AMONG CHROMIUM-MOLYBDENUM PRESSURE VESSEL STEELS (2)

The step cooling is designed to simulate temper embrittlement conditions corresponding to 1000 hours of exposure at 900 F.



EFFECT OF J-FACTOR ON FATT OF 2-1/4Cr-1Mo STEEL PLATES IN NONEMBRITTLED, FURNACE-COOLED, AND STEP-COOLED CONDITIONS (2)

WQ = water quenched from tempering temperature; FC = furnace cooled from tempering temperature at 50 C/hr; SC = FC plus step cooling. FATT corresponds to 50% shear-fracture appearance transition temperature. The step-cooling treatment is chosen to simulate temper embrittlement due to 1000-hour exposure at 900 F.



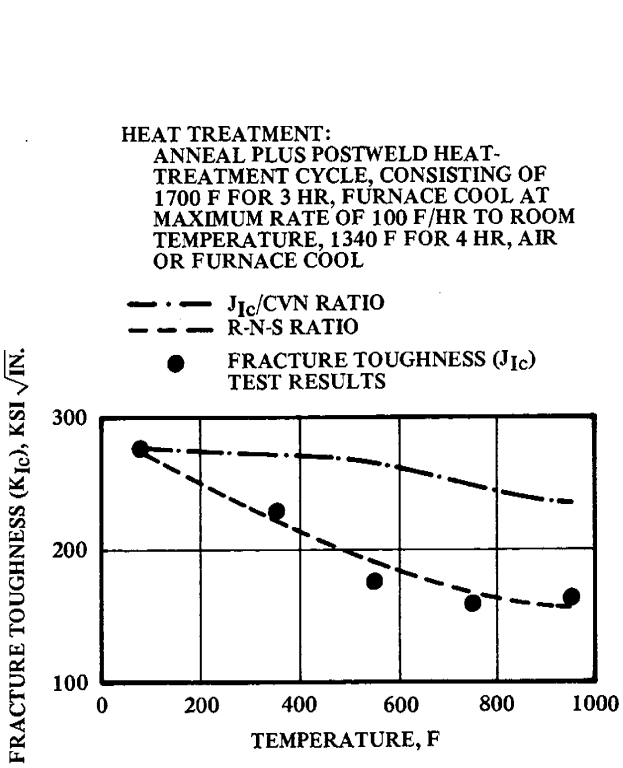
EFFECT OF SILICON CONTENT ON THE SHIFT OF FATT (ΔFATT) OF 2-1/4Cr-1Mo STEEL CAUSED BY STEP-COOLING (2)

FATT corresponds to 50% shear fracture appearance transition temperature. The step-cooling treatment is chosen to simulate temper embrittlement due to 1000-hour exposure at 900 F.

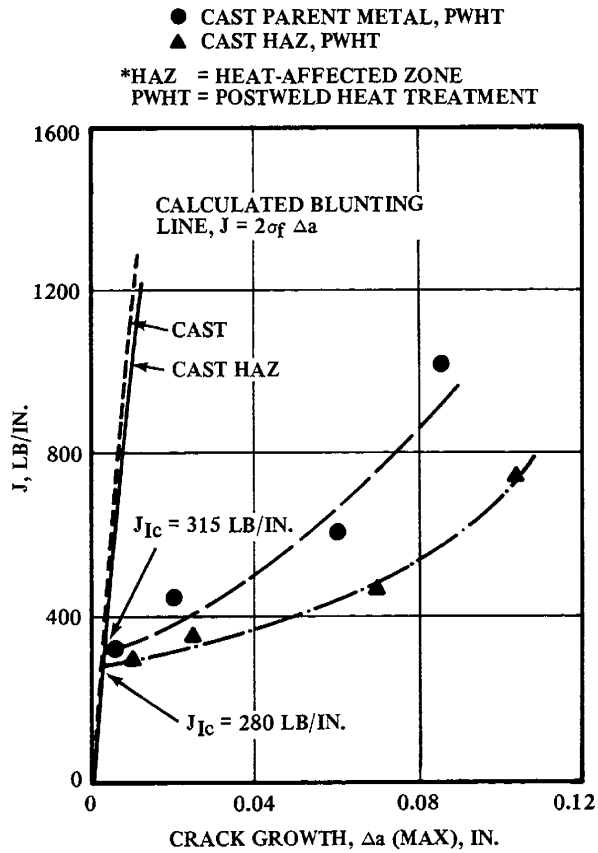
EFFECT OF HEAT TREATMENT ON FRACTURE TOUGHNESS[a] (64)

SI No.	Material Condition	Fracture Toughness (J_{Ic}), lb/in.	Equivalent Fracture Toughness (K_{Ic}), ksi $\sqrt{in.}$
A	Annealed 1700 F for 3 hr, furnace cooled at maximum rate of 55.5 C/hr to less than 900 F	1050	187
B	Annealed + postweld heat treated (PWHT): Condition A above + 1340 F for 4 hr, air cool or furnace cool	1160–2343	196–275
D	Anneal + PWHT + embrittlement treatment: Condition B above + 950 F for 1000 hr, air cool	1400	215
E	Normalize and temper: 1700 F for 3 hr, air cool, 1340 F for 4 hr, air cool	1310	208

(a) Room-temperature data.

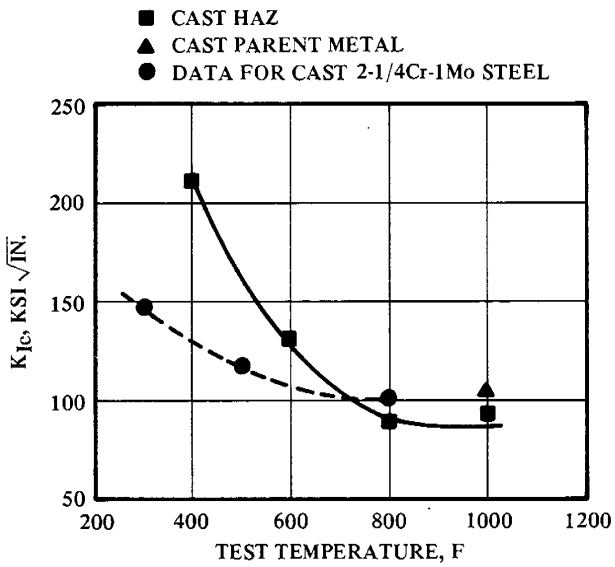


FRACTURE TOUGHNESS OF 2-1/4Cr-1Mo STEEL AT VARIOUS TEMPERATURES (64)

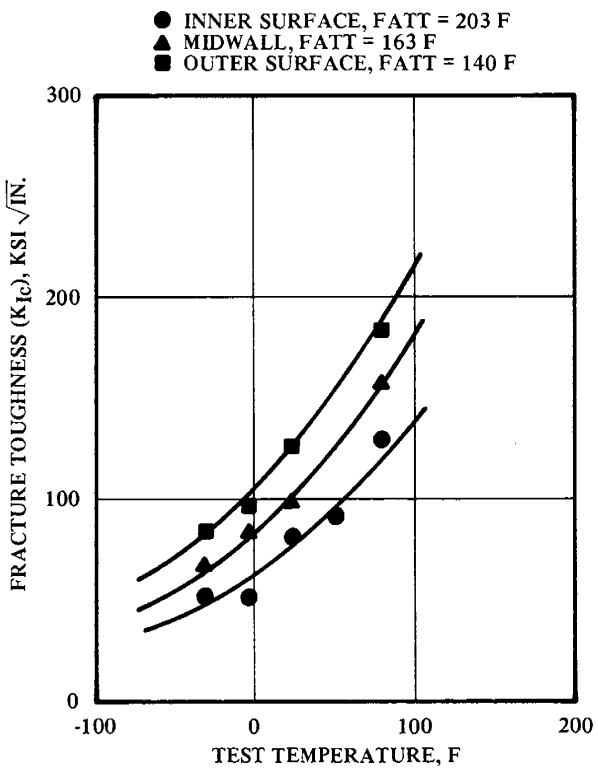


J-R CURVE FOR CAST PARENT METAL AND CAST HAZ OF 2-1/4Cr-1Mo STEEL (66)

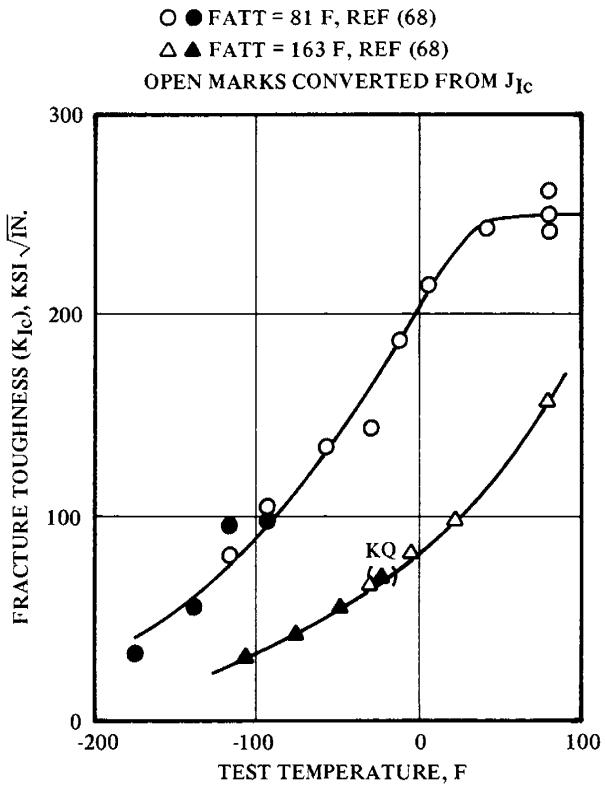
FRACTURE



VARIATION OF K_{Ic} WITH TEST TEMPERATURE IN CAST PARENT METAL AND CAST HAZ REGION OF 2-1/4Cr-1Mo STEEL WELDMENTS (66)

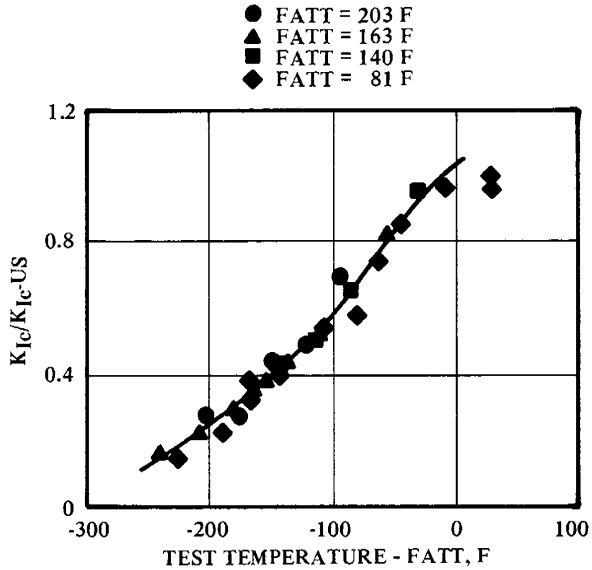


TEMPERATURE DEPENDENCE OF FRACTURE TOUGHNESS OF 2-1/4Cr-1Mo STEEL AFTER 30,000-HOUR SERVICE AT TEMPERATURES BETWEEN 630 AND 810 F (67)



TEMPERATURE DEPENDENCE OF FRACTURE TOUGHNESS OF TWO TEMPER-EMBRITTLED 2-1/4Cr-1Mo STEEL FORGINGS (67,68)

FATT = 50% SHEAR FRACTURE APPEARANCE TRANSITION TEMPERATURE
K_{Ic} = FRACTURE TOUGHNESS AT A PARTICULAR TEMPERATURE
K_{Ic}-US = FRACTURE TOUGHNESS CORRESPONDING TO THE UPPER SHELF TEMPERATURE



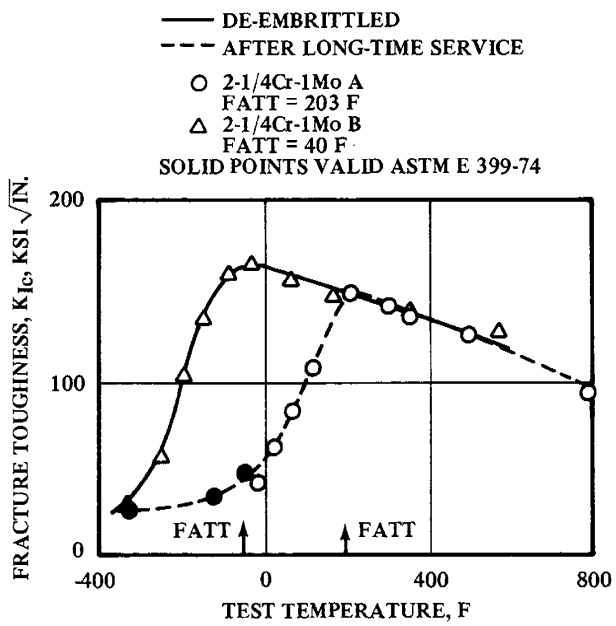
RELATION BETWEEN FRACTURE TOUGHNESS, K_{Ic}/K_{Ic}-US AND EXCESS TEMPERATURE FOR 2-1/4Cr-1Mo STEEL FORGINGS (67)

FRACTURE-TOUGHNESS TEST RESULTS AT 1000 F FOR 2-1/4Cr-1Mo STEEL[a] (69)

Material	J_{Ic}, in.-lb/in.^{2}	K_{Ic}[b], ksi $\sqrt{\text{in.}}$	Tearing Modulus (T)[c]
Composite[d]	1163	164	366
Composite	1119	160	188
Composite	843	139	202
Composite	320	86	177
Composite	174	63	95
Weld	1132	161	312
Weld	512	109	215

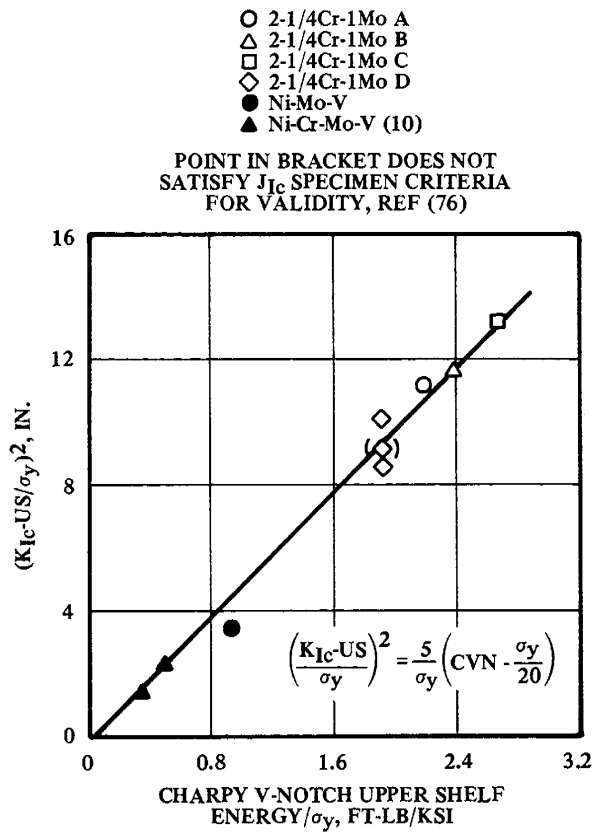
(a) Material taken from a steam pipe that had been in operation for 15 years. The operating temperature was approximately 1000 F.
(b) $K_{Ic} = \sqrt{J_{Ic}E}$ where J_{Ic} is determined from fracture-toughness testing and E is Young's modulus.
(c) $T = (E/\sigma^2_f)\, dJ/da$ where σ_f (flow stress) = $1/2(\sigma_y + \sigma_u)$, σ_y is yield strength, σ_u is ultimate tensile strength and dJ/da is the slope of the J-integral versus crack extension (a) curve (J_R curve).
(d) Composite specimen with the cracking plane along the fusion line. The specimen consisted of base metal, weld metal, fusion line, and heat-affected zone.

FRACTURE



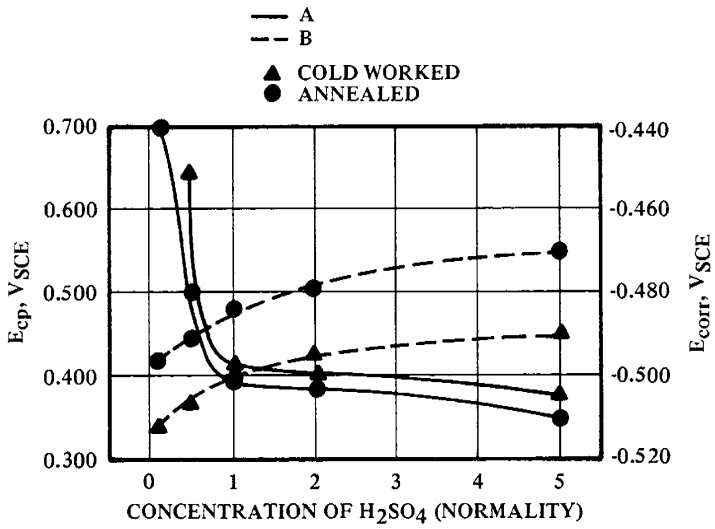
FRACTURE TOUGHNESS VERSUS TEMPERATURE BEHAVIOR OF 2-1/4Cr-1Mo STEEL SHOWING TEMPER-EMBRITTLEMENT BEHAVIOR DUE TO LONG-TIME SERVICE IN A HYDRODESULFURIZING REACTOR (2)

FATT is the 50 percent shear-fracture appearance transition temperature.

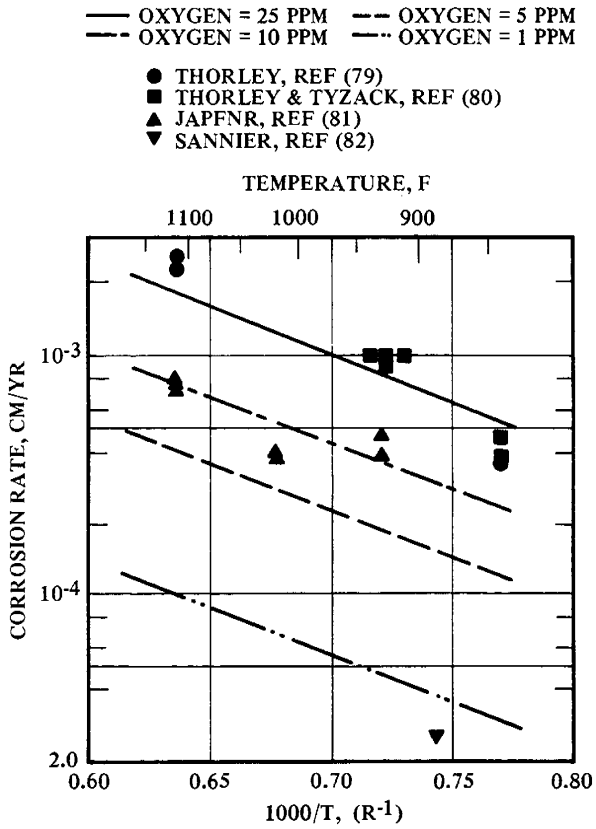


RELATIONSHIP BETWEEN K_{Ic} AND CVN ENERGY IN THE UPPER SHELF REGION (70)

K_{Ic}-US is the upper shelf fracture toughness in ksi $\sqrt{in.}$; σy is the yield strength in ksi; CVN is the Charpy V-Notch Upper Shelf Energy in ft-lb.



EFFECT OF SULPHURIC ACID CONCENTRATION ON PASSIVATION POTENTIAL E_{CP} (CURVES A) AND CORROSION POTENTIAL E_{CORR} (CURVES B) FOR ANNEALED AND COLD WORKED 2-1/4Cr-1Mo STEEL (77)

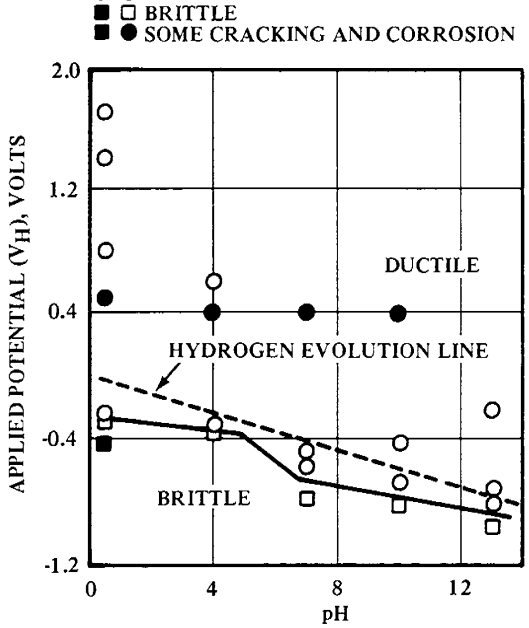


SODIUM CORROSION OF 2-1/4Cr-1Mo STEEL (78)

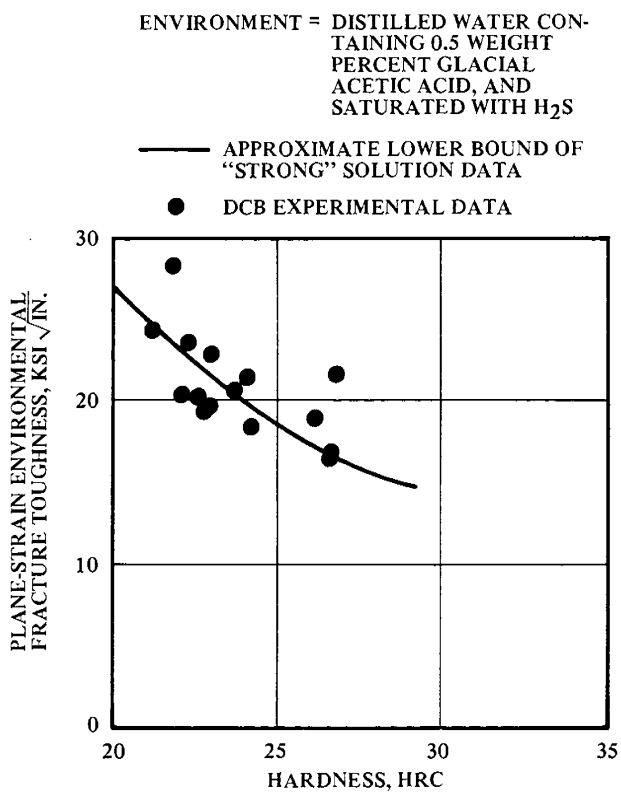
COMPARATIVE MECHANICAL PROPERTIES FOR 2-1/4Cr-1Mo STEEL IN BOILING 50 PERCENT NaOH AT DIFFERENT POTENTIALS (83)

Potential[a], mV	Time, hr	Ultimate Tensile Strength, ksi	τ Fracture, ksi	Reduction in Area, percent	Ratio of . . . Environment/Air: Ultimate Tensile Strength	τ Fracture	Time	Reduction in Area
-1100	29.90	66.105	52.188	40.15	0.888	1.084	0.756	0.49
-912	5.50	29.828	0.548	95.00	0.400	0.011	0.139	1.17
-912	7.50	37.950	–	95.71	0.509	–	0.189	1.18
-912	7.25	68.886	11.597	93.02	0.925	0.241	0.183	1.15
-912[b]	6.20	65.505	42.110	46.22	0.879	0.565	0.157	0.57
-825	22.50	60.306	41.750	39.10	0.804	0.867	0.569	0.48
-800	29.25	63.875	44.650	61.20	0.856	0.927	0.740	0.73
-750	27.40	64.365	30.153	42.98	0.864	0.626	0.693	0.53
-700	30.50	56.827	36.183	63.90	0.763	0.751	0.772	0.79
-700	21.83	62.046	41.750	57.00	0.836	0.867	0.555	0.70
-600	27.60	61.002	38.851	70.80	0.819	0.807	0.698	0.87
-600	30.90	63.875	40.590	59.58	0.856	0.843	0.782	0.74
-400	23.25	63.875	33.052	63.50	0.836	0.867	0.588	0.78
Air	39.50	74.455	48.129	80.80	1.000	1.000	1.000	1.00

(a) Potential, mV, with respect to Hg/HgO electrode.
(b) Higher strain rate: 10^{-5}/sec.



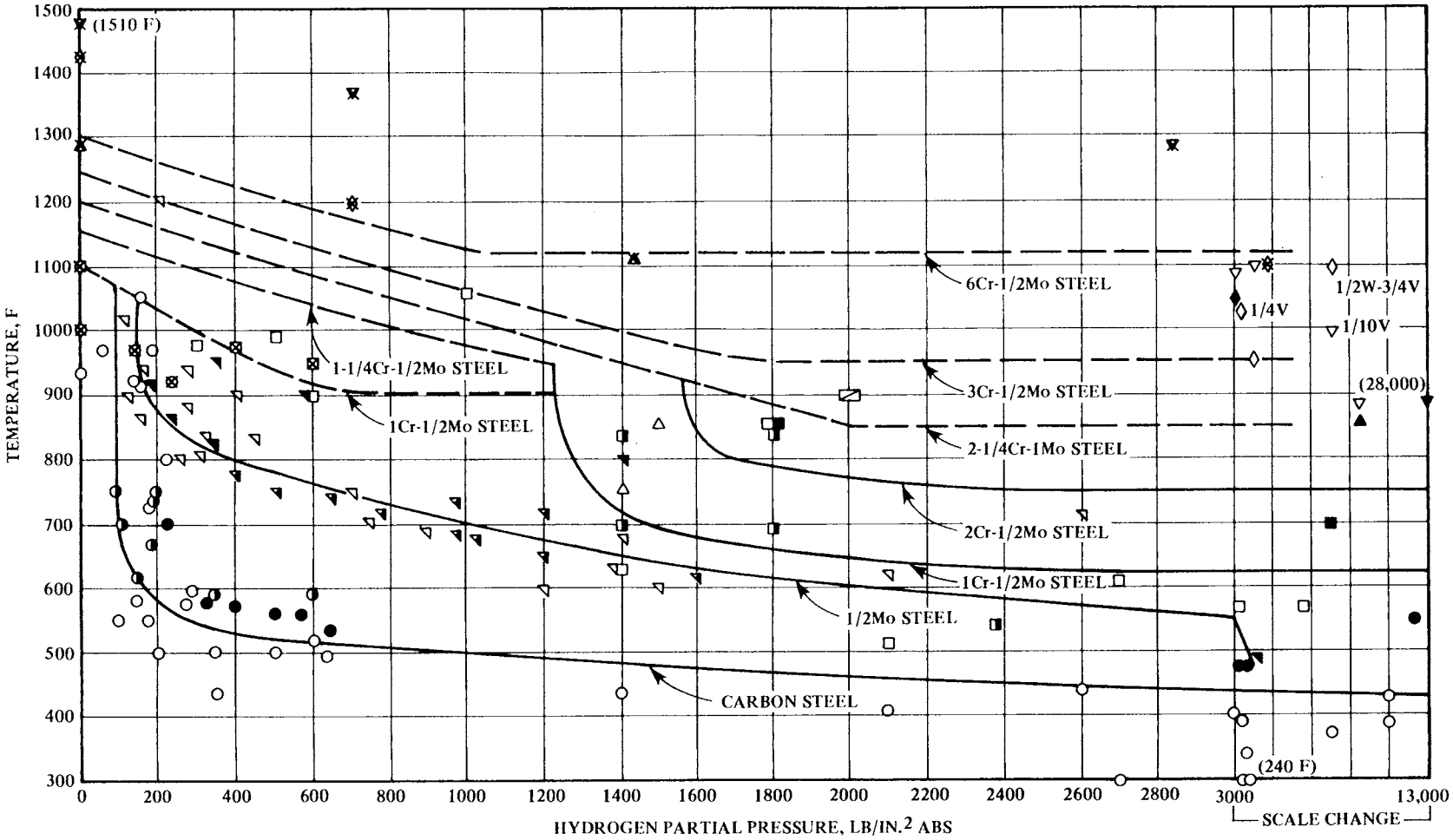
FRACTURE APPEARANCE OF SLOW-STRAIN-RATE TEST SPECIMENS AS A FUNCTION OF POTENTIAL AND pH (84)



THRESHOLD K_{Iscc} OF QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo IN A SULFIDE ENVIRONMENT (85)

HYDROGEN EMBRITTLEMENT

2¼ Cr-1 Mo STEELS



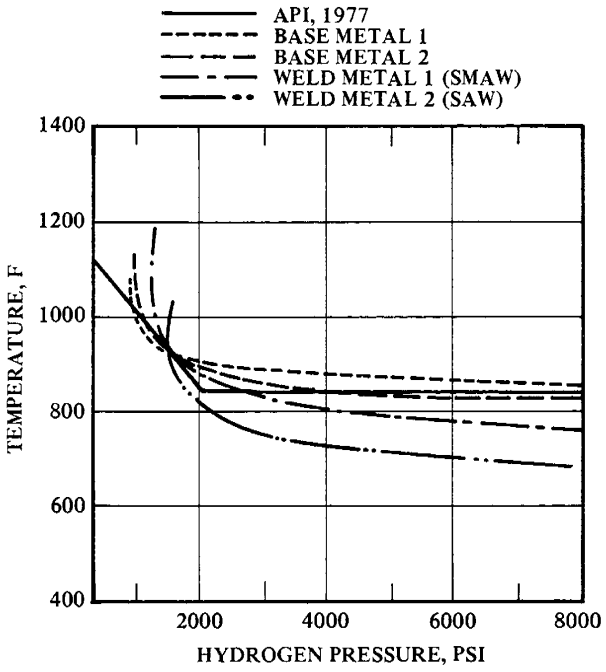
LEGEND

SURFACE DECARBURIZATION — — —
INTERNAL DECARBURIZATION ——— (HYDROGEN ATTACK)

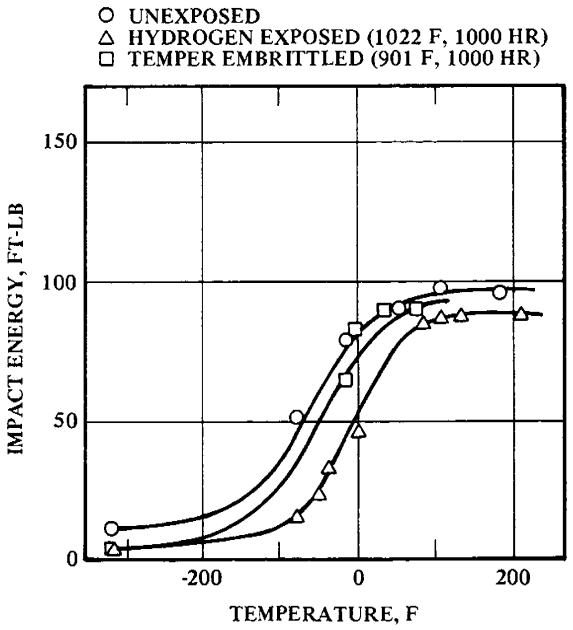
	LOW CARBON	1/2Mo	1Cr-1/2Mo	2Cr-1/2Mo	2-1/4Cr-1Mo	3Cr-1/2Mo	6Cr-1/2Mo
SATISFACTORY	○	◁	□	△	□	◇	▽
HYDROGEN ATTACK	●	◀	■	▲	■	◆	▼
DECARBURIZATION	⊠	◁	⊠	⊠	⊠	⊠	⊠
SEE FOOTNOTES	◑	◀	◨	▲	◨	◆	▼

NOTES:
1. AUSTENITIC STAINLESS STEELS ARE SATISFACTORY AT ALL TEMPERATURES AND PRESSURES.
2. HALF-FILLED SYMBOLS REPRESENT MIXED BEHAVIOR; SEE REFERENCE (86) FOR DETAILS.

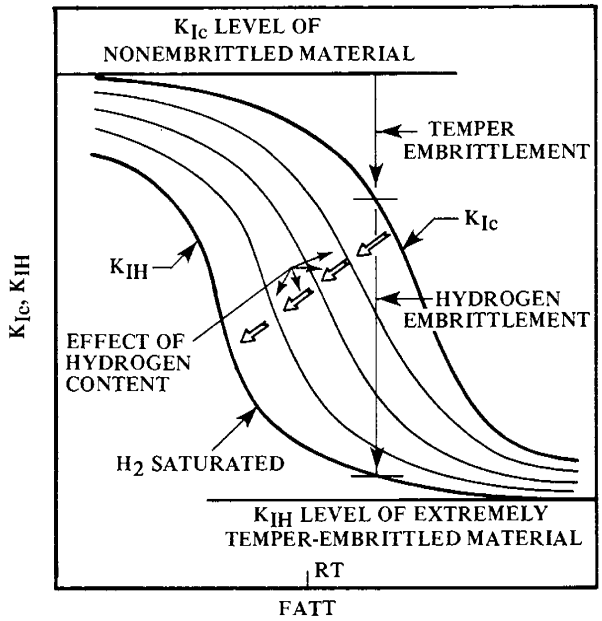
NELSON CURVES -- OPERATING LIMITS FOR STEELS IN HYDROGEN SERVICE (86)



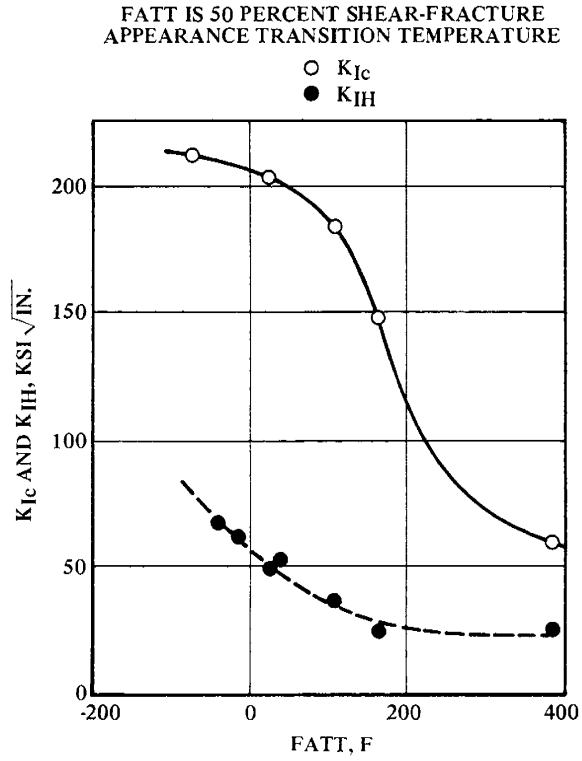
SUGGESTED CURVES OF "USEFUL LIFE" DRAWN FOR THE BASE METALS AND THE WELD METALS. THE API (1977) NELSON CURVE FOR 2-1/4Cr-1Mo STEEL IS ALSO INCLUDED IN THE FIGURE. THE SUGGESTED CURVES ARE BASED ON A HYDROGEN ATTACK INDUCED VOLUMETRIC INCREASE OF THE STEEL OF 0.1 PERCENT IN 50,000 HOURS (87)



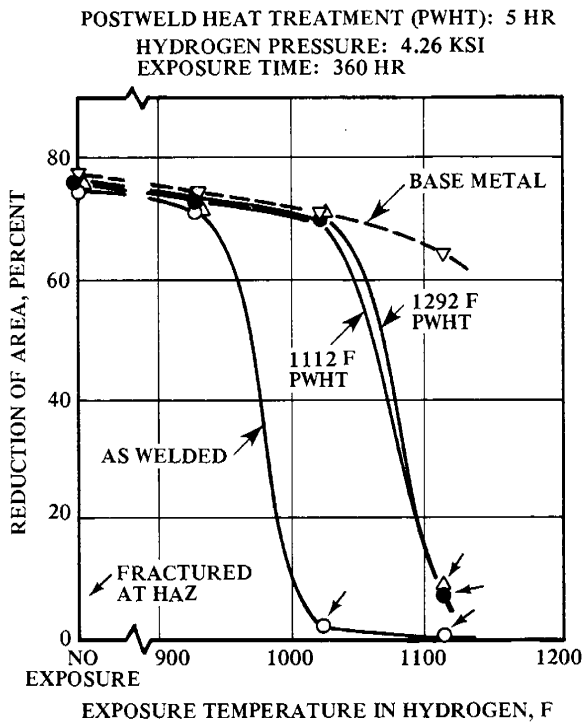
CHARPY V-NOTCH TRANSITION CURVES OF QUENCHED-AND-TEMPERED 2-1/4Cr-1Mo BASE STEEL BEFORE AND AFTER EXPOSURE TO HYDROGEN AND CHARPY V-NOTCH CURVES AFTER TEMPER EMBRITTLEMENT (88)



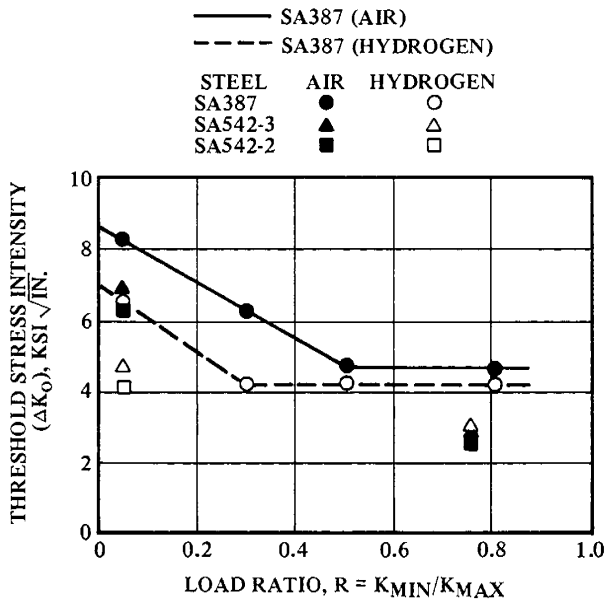
SCHEMATIC ILLUSTRATION OF THE CORRELATION BETWEEN ROOM TEMPERATURE K_{Ic} AND K_{IH} AND FATT (2)



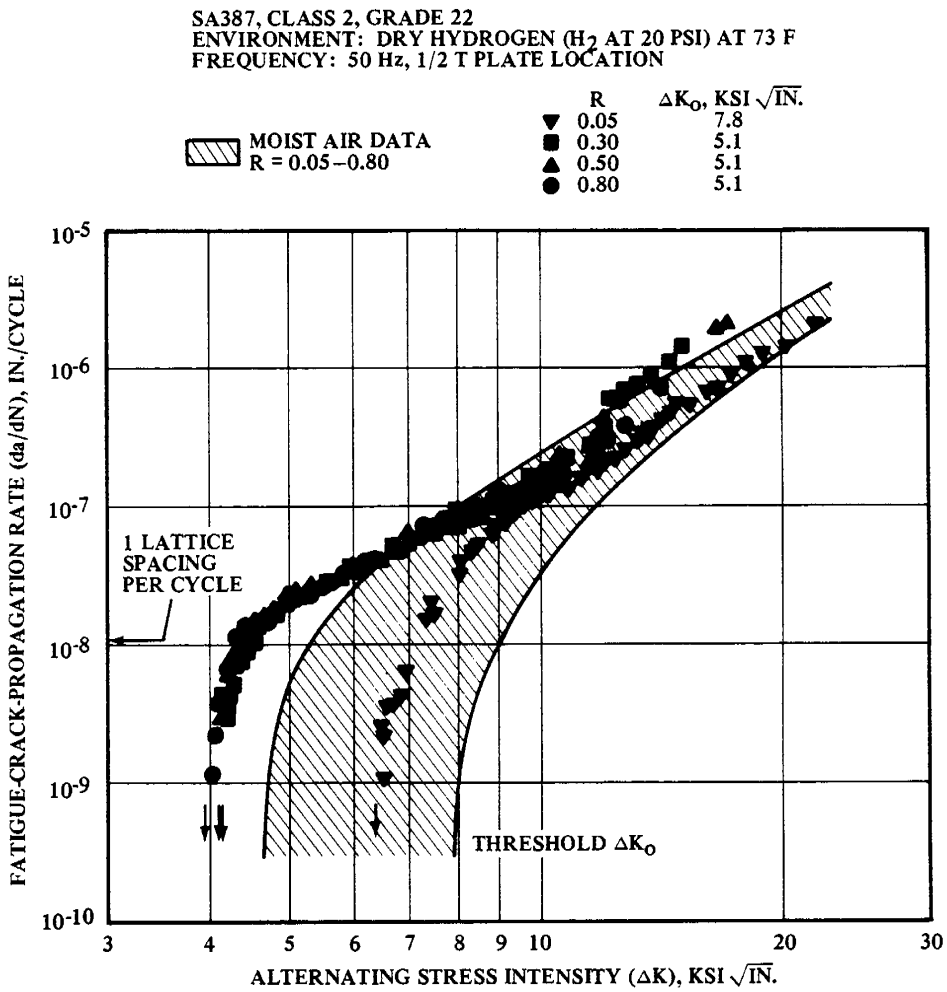
ROOM TEMPERATURE (K_{Ic} AND K_{IH}) VERSUS FATT OF 2-1/4Cr-1Mo STEEL (2)



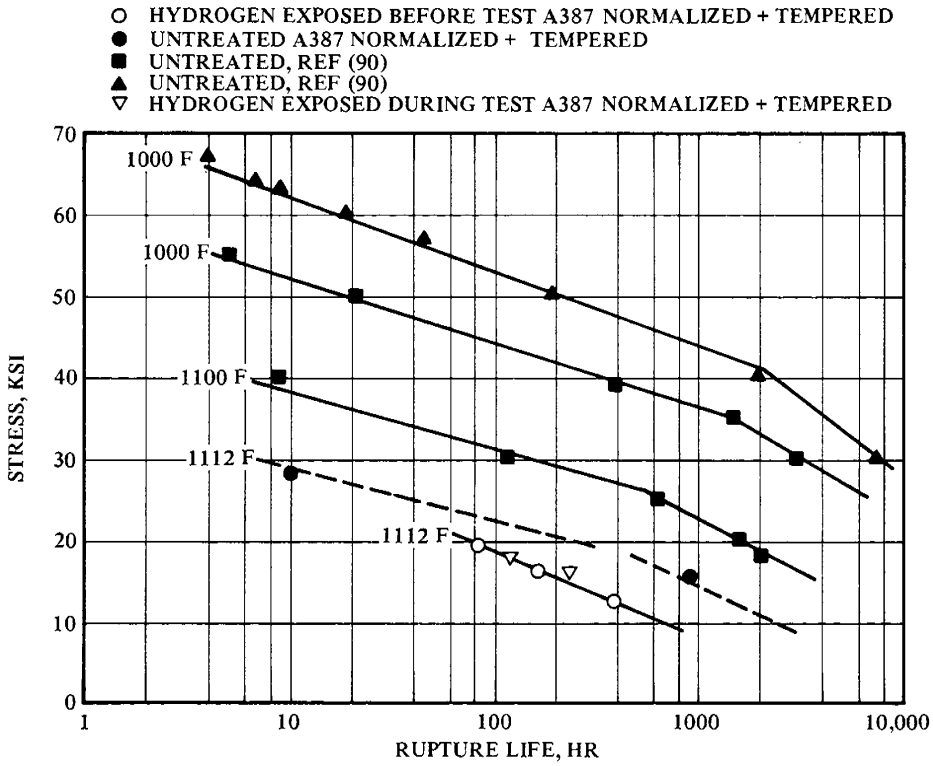
EFFECT OF POSTWELD HEAT TREATMENT ON RESISTANCE TO HYDROGEN ATTACK OF 2-1/4Cr-1Mo WELD JOINT (2)



VARIATION OF THRESHOLD STRESS INTENSITY WITH LOAD RATIO FOR 2-1/4Cr-1Mo STEELS TESTED AT 50 Hz IN AIR AND HYDROGEN GAS (51)

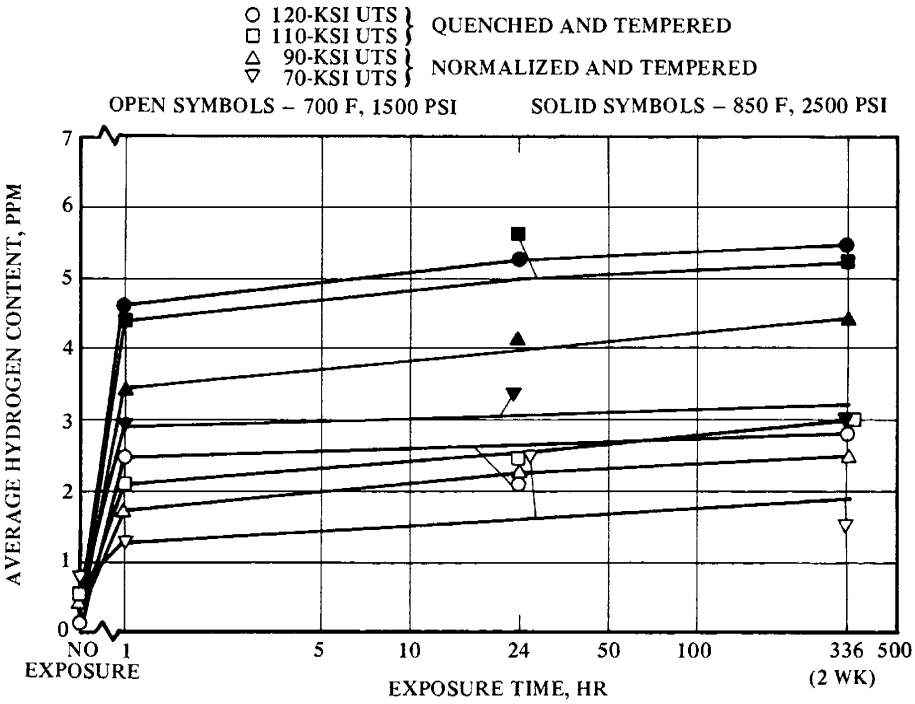


VARIATION OF FATIGUE CRACK PROPAGATION RATES, da/dN, WITH ALTERNATING STRESS INTENSITY, ΔK, IN 2-1/4Cr-1Mo STEEL TESTED AT 50 Hz IN DEHUMIDIFIED GASEOUS HYDROGEN AS A FUNCTION OF LOAD RATIO, R (51)

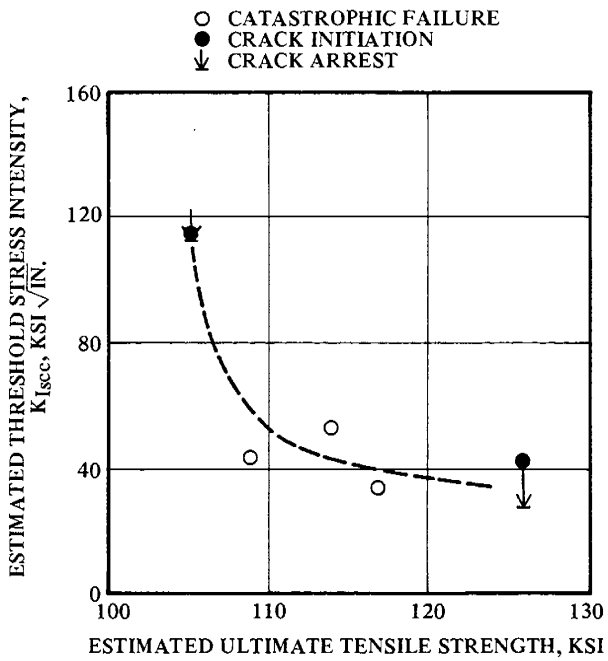


STRESS-TO-RUPTURE DATA FOR ASTM A387 (NORMALIZED AND TEMPERED) AND OTHER 2-1/4Cr-1Mo STEELS (89)

The open circular symbols correspond to pre-exposure of samples for 20 days at 600 C (1112 F) under a hydrogen pressure of 2000 psig. The inverted open symbols correspond to in situ exposures during the tests at 600 C (1112 F) in 2000 psig hydrogen gas.

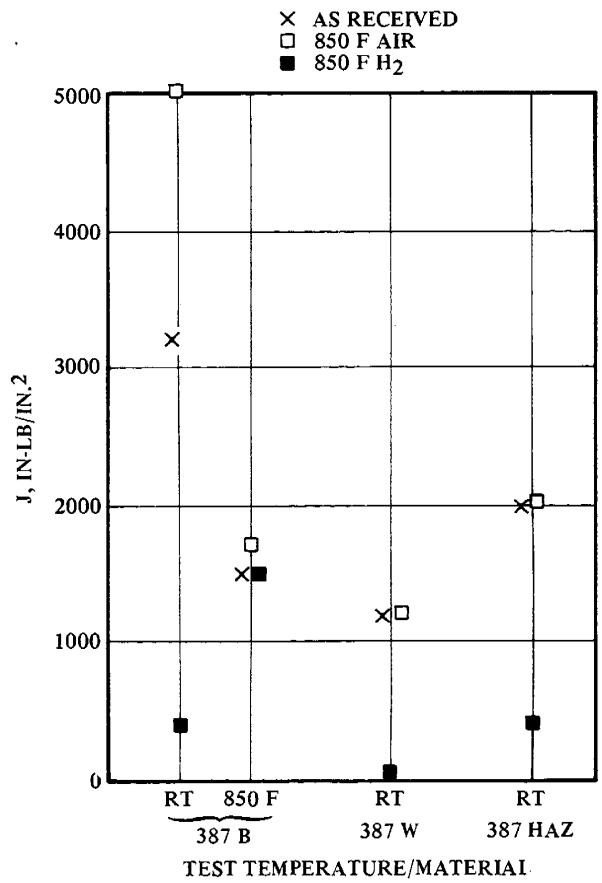


HYDROGEN CONTENT VERSUS EXPOSURE TIME FOR 2-1/4Cr-1Mo STEEL COUPONS EXPOSED TO HIGH-TEMPERATURE, HIGH-PRESSURE HYDROGEN GAS (91)



K_{Iscc} AS A FUNCTION OF THE ULTIMATE TENSILE STRENGTH FOR 2-1/4Cr-1Mo STEEL (6)

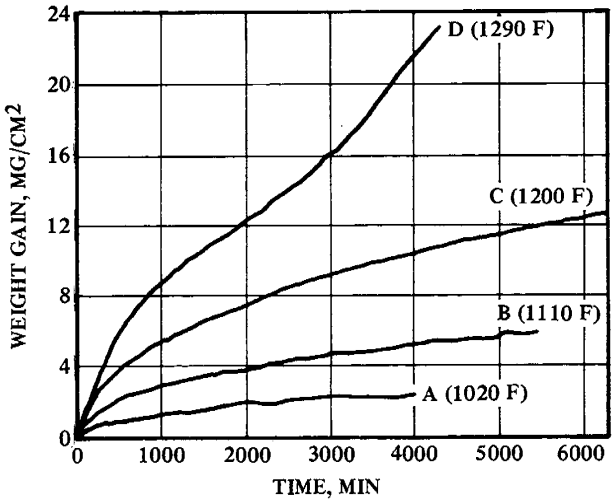
The K_{Iscc} was estimated from tests in 50 psig H_2S on 2T CT side-grooved specimens and the ultimate tensile strength from Rockwell hardness measurements.



ESTIMATED J_{Ic} VERSUS TEMPERATURE AND MATERIAL FOR VARIOUS EXPOSURE CONDITIONS FOR 2-1/4Cr-1Mo STEEL (ASTM A387, GRADE 22, CLASS 2) (92)

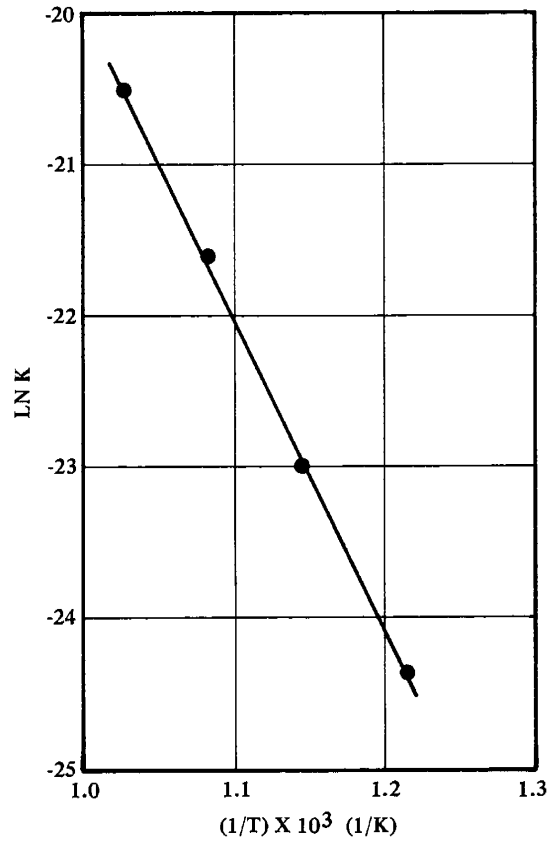
Compact tension specimens were pre-exposed either in air or 3500 psi hydrogen gas for 6 months at 850 F, and subsequently fracture tested in air at room temperature. B, W, and HAZ refer to base metal, weld metal and heat-affected zones, respectively. The cross symbols refer to tests conducted on as-received material without any pre-exposure.

OXIDATION



OXIDATION OF Fe-2-1/4Cr-1Mo STEEL IN DRY-FLOWING OXYGEN FOR PERIODS OF UP TO 100 HOURS (93)

ACTIVATION ENERGY = 167 kJ/MOLE



ACTIVATION ENERGY PLOT FOR Fe-2-1/4Cr-1Mo OXIDIZED BETWEEN 1020 AND 1290 F (93)

VALUES OF k IN THE EQUATION $W^n = kt$, ASSUMING n = 2 (93)

Temperature, F	Time Period Over Which Analysis is Valid, min	Parabolic Rate Constant (k), $g^2\ cm^{-4}\ sec^{-1}$ ($x\ 10^{-10}$)
1020	400 < t	0.26
1110	400 < t	1.03
1200	210 < t	4.12
1290	600 < t < 2100	12.3

W = weight gain, gm/cm^2; t = oxidation time, sec.

EQUATION FOR OXIDE SCALE THICKNESS FOR 2-1/4Cr-1Mo STEEL IN AIR (94)

$$h = 1.26 \times 10^{-3}\, t^{1/2} \exp\left(\frac{12210}{RT}\right)$$

where

h = oxide thickness, meters
t = time, hours
T = temperature, K
R = gas constant, 8.3 J/mole · K.

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